

CONTAINER TERMINAL OPERATIONS ANALYTICS

Data Analytics and Business Intelligence Project Report

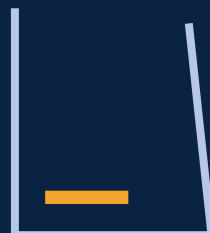
Data preparation - Excel - Power Query - SQL - Python - Semantic modeling -
DAX - Power BI

Damietta Terminal Analytics

Primary comparison: Q1 2025 versus Q1 2026

Data scope: Available operational records covering the 2025–
2026 period

Data Analytics Project Report
Version 1.0 - 18 August 2026



Container Terminal Operations Analytics

Complete Project Documentation

Project type	Data Analytics and Business Intelligence Project
Primary business scope	Q1 2025 versus Q1 2026, with supporting analysis across available 2025–2026 records
Primary platform	Microsoft Power BI Desktop and Power Query
Supporting tools	Excel, SQL Server, Python/Colab, DAX
Documentation status	Reorganized, reconciled, validated, and fully traceable final documentation package
Analytical status	Final analysis retained without recomputation or changes to the established conclusions

Project Team

The project was completed through the coordinated contributions of seven team members who participated across the different stages of data preparation, analysis, modeling, visualization, validation, documentation, and final project integration.

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Document Control and Package Contents

Document title	Container Terminal Operations Analytics: Complete Project Documentation
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Executive Summary

This project presents the development of a complete data analytics and business intelligence solution for container-terminal operations. The work transforms operational source records into a structured analytical environment that supports data preparation, validation, modeling, KPI development, dashboarding, and management reporting.

The analysis focuses on the main operational areas affecting terminal performance, including container and TEU flow, dwell time, vessel activity, equipment workload, cargo and commodity patterns, port connectivity, weight characteristics, and data quality. Multiple analytical tools were used throughout the project, but all workstreams were developed as part of one coordinated process using a common data foundation and shared validation principles.

The final results show that terminal performance should not be assessed through throughput alone. Higher container and TEU activity must be considered together with dwell performance, equipment productivity, vessel waiting and berth time, workload concentration, and the reliability of the underlying operational data. The analysis also demonstrates the importance of distinguishing reporting-period flow from current inventory and of validating source values before they are used in management KPIs.

The completed solution includes a governed semantic model and a nine-page Power BI dashboard suite that progresses from executive-level performance monitoring to detailed operational analysis and container-level traceability. Together, the project deliverables provide a structured decision-support framework for understanding terminal performance, identifying areas that require further investigation, and supporting more consistent and evidence-based operational review.

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1. Project Overview, Objectives, and Scope

This project developed a complete data analytics and business intelligence solution for container-terminal operations using the available operational records covering the 2025–2026 reporting period. The objective was to transform raw operational data into a structured, validated, and management-ready analytical environment capable of supporting performance monitoring, operational assessment, and informed decision-making.

The project covered the full analytical lifecycle, including dataset research and acquisition, data collection, data profiling, cleaning and preprocessing, data-quality assessment, data dictionary development, SQL-based exploration and validation, Power Query transformation, fact and dimension table creation, semantic data modeling, relationship management, DAX measure development, KPI definition, dashboard design, documentation, and final quality review.

The original source dataset contains 98,052 operational rows across 33 source fields. Following data preparation, cleaning, and validation procedures, the analytical layer contains 95,870 operation records. These totals represent different stages of the data-processing workflow and are therefore maintained separately to preserve traceability and analytical governance.

The analytical solution evaluates terminal performance across several interconnected areas, including container volume and TEU movement, dwell time, vessel activity, berth and turnaround performance, equipment workload, cargo and commodity characteristics, port connectivity, operational trends, and data-quality conditions. Selected analyses also provide direct comparisons between Q1 2025 and Q1 2026 to support period-over-period performance assessment.

The final project deliverables include the complete project report, validated analytical outputs, data models, KPI definitions, dashboards, supporting technical documentation, and a separate management presentation. Together, these deliverables provide both the technical foundation and the management-level interpretation required to understand terminal performance and identify areas requiring further operational investigation.

1.1. Project objective

The project objective was to develop a management-ready, technically governed, and reliable analytical solution for container-terminal operations. The work began with raw operational extracts rather than a prestructured analytical warehouse, requiring the team to establish the complete analytical workflow from source data to final reporting.

Accordingly, the project encompassed business and operational interpretation, dataset assessment and profiling, data cleaning and standardization, data-quality verification, key and relationship definition, fact and dimension table creation, dimensional and semantic modeling, KPI and measure development, analytical validation, dashboard development, and management-oriented reporting. Particular emphasis was placed on ensuring that the analytical outputs were consistent, traceable, and supported by clearly defined business logic.

The resulting solution was designed not simply to visualize operational data, but to convert the available records into a structured decision-support environment. It enables terminal performance to be examined across operational volume, container flow, dwell and turnaround time, vessel activity, equipment workload, trade connectivity, and data-quality conditions, while maintaining validation controls that support reliable interpretation and management decision-making.

1.2. Central business question

The final presentation states the central question as:

How efficiently is the terminal converting vessel calls and container moves into throughput, and where is that efficiency being lost?

The question is decomposed into four decision areas:

1. **Volume and flow:** entries, exits, TEU, vessel calls, voyages, and operator workload.
2. **Efficiency:** completed dwell, pre-berth waiting, berth occupancy, vessel turnaround, and validated QC productivity.
3. **Risk and data quality:** invalid date sequences, unknown mappings, weight readiness, damage condition, and conflicting QC-hour values.
4. **Connectivity:** loading/discharge ports, countries, regions, and trade-lane concentration.

1.3. Evidence-supported objectives

1. Preserve the source through a traceable staging layer.
2. Standardize headers, types, text, blanks, dates, and operational codes.
3. Convert repeated descriptors into reusable dimensions.
4. construct facts at explicit container-operation, QC-group, and vessel-call grains.
5. Establish entry- and exit-date roles through a governed date table.
6. Prevent repeated QC-hour values from being summed or averaged without grain validation.
7. Create governed measures for flow, inventory, dwell, TEU, weight, vessels, quality, equipment, and connectivity.
8. Build validation views before management dashboards.
9. Deliver a final presentation that connects problem, insight, story, and action.

1.4. Analytical Scope

The project analysis covers the available container-terminal operational records associated with the 2025 and 2026 reporting period. The full source extract contains 98,052 operational rows and 33 original source fields, while the cleaned analytical dataset contains 95,870 validated operation records after data preparation and quality-control procedures.

These record counts represent different stages of the project data pipeline and are therefore preserved as distinct source and analytical layers. The original source extract reflects the broader operational data received for the project, whereas the cleaned analytical layer represents the records retained for structured analysis, validation, modeling, KPI development, and reporting. The two totals should not be interpreted as interchangeable measures, since each corresponds to a different stage of data processing and analytical governance.

Within this broader scope, selected comparative analyses focus specifically on Q1 2025 versus Q1 2026 to support period-over-period performance evaluation. The project documentation therefore distinguishes clearly between the complete 2025–2026 operational data scope and the narrower Q1 comparative views used for selected performance assessments.

2. Dataset, Data Sources, and Business Context

2.1. Data Landscape

The project is based on operational data covering the principal activities of a container-terminal environment. For analytical purposes, the available data can be organized into several connected business domains, including vessel operations, container movements, yard activity, cargo and risk characteristics, port connectivity, and equipment utilization. These domains provide the operational context required to interpret terminal performance from multiple perspectives rather than as a single isolated measure.

The project data dictionary contains 75 documented field definitions distributed across seven logical sections. These definitions provide business meaning, expected usage, and interpretation guidance for the available fields and support consistent understanding of the data throughout the analytical workflow. The data dictionary therefore serves as an important governance reference for data preparation, modeling, KPI development, and validation.

The implemented Power BI analytical baseline is built from a Terminal Operating System export containing 33 source fields. These fields represent the core operational attributes required for the main analytical model and are used to construct the fact tables, dimension tables, relationships, measures, and reporting views developed throughout the project.

The project therefore distinguishes between the broader documented data landscape and the specific source fields implemented in the analytical model. This distinction supports transparency and ensures that documented business definitions are not automatically interpreted as fields directly used in every analytical layer.

Figure 2.1 summarizes the overall analytical workflow followed throughout the project. The process begins with raw operational data and progresses through exploration, data cleaning and preparation, KPI calculation, validation, insight development, and dashboard construction. This sequence reflects the governed analytical approach used to transform operational records into structured and decision-oriented business intelligence outputs.

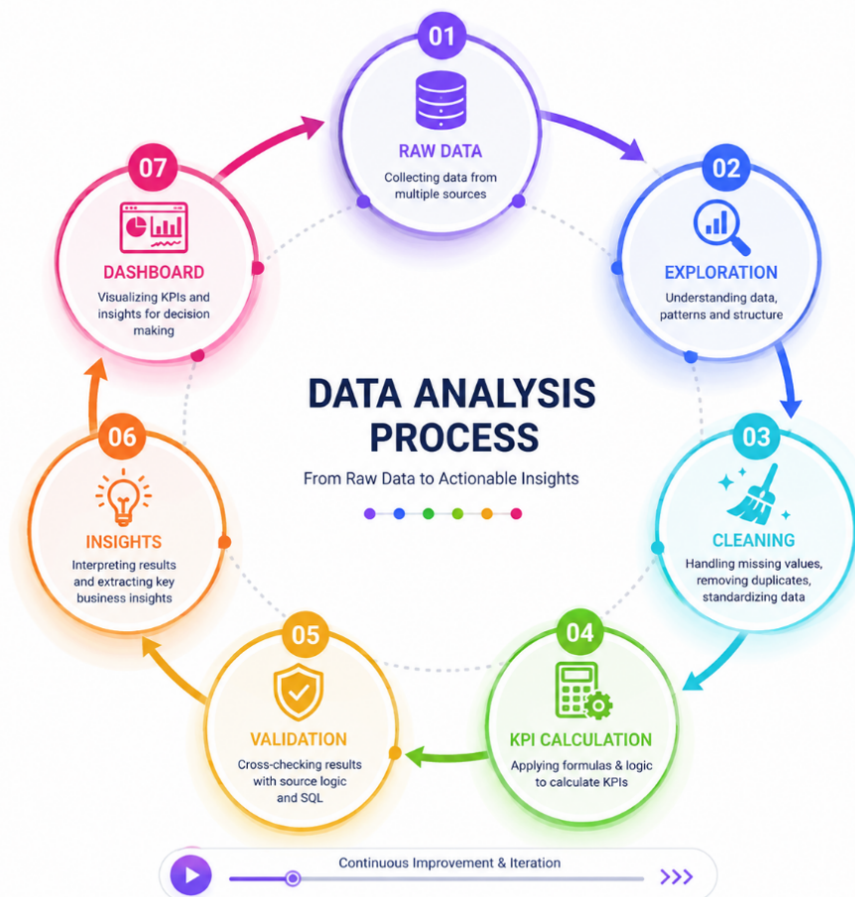


Figure 2.1.: Data-analysis workflow showing the progression from raw operational data through exploration, cleaning, KPI development, validation, insight generation, and dashboard development.

2.2. Source Fields Used in the Analytical Model

The implemented analytical baseline is built from 33 source fields extracted from the Terminal Operating System (TOS). These fields provide the operational foundation for subsequent data preparation, validation, dimensional modeling, KPI development, and dashboard analysis.

The interpretations shown in Table 2.1 reflect the business meaning applied throughout the project documentation. Where a field name or source abbreviation does not provide sufficient information to establish an exact operational definition, the interpretation is intentionally stated with appropriate caution rather than inferred without evidence.

Table 2.1.: Implemented TOS source fields and business interpretation

Field	Business interpretation used in the project
Size Type	Combined container size and equipment-type classification, including ISO-style size/type information where applicable.
Cntr Len	Nominal container length used to distinguish major equipment sizes and support container-capacity and TEU-related analysis.

Continued on next page

Table 2.1 continued from previous page

Field	Business interpretation used in the project
Container No.	Operational container identifier. The identifier is not assumed to be globally unique across multiple terminal visits or operational lifecycles.
OPR	Operator or carrier code associated with the container record. The precise commercial or contractual role represented by the source code should be interpreted according to the originating operational system.
Status	Source operational status assigned to the container or handling record.
Out Date	Date and time associated with the container's recorded exit or completion of the observed terminal lifecycle.
Calling Seq.	Source vessel-call sequence identifier used to associate operational records with a particular call or sequence of calls.
Calling Year	Year assigned to the corresponding vessel call. The implemented project records cover 2025 and 2026.
Cargo Type	Broad classification describing the cargo category associated with the container operation.
Cntr H&W	Source dimensional classification representing container height and width characteristics.
Cntr Type	Container equipment-type classification used to distinguish different container configurations.
Commodity	Source commodity description or commodity category associated with the cargo.
Damage Condition	Recorded condition indicating the presence or classification of container damage.
Delivery	Source delivery classification or delivery-related operational status.
EQU No.	Identifier of the operational equipment associated with the recorded handling activity.
QC Working Hours	Source value representing quay-crane working hours. This field was subsequently subjected to validation at the appropriate vessel-call and equipment grain before use in productivity analysis.
Dispatch Mode (In)	Source classification describing the dispatch or handling mode associated with the inbound movement.
FE	Full/empty classification indicating whether the recorded container was loaded or empty.
IMDG	Source dangerous-goods value associated with International Maritime Dangerous Goods classification or identification.

Continued on next page

Table 2.1 continued from previous page

Field	Business interpretation used in the project
In Date	Date and time associated with the recorded entry of the container into the terminal.
LCL FCL	Classification distinguishing Less-than-Container Load (LCL) and Full-Container Load (FCL) conditions where recorded.
Mode	Source operational-mode classification describing the movement or handling context represented by the record.
POD	Port of discharge code associated with the container or voyage movement.
POL	Port of loading code associated with the container or voyage movement.
S/A	Source abbreviation retained from the operational extract. Its exact business definition was not sufficiently established and is therefore not assigned an unsupported interpretation.
Stack Days	Source-provided number of days associated with container stacking or dwell within the terminal.
Trans Type (In)	Source classification describing the transport type associated with the inbound container movement.
Trans Type (Out)	Source classification describing the transport type associated with the outbound container movement.
User Voyage	Operational voyage reference used to associate records with a vessel voyage or terminal call context.
Vessel Code	Source identifier assigned to the vessel.
Vessel Name	Vessel name recorded in the operational source data.
Weight Group	Source-defined weight-band classification used to group container records by weight category.
WGT	Source-recorded weight value. The project preserves the value as supplied and does not assign a physical unit unless that unit is independently confirmed.

The 33 fields collectively support the main analytical domains developed in the project, including container characteristics, entry and exit flows, vessel activity, cargo classification, equipment workload, dwell behavior, transport modes, and port connectivity. During subsequent stages of the project, these source fields were cleaned, standardized, transformed, and reorganized into fact and dimension structures suitable for governed business intelligence analysis.

2.3. Operational Grain and Record Uniqueness

The operational dataset is structured at the level of individual imported records rather than assuming that the container number uniquely identifies each observation. A single container may appear in mul-

multiple records because the same container can participate in different visits, movements, or operational lifecycles. For this reason, `Container No.` was not treated as a unique record key.

To provide a reliable technical identifier for each imported row, a generated `SourceRowKey` was introduced and later renamed `ContainerOperationKey`. This key ensures that every source record can be uniquely referenced throughout the data preparation, modeling, validation, and analytical workflow.

The source dataset contains 98,052 records, with 98,052 unique technical keys, zero duplicate technical keys, and zero blank technical keys. These results confirm that the generated key successfully provides row-level technical uniqueness and supports reliable record traceability.

This validation should not be interpreted as evidence of business-level uniqueness. It confirms only that each imported record has a distinct technical identifier. Business entities such as containers, vessels, operators, or voyages may legitimately appear multiple times depending on the operational grain and lifecycle represented by the dataset.

2.4. Operational Scope of the Source Data

The source dataset captures a compact but operationally meaningful representation of the container life-cycle within the terminal environment. It includes container identification and specification, vessel-call and voyage information, entry and exit timestamps, stack duration, cargo and commodity attributes, dangerous-goods indicators, damage conditions, equipment associations, port information, transport classifications, weight-related fields, and quay-crane working hours.

Together, these fields support analysis of container movement, dwell behavior, vessel activity, cargo characteristics, equipment workload, trade connectivity, and selected operational-performance indicators. The dataset therefore provides a practical analytical view of the main processes required for business intelligence and performance assessment.

At the same time, the source should not be interpreted as a complete digital representation of every terminal process. It does not independently provide a full berth-event history, detailed yard-slot geometry, truck queue and gate-event sequences, rail-call events, detailed inspection records, financial transactions, or complete historical inventory snapshots. These limitations define the analytical boundary of the project and should be considered when interpreting results that depend on operational events or attributes not explicitly captured in the available source data.

2.5. Reference Data and Port Enrichment

Two closely related port reference tables are used intentionally for different analytical roles within the project. Although their structures are similar, each table supports a distinct port dimension and must therefore be joined according to its intended business purpose.

- **PortMaster**: used to enrich loading-port information associated with `POL` and to support the construction of `DimPortPOL`.
- **POD_Master**: used to enrich discharge-port information associated with `POD` and to support the construction of `DimPortPOD`.

Because the two reference tables contain similar schemas and comparable port attributes, an incorrect join could technically execute without producing a syntax error while still generating analytically incorrect results. For this reason, the project applies an explicit role-based joining rule throughout the SQL and Power Query workflows:

`POL` → `PortMaster`

`POD` → `POD_Master`

This separation preserves the semantic distinction between port of loading and port of discharge, reduces the risk of incorrect enrichment, and supports reliable downstream modeling, relationship creation, and connectivity analysis.

2.6. Data Dictionary Coverage

The project data dictionary provides broader documentation than the 33 source fields implemented directly in the analytical model. It contains 75 documented field definitions that collectively describe the wider operational data landscape associated with the terminal environment.

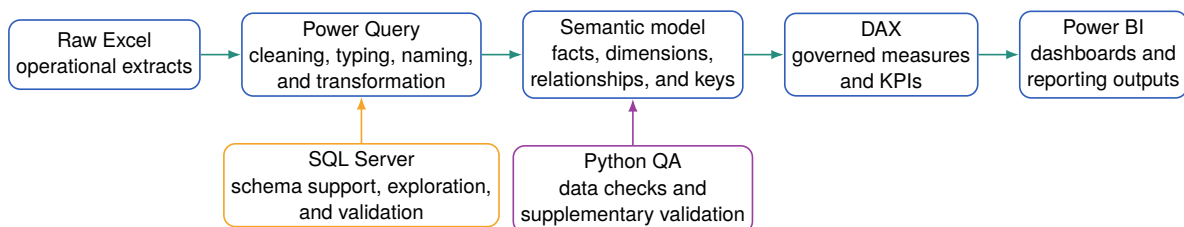
The documented fields extend across several functional areas, including vessel identifiers, vessel-call timestamps and vessel dimensions; container identification and physical characteristics; weight and safety-related information; stowage-related attributes; gate, yard, and logistics information; and documentation, routing, and verification fields. This broader coverage provides important business context for understanding both the implemented analytical model and additional operational attributes that may support future extensions of the project.

Appendix [A](#) provides a structured summary of all 75 documented data-dictionary entries, organized by operational domain to support business interpretation, data governance, and analytical traceability.

3. End-to-End Analytical Workflow and Architecture

3.1. Layered Implementation

The project implemented data cleaning and analytical preparation as a structured multi-stage pipeline rather than as a single isolated step. The workflow progressed from raw operational extracts to transformed and validated analytical outputs through coordinated stages of preparation, modeling, measure development, and reporting.



Integrated implementation statement

The project followed a coordinated layered implementation in which Power Query, SQL Server, Python QA, semantic modeling, DAX, and Power BI functioned as complementary parts of the same analytical workflow. Final KPIs and dashboard outputs were developed and validated through this integrated process to ensure consistency, traceability, and a unified interpretation of the underlying operational data.

3.2. Chronological Workflow

The project followed a structured sequence of analytical and technical activities, progressing from source understanding and business interpretation through data preparation, modeling, validation, visualization, and final reporting.

1. Review the source fields, operational terminology, and business context of the container-terminal data.
2. Define the principal analytical questions, comparison objectives, KPI logic, and expected management insights.
3. Prepare the operational source data and supporting reference datasets for analysis.
4. Build traceable staging queries to standardize field names, data types, formats, and transformation logic.
5. Create the required dimensions, technical identifiers, and supporting reference structures.
6. Construct fact tables according to clearly defined operational grains and analytical purposes.
7. Configure model relationships, role-playing date logic, and appropriate filtering behavior across the semantic model.
8. Develop DAX measures, KPI calculations, comparative indicators, and validation measures.

9. Apply SQL Server for schema creation, structural checks, analytical exploration, joins, aggregations, and result verification.
10. Apply Python for supplementary data-quality assessment, workbook-level checks, validation, and exploratory analysis.
11. Validate key analytical elements, including TEU calculations, weight classifications, QC working hours, technical keys, reference-data matches, and date-sequence consistency.
12. Develop dashboard pages, drill-down and analytical navigation paths, visual layouts, and the final management-oriented reporting structure.

This chronological workflow ensured that each analytical stage was supported by the preceding preparation and validation activities, thereby improving consistency, traceability, and confidence in the final project outputs.

3.3. Dimension Generation in Power Query

The project dimensions were generated as referenced queries from the governed staging layer rather than being maintained as separate manually prepared Excel sheets. This approach ensures that the same cleaning, standardization, and transformation logic applied to the source data is inherited consistently by the dimensional structures.

Generating dimensions in Power Query also improves refreshability and reproducibility. When new operational records or previously unseen categories are introduced into the source data, the referenced dimension queries can capture those values automatically during refresh, subject to the defined transformation and validation rules. This reduces manual maintenance and lowers the risk of inconsistencies between the fact tables and their related dimensions.

The approach also supports consistent key creation, controlled de-duplication, standardized naming, and traceable lineage from the source data through the staging layer to the final semantic model. As a result, the dimensions remain aligned with the operational data used by the analytical model.

Manually maintained Excel files were reserved for governed reference data that originates outside the transactional source and requires controlled enrichment, such as the port master tables used for loading-port and discharge-port information. This distinction separates dynamically generated analytical dimensions from externally maintained business reference data.

3.4. Data Storage and Refresh

The implemented analytical solution uses Power BI Import mode for the available operational dataset of 98,052 source records. Given the size and structure of the current dataset, Import mode provides an appropriate balance of performance, simplicity, and analytical responsiveness for the project environment.

The current solution is designed around a full-refresh process in which the source data, staging transformations, dimensions, fact tables, relationships, and measures are refreshed together as a consistent analytical model. This approach supports reproducibility and ensures that the reporting layer reflects the latest validated version of the available operational data.

More advanced data-management approaches, including incremental refresh, warehouse-managed surrogate keys, historical snapshot structures, and Microsoft Fabric-based architecture, are identified as potential future enhancements rather than components of the current implementation. These options may become appropriate as data volume, refresh frequency, historical retention requirements, or enterprise integration needs increase.

Accordingly, the implemented storage and refresh strategy reflects the scale and objectives of the current project while preserving a clear path for future architectural expansion.

3.5. Naming Conventions and Model Organization

The project applies consistent naming conventions across the data-preparation and analytical layers to improve readability, maintainability, and traceability. Queries and model tables use standardized prefixes that indicate their functional role within the solution. In particular, staging queries use the `Stg` prefix, dimension tables use `Dim`, and fact tables use `Fact`. This convention makes the structure of the model easier to understand and helps distinguish transformation, descriptive, and transactional components.

Column names were also standardized using concise, model-friendly naming patterns to improve consistency across Power Query, the semantic model, and DAX calculations. Where appropriate, source-field names were cleaned or simplified while preserving their business meaning.

DAX measures are centralized within the dedicated `_Measures` table rather than being distributed across individual fact or dimension tables. The measures are further organized into analytical display folders, allowing related KPIs and calculations to be grouped logically by subject area.

These conventions provide a structured and consistent model organization that supports easier navigation, development, validation, maintenance, and future extension of the analytical solution.

4. Excel-Based Q1 KPI Analysis

4.1. Purpose and Scope

The Excel workstream was developed to provide a focused management-level comparison of terminal performance between Q1 2025 and Q1 2026, covering the January–March period of each year. Its purpose was to organize key operational indicators into a concise comparative dashboard that allows changes in terminal activity, capacity utilization, productivity, dwell performance, vessel activity, and cargo-flow characteristics to be assessed consistently across the two periods.

The workstream followed a structured analytical process beginning with data exploration and preparation, followed by cleaning, KPI calculation, validation, comparative analysis, insight development, and dashboard construction. Particular attention was given to ensuring that the reported indicators were calculated consistently for both comparison periods so that changes represented meaningful operational differences rather than variations in calculation methodology.

The Excel analysis complements the broader project by providing a dedicated Q1 performance view. It translates detailed operational records into management-oriented indicators and visual comparisons that highlight both improvements and areas requiring further investigation. The dashboard therefore serves as a concise analytical summary of quarter-over-quarter operational performance while remaining aligned with the project's common data foundation, KPI definitions, and validation principles.

4.2. Documented KPI Definitions

The Excel workstream established a defined set of operational KPIs to support a consistent comparison between Q1 2025 and Q1 2026. Each KPI was calculated using a documented business rule so that performance differences could be interpreted consistently across the two periods.

Table 4.1.: Excel workstream KPI definitions

KPI	Documented calculation and interpretation
Capacity utilization	Calculated as actual monthly TEU divided by the assumed monthly design capacity, where the documented annual terminal design capacity of 500,000 TEU is divided by 12. The 500,000-TEU value is treated explicitly as a planning assumption for comparative analysis rather than as a directly derived operational measure.
Quay-crane productivity	Calculated as container moves divided by recorded quay-crane working hours. The metric is used as a weighted monthly productivity indicator to evaluate the relationship between handled volume and crane working time.
Dwell time	Calculated using average <code>Stack Days</code> for container records with a valid <code>Out Date</code> . The measure is compared at both monthly and aggregated Q1 levels to assess changes in the average time containers remain within the terminal.
Import/export ratio	Calculated as import moves divided by export moves using a consistent classification and comparison method for both Q1 periods. The ratio provides an indication of changes in the balance between inbound and outbound container activity.
Vessel activity	Evaluated using container-voyage activity, voyage counts, and vessel-call counts across the respective Q1 periods. These indicators provide complementary views of vessel-related operational activity and support interpretation of changes in terminal traffic and workload.

These KPI definitions establish a consistent analytical basis for the Excel comparative dashboard and support interpretation of changes in terminal performance between Q1 2025 and Q1 2026.

4.3. Q1 KPI Summary

The Q1 comparison provides a consolidated view of the principal performance indicators for January–March 2025 and January–March 2026. The results show a mixed operational picture: terminal activity and capacity utilization increased, while crane productivity and dwell-time performance deteriorated. Vessel activity also declined, and the import/export balance shifted substantially between the two periods.

Table 4.3.: Q1 2025 versus Q1 2026 KPI summary

KPI	2025 Q1	2026 Q1	Abs. change	Change	Interpretation
Capacity utilization	67.0%	70.8%	+3.8 pp	+5.7%	Higher utilization of the documented design-capacity assumption.
Crane productivity	21.1	18.9	-2.2	-10.5%	Productivity declined; higher values indicate stronger crane performance.
Dwell time (days)	7.62	8.26	+0.64	+8.4%	Average dwell increased; lower dwell time is operationally preferable.
Container-voyages	47,500	49,621	+2,121	+4.5%	Increased container-related voyage activity.
Voyages	121	112	-9	-7.4%	Fewer voyages were recorded during Q1 2026.
Vessel calls	127	114	-13	-10.2%	Vessel-call activity declined between the two Q1 periods.
Import/export ratio	0.68x	1.88x	+1.20x	+177.1%	Significant shift in the balance between import and export activity.

Overall, the Q1 results indicate that the terminal handled a higher level of container-related activity and operated at a greater share of the assumed design capacity in 2026. However, this increase was accompanied by lower quay-crane productivity and longer average dwell time, suggesting that the growth in handled activity did not translate into equivalent improvements in operational efficiency.

The reduction in voyages and vessel calls, together with the increase in container-voyage activity, also suggests that operational volume was distributed across fewer vessel movements during Q1 2026. In addition, the substantial change in the import/export ratio indicates a major shift in cargo-flow composition that may have influenced yard workload, dwell behavior, and equipment demand. These results should therefore be interpreted collectively rather than as isolated KPI movements.

4.4. Monthly KPI Results

The monthly comparison provides a more detailed view of how terminal performance evolved across January, February, and March of 2025 and 2026. The results show changes in handled moves, TEU volume, quay-crane working hours, productivity, capacity utilization, dwell time, voyage activity, vessel calls, and import/export balance. Reviewing these indicators at the monthly level helps identify whether the Q1 changes were consistent throughout the quarter or concentrated in specific months.

Table 4.5.: Monthly KPI results for Q1 2025 and Q1 2026

Month	Moves	TEU	QC hours	Moves/h	Capacity	Dwell	Cont.-voyages	Voyages	Calls	I/E
2025-01	13,513	22,603.5	640.7	21.1	54.2%	8.7	13,488	37	38	0.78x
2025-02	14,512	26,017.0	685.5	21.2	62.4%	7.2	14,462	38	41	0.67x
2025-03	19,636	35,116.0	930.9	21.1	84.3%	7.2	19,550	46	48	0.62x
2026-01	14,500	25,950.3	766.7	18.9	62.3%	7.7	14,430	33	35	2.04x
2026-02	17,935	31,908.5	955.1	18.8	76.6%	8.2	17,899	41	41	1.66x
2026-03	17,409	30,646.0	915.6	19.0	73.6%	8.8	17,292	38	38	2.01x

The monthly results reinforce the broader Q1 findings. In 2025, handled activity and capacity utilization increased steadily across the quarter, reaching their highest levels in March. Quay-crane productivity remained relatively stable at approximately 21 moves per hour, while average dwell declined from 8.7 days in January to 7.2 days in both February and March.

In 2026, handled volumes and capacity utilization were higher than the corresponding early-quarter levels in 2025, particularly in February and March. However, quay-crane productivity remained consistently lower at approximately 19 moves per hour. Average dwell also increased progressively from 7.7 days in January to 8.8 days in March, indicating growing container residence time within the terminal as the quarter progressed.

The import/export ratio shows one of the most pronounced changes between the two periods. During Q1 2025, the ratio remained below 1.0x in each month, whereas all three months of Q1 2026 exceeded 1.0x. This indicates a substantial shift in cargo-flow composition and provides important context when interpreting changes in yard workload, dwell behavior, and overall terminal operations.

Monthly voyage and vessel-call counts also vary across the quarter. These changes should be considered together with container volume and TEU performance, since higher handled activity does not necessarily correspond to a higher number of vessel calls. The monthly view therefore provides additional operational context for the aggregated Q1 KPI results.

4.5. Trade-Mix Analysis

The Q1 trade-mix analysis compares import and export activity between 2025 and 2026 using both container moves and TEU volume. The results show a substantial change in the composition of terminal traffic between the two periods.

Table 4.6.: Q1 import/export analysis

Year	Import moves	Export moves	Import share	Export share	I/E	Import TEU	Export TEU
2025	19,269	28,392	40.4%	59.6%	0.68x	32,331	51,405
2026	32,541	17,303	65.3%	34.7%	1.88x	57,498	31,007

The comparison indicates a clear shift from export-dominant activity in Q1 2025 to import-dominant activity in Q1 2026. In 2025, exports represented 59.6% of recorded moves, while imports accounted for 40.4%, producing an import/export ratio of 0.68x. In 2026, the pattern reversed: imports increased to 65.3% of moves, exports declined to 34.7%, and the import/export ratio increased to 1.88x.

The TEU results show the same directional change. Import TEU increased from 32,331 in Q1 2025 to 57,498 in Q1 2026, while export TEU decreased from 51,405 to 31,007. This confirms that the change was not limited to container counts but was also reflected in equivalent container capacity.

The trade-mix shift is operationally important because changes in the balance between imports and exports can influence yard occupancy, container dwell patterns, equipment demand, stacking requirements, and the timing of terminal movements. It therefore provides important context for interpreting the simultaneous changes observed in capacity utilization, dwell time, and operational workload.

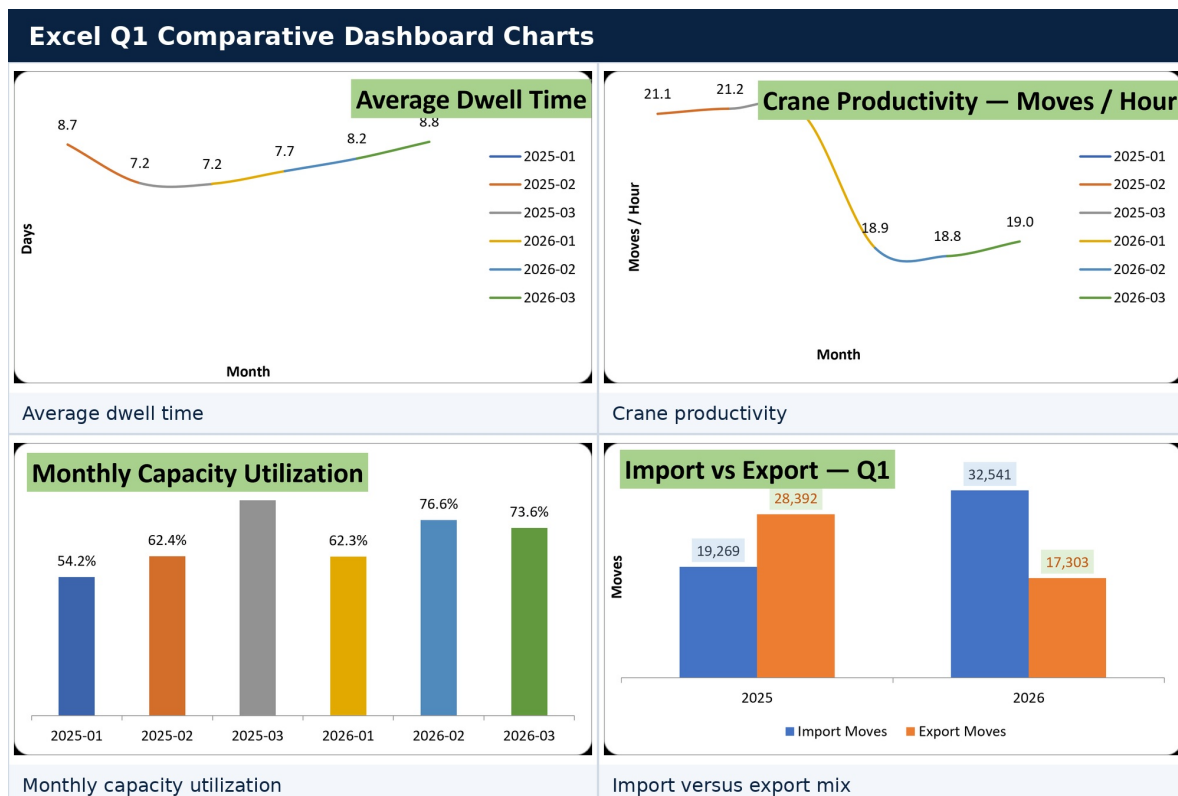


Figure 4.1.: Excel-based Q1 performance visualizations covering average dwell time, quay-crane productivity, capacity utilization, and import/export mix.

The combined visualizations provide a concise management view of the main Q1 performance changes. Together, they show that increased capacity utilization and a major shift toward import activity occurred alongside lower quay-crane productivity and higher average dwell time, reinforcing the need to interpret terminal performance across multiple connected KPIs rather than through volume indicators alone.

4.6. Interpretation of the Q1 Excel Analysis

The Excel-based Q1 analysis identifies three principal operational findings: quay-crane productivity decreased, average dwell time increased, and the trade mix shifted substantially toward imports. These findings are retained as part of the documented project interpretation because they are supported by the comparative KPI results for Q1 2025 and Q1 2026.

The reduction in crane productivity indicates that a greater amount of quay-crane working time was required relative to the number of container moves handled during Q1 2026. At the same time, the increase in average dwell time suggests that containers remained within the terminal for longer periods on average. When considered together, these indicators point to a potential reduction in operational efficiency despite the increase in overall container-related activity and capacity utilization.

The change in the import/export ratio provides additional context for these results. The shift from an export-dominant pattern in Q1 2025 to an import-dominant pattern in Q1 2026 may have influenced yard utilization, stacking requirements, equipment demand, and container residence patterns. The trade-mix change should therefore be considered when interpreting the observed movement in dwell time and operational workload.

Capacity utilization is interpreted using the documented annual design-capacity assumption of 500,000 TEU. Accordingly, the reported utilization values represent performance relative to this planning assumption and should not be interpreted as a directly measured physical capacity limit unless the assumed design

capacity is independently confirmed.

The conclusions in this section are based on the documented KPI methodology and the reported analytical results. They are intended to support management-level interpretation of the Q1 comparison while maintaining appropriate distinction between observed performance patterns and conclusions that would require additional operational evidence.

5. Data Preparation, Power Query, and Reference Data

5.1. Staging Query

The data-preparation workflow begins with `StgContainerOperations`, which serves as the governed staging layer for the operational source data. The purpose of this query is to preserve the original operational records while applying the initial transformations required to make the dataset suitable for downstream modeling and analysis.

The staging process includes header standardization, data-type assignment, text trimming and cleaning, removal of completely blank rows, and creation of technical control fields used later in the analytical model. By concentrating these transformations in a single reusable staging query, subsequent fact and dimension queries inherit a consistent set of data-preparation rules rather than repeating the same cleaning logic independently.

In the documented initial model, `Enable Load` was disabled for `StgContainerOperations`. This allowed the query to function as a reusable parent transformation layer without loading an additional copy of the staging table into the semantic model. The approach reduces unnecessary duplication while preserving a clear and traceable transformation path from the source extract to the final analytical structures.

Queries [19] <

✕ ✓ fx = Table.AddColumn(

	1 ² ₃ SourceRowKey	A ^B _C SizeType	1 ² ₃ ContainerLength	A ^B _C ContainerNum
1		1 22G1		20 AAI V3111711
2		2 22G1		20 AAI V3113933
3		3 22G1		20 AAI V3113135
4		4 22G1		20 AAI V3131593
5		5 22G1		20 AAI V3131771
6		6 22G1		20 AAI V3131771
7		7 22G1		20 AAI V3131937
8		8 22G1		20 AAI V3151153
9		9 22G1		20 AAI V3151153
10		10 22G1		20 AAI V3151171
11		11 22G1		20 AAI V3151171
12		12 22G1		20 AAI V3151195
13		13 22G1		20 AAI V3151195
14		14 22G1		20 AAI V3151151
15		15 22G1		20 AAI V3151151
16		16 22G1		20 AAI V3151173
17		17 22G1		20 AAI V3151173
18		18 22G1		20 AAI V3151199
19		19 22G1		20 AAI V3151199
20		20 22G1		20 AAI V3151193
21		21 22G1		20 AAI V3151193
22		22 22G1		20 AAI V3151317

Activate Windows

Figure 5.1.: Power Query preview of StgContainerOperations, showing standardized field names and the generated SourceRowKey used for row-level traceability.

5.2. Header Standardization

Source-system field names were standardized during the staging process to create concise, consistent, and model-friendly column names. This improves readability across Power Query, the semantic model, DAX expressions, validation logic, and technical documentation while preserving the original business meaning of each field.

Representative source-to-model mappings are shown in Table 5.1.

Table 5.1.: Representative source-header standardization

Source header	Standardized model header
Container No.	ContainerNumber
Calling Seq.	CallingSequence
Calling Year	CallingYear
Cntr Len	ContainerLength
Cntr H&W	ContainerHeightWidth
EQU No.	EquipmentNumber
QC Working Hours	QCWorkingHoursSource
In Date	InDate
Out Date	OutDate
Stack Days	StackDays
WGT	WeightValue

The standardization process also removes spaces, punctuation, and source-specific abbreviations where appropriate, making field names easier to reference programmatically and reducing ambiguity during model development. Importantly, the renaming process changes only the technical representation of the fields; the underlying operational meaning is preserved and documented throughout the project.

5.3. Generated Data Controls

The project introduced several generated fields and validation controls to support row-level traceability, counting, dwell-time analysis, and date-sequence verification. These fields were created during the transformation stage so that data-quality conditions could be identified explicitly rather than handled implicitly within later calculations.

5.3.1. Technical Row Key and Unit Count

A sequential index beginning at 1 was generated to provide a unique technical identifier for every imported operational record. The field was initially created as `SourceRowKey` and was later renamed `ContainerOperationKey` to better reflect its role within the analytical model.

The technical key provides row-level uniqueness and traceability throughout the transformation and modeling workflow. It is intended to identify individual imported records and should not be interpreted as a business-level container identifier.

A constant field, `UnitCount`, with a value of 1 was also added to every operational row. This field provides a simple and consistent mechanism for counting records through aggregation and can be used in measures where the analytical requirement is to count container-operation rows at the fact-table grain.

5.3.2. Completed Dwell Calculation

A dedicated completed-dwell field was created to calculate the elapsed time between container entry and exit only when the required dates form a valid completed sequence.

Listing 5.1: Power Query expression for completed dwell

```
if [InDate] = null or [OutDate] = null then null
else if [OutDate] < [InDate] then null
else Duration.TotalDays([OutDate] - [InDate])
```

The calculation returns the total elapsed duration in days when both timestamps are available and the recorded exit occurs on or after the entry timestamp. Records with a missing entry date, missing exit date, or an exit timestamp earlier than the entry timestamp remain blank.

Retaining these cases as blank is an important analytical control. A value of zero would imply a valid completed movement with no elapsed dwell time, whereas a blank value correctly indicates that completed dwell cannot be calculated reliably from the available record.

5.3.3. Date-Sequence Validation

A complementary status field was generated to classify each operational record according to the validity and completeness of its entry and exit timestamps.

Listing 5.2: Date-sequence status logic

```
if [InDate] = null then "Missing In Date"
else if [OutDate] = null then "Active or Missing Out Date"
else if [OutDate] < [InDate] then "Invalid: Out Before In"
else "Valid Completed Record"
```

This control separates records into four interpretable conditions: missing entry information, records without a recorded exit, invalid chronological sequences, and valid completed operations. The classification supports data-quality monitoring and ensures that incomplete or invalid records can be excluded from calculations that require a completed operational lifecycle.

Together, ContainerOperationKey, UnitCount, completed dwell, and date-sequence status provide foundational controls for record counting, traceability, lifecycle analysis, and validation within the analytical model.

5.4. Dimension-Construction Pattern

A consistent and repeatable procedure was used to construct the analytical dimensions from the governed staging layer. This approach ensured that dimension tables were created using the same cleaning, standardization, and validation principles applied throughout the project.

The general dimension-construction process consisted of the following steps:

1. Reference the appropriate governed staging query rather than duplicating the original source.
2. Retain only the descriptive attributes required for the dimension and its analytical purpose.
3. Apply consistent data types, text cleaning, trimming, and standardization rules.
4. Standardize missing or blank descriptive values to UNKNOWN where this was appropriate for the corresponding business attribute.
5. Remove duplicate attribute combinations so that each dimension contains a controlled set of descriptive members at its defined grain.
6. Generate a technical index key to provide a stable model identifier for each dimension member.
7. Perform a left-outer merge between the operational staging data and the corresponding dimension using the defined business matching fields.
8. Expand only the generated dimension key into the fact-table preparation workflow, avoiding unnecessary duplication of descriptive attributes.

9. Create a match-status control to identify records that successfully matched the dimension and records requiring further data-quality review.
10. Verify that the merge preserves the expected fact-table row count and does not introduce unintended duplication or record loss.
11. After validation, remove repeated descriptive columns from the fact table when those attributes are already governed through the related dimension.

This pattern supports a cleaner star-schema structure by separating descriptive business attributes from transactional facts while preserving traceability back to the source data. Applying the same construction method across dimensions also improves consistency, refreshability, relationship integrity, and future maintainability of the semantic model.

Power Query Dimension Evidence

The figure displays six screenshots of the Power Query Editor, each showing a different dimension table. The dimensions are:

- DimOperator**: Shows a table with columns like OperatorCode, OperatorName, and OperatorType. The 'Queries' pane on the left shows the source data for this dimension.
- DimCargo**: Shows a table with columns like CargoType, CargoName, and CargoWeight. The 'Queries' pane on the left shows the source data for this dimension.
- DimContainerSpecification**: Shows a table with columns like ContainerType, ContainerName, and ContainerSize. The 'Queries' pane on the left shows the source data for this dimension.
- DimEquipment**: Shows a table with columns like EquipmentType, EquipmentName, and EquipmentStatus. The 'Queries' pane on the left shows the source data for this dimension.
- DimTransportProfile**: Shows a table with columns like TransportType, TransportName, and TransportStatus. The 'Queries' pane on the left shows the source data for this dimension.
- DimDamageCondition**: Shows a table with columns like DamageType, DamageName, and DamageSeverity. The 'Queries' pane on the left shows the source data for this dimension.

Figure 5.2.: Power Query evidence for six conformed dimensions.

5.5. Role-Specific Port Dimensions

The analytical model uses separate port dimensions for the loading and discharge roles in order to preserve the distinct business meaning of POL and POD. Although both fields describe ports, they represent different positions within the container movement and are therefore modeled independently.

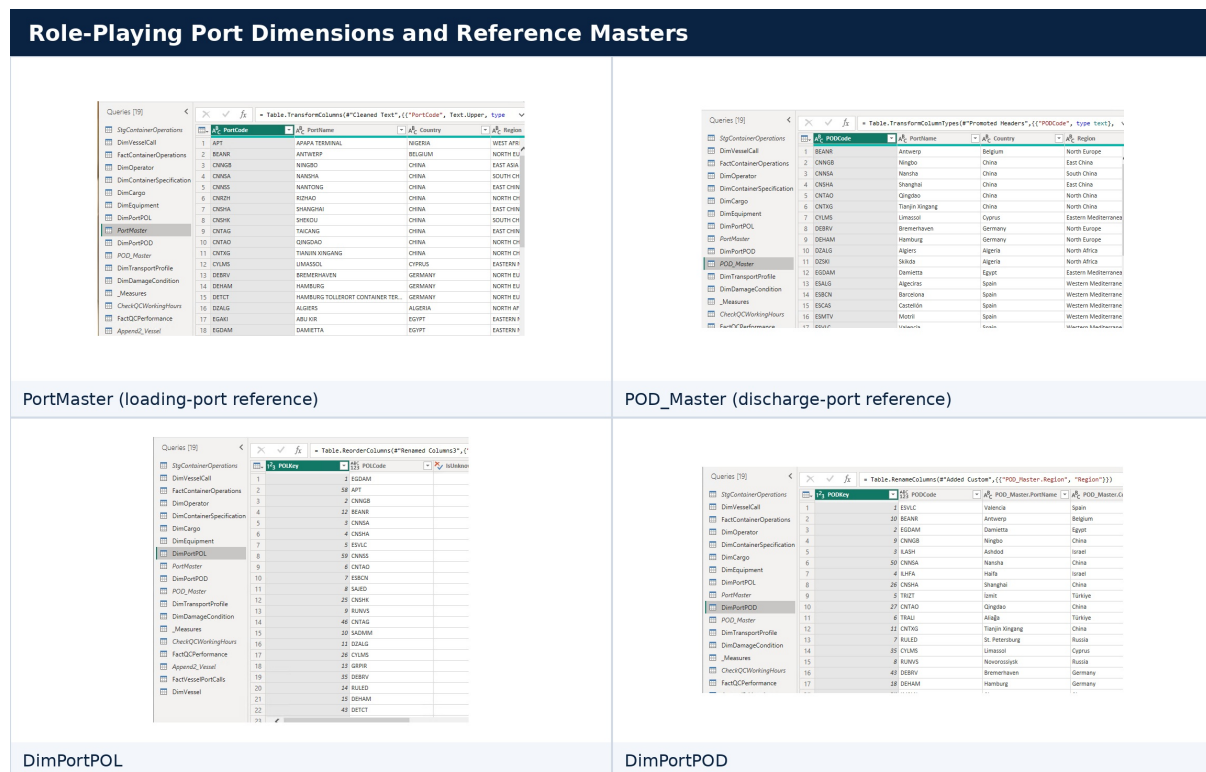


Figure 5.3.: Separate loading-port and discharge-port reference structures used to support the POL and POD analytical roles.

The loading-port attributes are enriched through PortMaster and incorporated into DimPortPOL, while discharge-port attributes are enriched through POD_Master and incorporated into DimPortPOD. After the required reference attributes are merged into their respective dimensions, the final semantic model avoids unnecessary direct relationships between the original port master tables and the analytical model.

This design reduces the risk of unintended cross-filtering between loading and discharge roles and keeps the semantic meaning of each port dimension explicit. As a result, analyses of origin and destination activity can be performed independently while still supporting port-level attributes such as port name, country, region, or other available reference information.

Maintaining separate role-specific dimensions also improves model clarity and supports reliable connectivity analysis by ensuring that POL is consistently interpreted as the loading-port role and POD as the discharge-port role throughout the model.

5.6. TEU Mapping and Validation

The project applies a standardized TEU conversion rule within DimContainerSpecification to translate container length into Twenty-Foot Equivalent Units (TEU). The approved mapping used throughout the analytical model is:

- 20-foot container = 1.00 TEU,
- 40-foot container = 2.00 TEU,
- 45-foot container = 2.25 TEU.

Container lengths that do not match one of the approved mappings are not assigned an assumed TEU value. Instead, the corresponding TEU factor remains blank so that unsupported or unexpected source values are not silently converted into analytical results.

A dedicated mapping-status field is retained as part of the data-quality controls. This field indicates whether each container specification was successfully matched to an approved TEU factor and supports identification of unmapped or exceptional records during validation.

This approach ensures that TEU calculations are based on explicit and governed conversion rules, while preserving transparency for records that require further review.

5.7. Current-State Monitoring and Controls

The analytical model includes dedicated controls for identifying and monitoring the latest known container state. The derived fields `InDateOnly` and `OutDateOnly` provide date-level versions of the original entry and exit timestamps and support consistent relationships with the calendar dimension.

A container record with a valid entry date and no recorded exit date is treated as provisionally active within the available dataset. This status represents the latest known state based on the source extract and should not automatically be interpreted as confirmation that the container remains physically present in the terminal beyond the reporting cutoff.

To support current-state monitoring, `ActiveAgeDays` measures the elapsed time associated with provisionally active records. The resulting values are grouped into ordered operational age bands:

- 0–3 days,
- 4–7 days,
- 8–14 days,
- 15–30 days, and
- 31+ days.

These age groups provide a structured view of active-container duration and help identify records requiring additional operational attention, particularly containers remaining active for extended periods.

For reproducible production reporting, the current-state calculations should use a controlled extract or reporting timestamp rather than the local system refresh time. A fixed reporting reference ensures that the same source extract produces consistent age calculations regardless of when or where the analytical model is refreshed.

5.8. Vessel Data Extension

The analytical model was later extended to incorporate a more detailed vessel-focused data structure. This extension introduced `Append2_Vessel`, `DimVessel`, and `FactVesselPortCalls`, while `DimVesselCall` provides the linkage between detailed vessel-call information and the container-operations model.

The vessel extension was designed to support analysis at a more appropriate operational grain for vessel activity. A cleaned vessel business key was created by combining the standardized vessel code and vessel name, providing a consistent identifier for matching vessel-related records across the model.

`FactVesselPortCalls` is structured at one row per vessel call, allowing vessel-level measures such as call volume, turnaround time, berth occupancy, pre-berth waiting, vessel dimensions, and operator-level performance to be analyzed without relying on repeated container-level records.

`DimVessel` stores vessel-level descriptive attributes, while `DimVesselCall` supports the analytical relationship between individual vessel calls and the container operations associated with those calls. This separation improves model clarity, preserves the correct analytical grain, and enables vessel-performance analysis to coexist with container-level analysis within the same semantic model.

Vessel and Vessel-Call Extension

Queries [19]

Table.TransformColumnTypes(#"Navigation 1", {"ATB", type datetime}, {"Vessel Name", type text}, {"Vessel Code", type text}, {"Flag", type text}, {"Year Built", type text}, {"Operator", type text}, {"Owner", type text})

	ATB	Vessel Name	Vessel Code	Flag	Year Built	Operator	Owner
1	1/2/2025 11:45:00 AM	MITO	MITO	PORTUGAL		2006 ONE	ONEE
2	1/2/2025 2:00:00 PM	MSC DARIEN	DARI	PORTUGAL		2003 MSC	MSCE
3	1/3/2025 12:15:00 AM	YM EVOLUTION	YMEV	TAIWAN		2014 YML	YMES
4	1/4/2025 1:45:00 AM	MOL ENDOWMENT	MOND	PANAMA		2007 ONE	ONEE
5	1/4/2025 12:00:00 PM	LUCY BORCHARD	LUCY	ANTIGUA		2006 BOR	BAGA
6	1/5/2025 9:25:00 AM	OMA CDM MONTREAL	MOBL	MALTA		2007 CIAA	CNAE
7	1/5/2025 11:45:00 PM	MITO	MITO	PORTUGAL		2006 ONE	ONEE
8	1/11/2025 9:00:00 AM	LOUISE BORCHARD	LOBO	ANTIGUA		2006 BOR	BAGA
9	1/12/2025 8:55:00 AM	LDR SAKINE	SAKN	PANAMA		1993 LID	KDMR
10	1/12/2025 11:25:00 AM	CHATTANOOGA	CHAT	SINGAPORE		2012 ONE	ONEE
11	1/15/2025 10:15:00 PM	COSCO ANTWERP	ANTW	UNITED KINGDOM		2001 COL	COOL
12	1/16/2025 10:30:00 AM	ASTRID 1	ASTR	ANTIGUA		2008 BMR	BMEL
13	1/16/2025 11:55:00 AM	LUCY BORCHARD	LUCY	ANTIGUA		2006 BOR	BAGA
14	1/19/2025 10:00:00 PM	SCHOUSTON	SHOU	MARSHALL ISLANDS		2010 ONE	YMES
15	1/20/2025 12:15:00 AM	ZEYNEP	ZYNP	TUVALU		2003 LID	KDMR
16	1/20/2025 5:35:00 PM	COSCO SHANGHAI	SHAI	UNITED KINGDOM		2001 COL	COOL
17	1/21/2025 1:45:00 PM	OOCL KOBE	KOPE	HONG KONG		2007 OOL	OOCL
18	1/29/2025 11:20:00 AM	LOUISE BORCHARD	LOBO	ANTIGUA		2006 BOR	BAGA

Append2_Vessel source preview

Queries [19]

Table.ReorderColumns(#"Append2_Vessel", {"VesselKey", type text}, {"VesselBusinessKey", type text}, {"VesselCode", type text}, {"VesselName", type text})

	VesselKey	VesselBusinessKey	VesselCode	VesselName
1	2	LANG LANGNESS	LANG	LANGNESS
2	2	BZCH BZ CHONGFU	BZCH	BZ CHONGFU
3	3	NITA NITA A	NITA	NITA A
4	4	REMA REMAH	REMA	REMAH
5	5	XHAI XIN XIN HAI 1	XHAI	XIN XIN HAI 1
6	6	ZYNP ZEYNEP	ZYNP	ZEYNEP
7	7	LIPE LIDER PERIHAN	LIPE	LIDER PERIHAN
8	8	BRNS BARENTS	BRNS	BARENTS
9	9	KUZNS KUZNETSOV	KUZN	S.KUZNETSOV
10	20	HUJ HUJ DA 9	HUJ	HUJ DA 9
11	11	KRCA KR CELEBES	KRCA	KR CELEBES
12	12	VESP VERSA SPIRIT	VESP	VERSA SPIRIT
13	13	JEON MSC JEONGMIN	JEON	MSC JEONGMIN
14	14	VALC VALENCE	VALC	VALENCE
15	15	LIAN XIN LIAN YUN GANG	LIAN	XIN LIAN YUN GANG
16	16	DLNS DELPHINUS	DLNS	DELPHINUS
17	17	CETS CETUS	CETS	CETUS
18	18	AURE AURELIA	AURE	AURELIA
19	19	XMEN XIN XIA MEN	XMEN	XIN XIA MEN
20	20	ERAS ERASMUS NINJA	ERAS	ERASMUS NINJA
21	21	ZLNG ZHONGCHENGCHANGUL	ZLNG	ZHONGCHENGCHANGUL
22	22	ELMA MSC ELMA	ELMA	MSC ELMA

DimVessel

Queries [19]

Table.Distinct(#"Removed Other Columns", {"CallingSequence", type text}, {"CallingYear", type text})

	VesselKey	VesselBusinessKey	CallingSequence	CallingYear
1	2	LANG LANGNESS	502	2025
2	2	BZCH BZ CHONGFU	86	2026
3	3	NITA NITA A	93	2026
4	38	REMA REMAH	114	2026
5	108	XHAI XIN XIN HAI 1	510	2025
6	229	REMA REMAH	9	2026
7	280	ZYNP ZEYNEP	78	2026
8	281	LIPE LIDER PERIHAN	75	2026
9	282	BRNS BARENTS	91	2026
10	283	KUZNS KUZNETSOV	91	2026
11	284	HUJ HUJ DA 9	17	2026
12	286	KRCA KR CELEBES	72	2026
13	287	VESP VERSA SPIRIT	73	2026
14	289	JEON MSC JEONGMIN	1	2026
15	240	LANG LANGNESS	64	2026
16	241	VALC VALENCE	7	2026
17	242	LIAN XIN LIAN YUN GANG	496	2025
18	243	XHAI XIN XIN HAI 1	102	2026
19	286	LANG LANGNESS	21	2026
20	289	DLNS DELPHINUS	50	2026
21	280	DLNS DELPHINUS	123	2026
22	282	DLNS DELPHINUS	77	2026

DimVesselCall

Figure 5.4.: Supplied evidence for the vessel-source extension, vessel dimension, and vessel-call dimension.

5.9. Queries Evidenced in the Final Model

The final Power Query environment contains 18 documented queries supporting data staging, dimensional preparation, fact-table construction, reference-data enrichment, validation, vessel analysis, and centralized measure organization.

Query / Table	Query / Table
StgContainerOperations	DimVesselCall
FactContainerOperations	DimOperator
DimContainerSpecification	DimCargo
DimEquipment	DimPortPOL
PortMaster	DimPortPOD
POD_Master	DimTransportProfile
DimDamageCondition	DimVessel
CheckQCWorkingHours	FactQCPerformance
Append2_Vessel	FactVesselPortCalls

These queries collectively support the implemented analytical architecture. The staging and fact queries manage operational records, the dimension queries provide descriptive analytical context, the port-master queries support reference-data enrichment, and the QC and vessel-related queries support specialized validation and operational analysis.

It is important to distinguish `DimDate` from the Power Query objects listed above. `DimDate` was created as a DAX calculated table within the semantic model and therefore is not included in the count of 18 Power Query queries.

6. SQL Server Workstream

6.1. Purpose and Scope

The SQL Server workstream provided a structured environment for validating the relational model, examining data quality, enforcing key relationships, and performing detailed operational analysis. The work covered schema inspection, primary- and foreign-key configuration, data-type alignment, referential-integrity checks, data cleaning, exploratory querying, comparative analysis, KPI verification, and validation of analytical results.

A total of 32 documented SQL tasks and analytical queries were developed across the project. Together, these queries supported both the technical structure of the database and the interpretation of container-terminal operations. The SQL analysis addressed container activity, vessel and voyage patterns, operator workload, cargo characteristics, port connectivity, terminal capacity, quay-crane performance, dwell time, vessel turnaround, trade mix, and geographic distribution.

The SQL workstream was therefore used as an important analytical and validation layer within the overall project workflow. Its purpose was not only to produce query results, but also to verify that the underlying data structure, relationships, and analytical logic were appropriate before conclusions were incorporated into the wider project reporting environment.

6.2. Primary-Key Discovery and Structural Validation

The SQL modeling process began by inspecting the database structure and identifying the primary keys associated with the implemented tables. The `INFORMATION_SCHEMA.TABLE_CONSTRAINTS` and `INFORMATION_SCHEMA.KEY_COLUMN_USAGE` system views were used to identify existing primary-key constraints and determine where additional structural controls were required.

Listing 6.1: Primary-key discovery query

```
SELECT
tc.TABLE_NAME,
tc.CONSTRAINT_NAME AS Primary_Key_Name,
kcu.COLUMN_NAME,
kcu.ORDINAL_POSITION
FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS tc
JOIN INFORMATION_SCHEMA.KEY_COLUMN_USAGE kcu
ON tc.CONSTRAINT_NAME = kcu.CONSTRAINT_NAME
AND tc.TABLE_NAME = kcu.TABLE_NAME
WHERE tc.CONSTRAINT_TYPE = 'PRIMARY KEY'
ORDER BY
tc.TABLE_NAME,
kcu.ORDINAL_POSITION;
```

The structural review identified tables requiring additional consideration for record uniqueness. In particular, `FactVesselPortCalls` and `CheckQCWorkingHours` were initially identified as lacking primary-key constraints.

Before assigning a key to FactVesselPortCalls, the selected vessel-voyage identifier was checked for null values:

Listing 6.2: Null-value validation for the vessel-voyage identifier

```
SELECT *  
FROM FactVesselPortCalls  
WHERE User_Voy IS NULL;
```

After confirming that the selected field contained no null values within the examined dataset, a primary-key constraint was added:

Listing 6.3: Primary-key creation for the vessel-call fact table

```
ALTER TABLE FactVesselPortCalls  
ADD CONSTRAINT PK_FactVesselPortCalls  
PRIMARY KEY (User_Voy);
```

This process demonstrates the broader project principle that keys were validated against the available records before relational constraints were introduced.

6.3. Data-Type Alignment

Relational integrity requires columns participating in primary-key and foreign-key relationships to use compatible data types. The SQL workflow therefore included systematic inspection of column names and data types across the dimensions, fact tables, and reference tables.

Representative metadata inspection used:

Listing 6.4: Representative column-type inspection

```
SELECT  
COLUMN_NAME,  
DATA_TYPE  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME = 'DimCargo';
```

The same inspection pattern was applied to the principal model tables, including:

- DimCargo,
- DimContainerSpecification,
- DimDamageCondition,
- DimEquipment,
- DimOperator,
- DimPortPOD,
- DimPortPOL,
- DimTransportProfile,
- DimVessel,
- DimVesselCall,
- FactContainerOperations,
- FactQCPerformance,
- FactVesselPortCalls,
- POD_Master, and
- PortMaster.

Where join columns used incompatible types, the corresponding columns were aligned before foreign-key constraints were created.

Listing 6.5: Representative data-type alignment

```
ALTER TABLE FactContainerOperations
ALTER COLUMN VesselCallKey INT;

ALTER TABLE FactContainerOperations
ALTER COLUMN OperatorKey TINYINT;

ALTER TABLE FactContainerOperations
ALTER COLUMN DimEquipment_EquipmentKey TINYINT;

ALTER TABLE FactContainerOperations
ALTER COLUMN DimDamageCondition_DamageConditionKey TINYINT;

ALTER TABLE FactQCPerformance
ALTER COLUMN VesselCallKey INT;

ALTER TABLE FactQCPerformance
ALTER COLUMN EquipmentKey TINYINT;
```

This step reduced the risk of failed joins, inconsistent relationships, and constraint-creation errors.

6.4. Foreign Keys and Referential Integrity

After key discovery and data-type alignment, foreign-key constraints were established to represent the relationships between fact tables, dimensions, and selected reference tables.

The principal relationships included:

- FactContainerOperations to DimVesselCall,
- FactContainerOperations to DimOperator,
- FactContainerOperations to DimContainerSpecification,
- FactContainerOperations to DimCargo,
- FactContainerOperations to DimPortPOL,
- FactContainerOperations to DimPortPOD,
- FactContainerOperations to DimEquipment,
- FactContainerOperations to DimDamageCondition,
- FactContainerOperations to DimTransportProfile,
- FactQCPerformance to DimVesselCall,
- FactQCPerformance to DimEquipment,
- DimVesselCall to DimVessel, and
- FactVesselPortCalls to the relevant vessel and operator structures.

Representative foreign-key creation is shown below:

Listing 6.6: Representative fact-to-dimension foreign keys

```
ALTER TABLE FactContainerOperations
ADD CONSTRAINT FK_FC0_VesselCall
FOREIGN KEY (VesselCallKey)
REFERENCES DimVesselCall (VesselCallKey);

ALTER TABLE FactContainerOperations
ADD CONSTRAINT FK_FC0_Operator
```

```

FOREIGN KEY (OperatorKey)
REFERENCES DimOperator (OperatorKey);

ALTER TABLE FactContainerOperations
ADD CONSTRAINT FK_FCO_ContainerSpecification
FOREIGN KEY (ContainerSpecificationKey)
REFERENCES DimContainerSpecification (ContainerSpecificationKey);

```

Unique constraints were also created where required to support reference relationships, including port codes, vessel business keys, and operator codes.

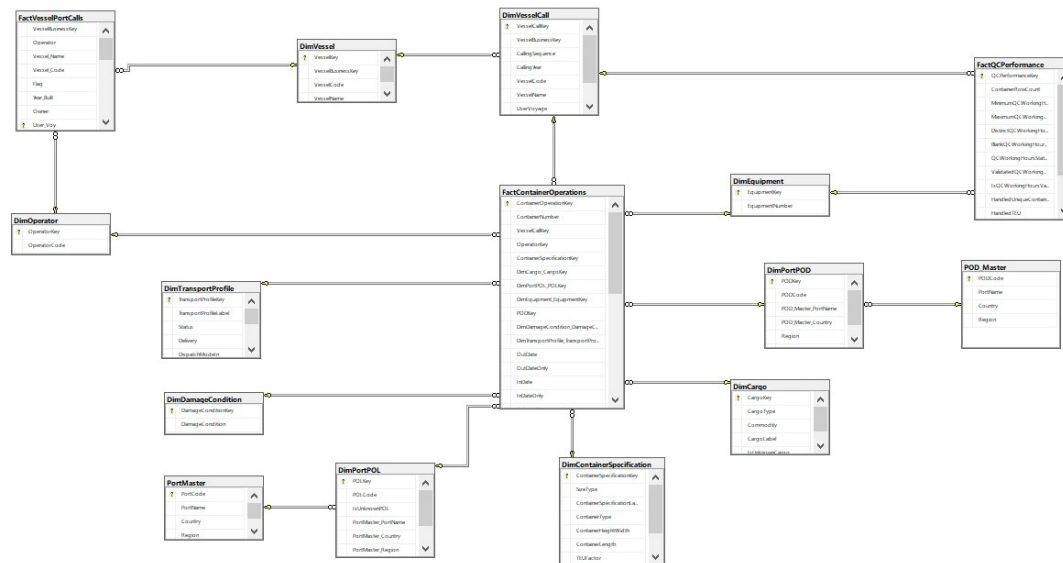


Figure 6.1: SQL Server entity-relationship diagram showing the implemented fact, dimension, and reference-table relationships.

The resulting relational structure provides an additional layer of integrity checking and supports consistent joins during analytical querying.

6.5. Documented Data Cleaning

SQL exploration was also used to identify records requiring additional data-quality treatment. Two records with ContainerOperationKey values 48676 and 48677 were identified as having insufficient usable information for the analytical layer and were removed.

Listing 6.7: Removal of unusable operational records

```

DELETE FROM FactContainerOperations
WHERE ContainerOperationKey IN (48676, 48677);

```

Additional cleaning activities included:

- trimming and standardizing text values,
- reviewing missing and blank values,
- standardizing categorical labels,
- applying explicit UNKNOWN classifications where appropriate,
- standardizing Danger Goods to Dangerous Goods, and
- standardizing Electrical Appliances to Electronic Devices.

These changes improved consistency across categorical analyses and reduced fragmentation caused by alternative labels representing the same business concept.

6.6. Operational Exploration and Analysis

The SQL analysis covered a broad range of terminal activities. The 32 documented queries can be grouped into the following major analytical areas:

1. database structure, keys, data types, and constraints;
2. record cleaning and data-quality validation;
3. operations, containers, vessels, and voyage counts;
4. vessel visit frequency;
5. operator workload and concentration;
6. full and empty container distributions;
7. commodity and cargo-type analysis;
8. loading-port and discharge-port activity;
9. monthly capacity utilization;
10. quay-crane productivity;
11. container dwell time;
12. vessel turnaround;
13. import/export composition; and
14. discharge-country share.

Appendix B provides the complete catalog of the 32 SQL tasks and queries.

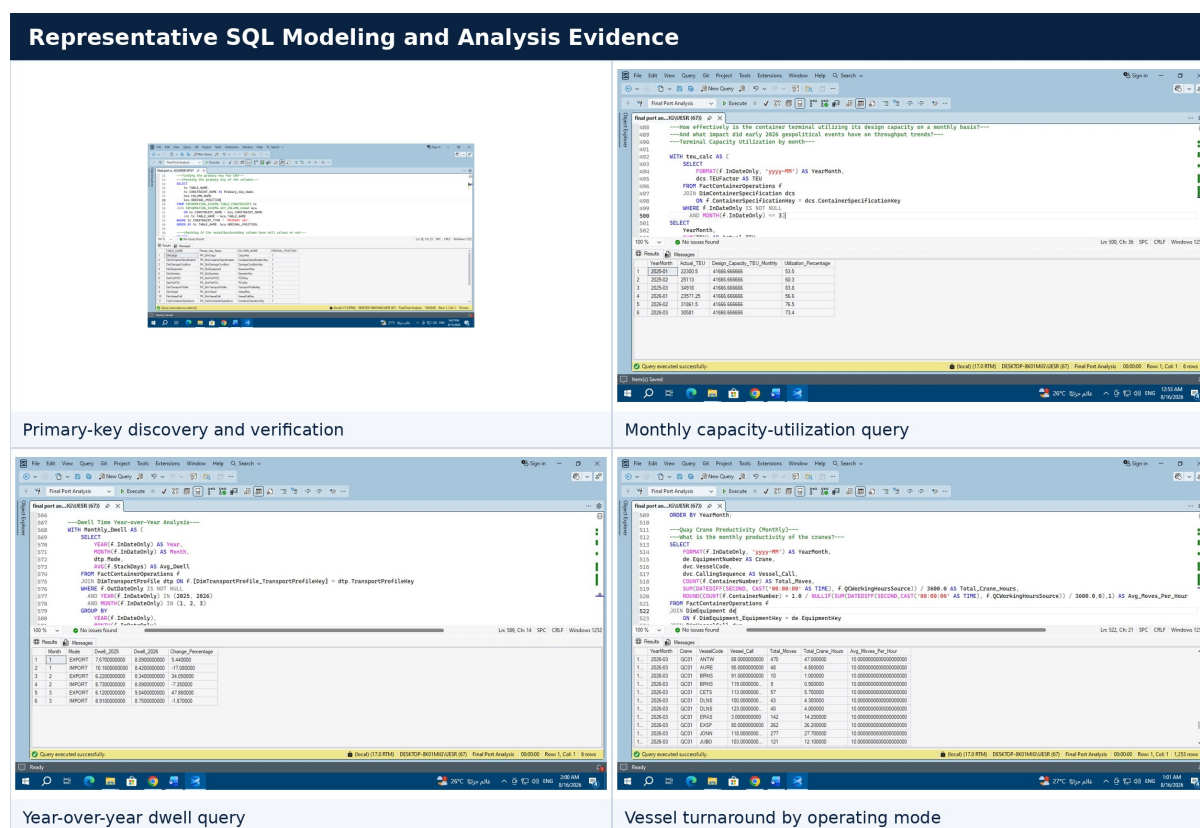


Figure 6.2.: Representative SQL analysis covering structural validation, capacity utilization, dwell analysis, and vessel-turnaround exploration.

6.7. Overall Operational Counts

The SQL analysis calculated the number of operational records, distinct containers, vessels, and voyages across the available 2025–2026 data.

Listing 6.8: Overall operational counts

```
SELECT
COUNT(*) AS [No. of operations],
COUNT(DISTINCT f.ContainerNumber) AS [No. of containers],
COUNT(DISTINCT dvc.VesselCode) AS [No. of vessels],
COUNT(DISTINCT dvc.UserVoyage) AS [No. of voyages]
FROM FactContainerOperations f
JOIN DimVesselCall dvc
ON f.VesselCallKey = dvc.VesselCallKey
WHERE dvc.CallingYear IN (2025, 2026);
```

The cleaned SQL analytical layer reports:

- 95,870 container-operation records,
- 67,738 distinct container identifiers,
- 81 distinct vessels, and
- 190 distinct voyages.

A year-level breakdown reported:

Table 6.1.: SQL operational counts by calling year

Metric	2025	2026
Operations	52,096	43,774
Containers	38,883	33,849
Vessels	51	48
Voyages	113	77

The results indicate lower activity in the available 2026 records than in 2025. However, the interpretation must remain within the scope of the available dataset and reporting period rather than being generalized automatically to full terminal activity outside the recorded data.

6.8. Vessel Activity

SQL queries were used to examine the frequency with which vessels appeared across the operational records.

The analysis identified several frequently represented vessels, including DELPHINUS, LOUISE BORCHARD, LUCY BORCHARD, 00CL KOBE, and ZEYNEP. Vessel frequency was also compared by year to determine how visit patterns changed between 2025 and 2026.

These queries provide an operational view of vessel recurrence and complement the more detailed vessel-call fact table developed later in the project.

6.9. Operator Workload

Operator analysis was performed using distinct container-voyage combinations to reduce the risk of counting repeated operational records as separate business movements.

The analysis identified MSC as the largest operator in the available data, followed by COE, ONE, and BOR.

Table 6.2.: Selected operator activity from SQL analysis

Operator	2025	2026	Combined
MSC	10,845	8,784	19,629
COE	9,595	5,520	15,115
ONE	6,669	7,626	14,295
BOR	6,410	4,295	10,705

The operator results demonstrate that workload is concentrated among a relatively small group of carriers while also showing changes in ranking between the two years.

6.10. Full and Empty Container Mix

The SQL analysis also examined the distribution between full and empty container records.

For 2025, approximately 71.7% of the analyzed records were classified as full and 28.3% as empty. For 2026, the corresponding shares were approximately 70.0% and 30.0%.

The similarity of these proportions indicates that the full-to-empty composition remained relatively stable within the analyzed records despite changes in overall activity levels.

6.11. Commodity and Cargo Analysis

Commodity analysis was conducted by operational mode to identify the principal cargo categories represented in import and export activity.

Among the most frequently represented export commodities were oranges, accessories, clothes, ceramic and porcelain products, phosphate, and strawberries. Major import commodities included wood, vaccines, electronic devices, accessories, and clothes.

The SQL analysis also examined cargo-type distribution. General cargo represented the largest category, followed by empty containers and reefer cargo. Dangerous-goods and over-dimensional cargo represented substantially smaller portions of the available operational records.

The commodity and cargo analysis supports understanding of the operational mix handled by the terminal and provides context for equipment utilization, reefer demand, yard planning, and cargo-specific handling requirements.

6.12. Loading-Port Analysis

Loading-port analysis used DimPortPOL to identify the origins associated with the largest numbers of unique container operations.

The analysis showed substantial activity associated with ports including Damietta, Piraeus, Hamburg, Antwerp, Novorossiysk, and Rotterdam, with rankings varying between 2025 and 2026.

Port attributes such as country and region were incorporated from the governed loading-port reference structure, allowing operational counts to be interpreted geographically as well as by port code.

6.13. Discharge-Port Analysis

Discharge-port activity was analyzed independently through DimPortPOD. The results identified Damietta as the dominant discharge location within the available dataset, followed by major ports such as Piraeus, Novorossiysk, Antwerp, and Hamburg.

The independent treatment of loading and discharge ports is important because a port can play different roles depending on the movement being analyzed. The SQL analysis therefore preserves the semantic distinction between POL and POD rather than treating them as interchangeable port attributes.

6.14. POL and POD Comparison

Comparing loading and discharge activity provides additional insight into directional trade patterns.

Selected ports demonstrated different operational profiles:

- Damietta showed substantially high activity in both roles, with stronger discharge activity in the analyzed records.
- Hamburg and Antwerp generally showed higher loading than discharge counts.
- Piraeus and Novorossiysk exhibited more balanced loading and discharge patterns.
- Several lower-volume ports appeared or changed position between the two reporting years.

These results provide evidence of directional differences within the recorded trade network but should not by themselves be interpreted as complete measures of national trade balance or port-level commercial specialization.

6.15. Capacity Utilization Analysis

Monthly capacity utilization was estimated by comparing calculated TEU activity against the documented annual design-capacity assumption of 500,000 TEU.

Listing 6.9: Monthly capacity-utilization calculation

```
WITH teu_calc AS (
  SELECT
    FORMAT(f.InDateOnly, 'yyyy-MM') AS YearMonth,
    dcs.TEUFactor AS TEU
  FROM FactContainerOperations f
  JOIN DimContainerSpecification dcs
  ON f.ContainerSpecificationKey =
    dcs.ContainerSpecificationKey
  WHERE f.InDateOnly IS NOT NULL
  AND MONTH(f.InDateOnly) <= 3
)
SELECT
  YearMonth,
  SUM(TEU) AS Actual_TEU,
  500000 / 12.0 AS Design_Capacity_TEU_Monthly,
  ROUND(
    SUM(TEU) / (500000.0 / 12) * 100,
    1
  ) AS Utilization_Percentage
FROM teu_calc
GROUP BY YearMonth
ORDER BY YearMonth;
```

The analysis uses a monthly planning baseline of approximately 41,666.7 TEU, obtained by dividing the assumed annual design capacity by 12.

Because the 500,000-TEU figure is a documented project assumption, the resulting utilization percentages should be interpreted as comparative indicators relative to that assumption rather than as independently verified physical capacity limits.

6.16. Quay-Crane Productivity Exploration

SQL queries were developed to examine quay-crane productivity at both detailed and monthly levels.

The initial exploratory method calculated moves per hour directly from QCWorkingHoursSource contained in the container-operation fact.

Listing 6.10: Exploratory monthly quay-crane productivity

```
SELECT
FORMAT(f.InDateOnly, 'yyyy-MM') AS YearMonth,
COUNT(f.ContainerNumber) AS Total_Moves,
SUM(
DATEDIFF(
SECOND,
CAST('00:00:00' AS TIME),
f.QCWorkingHoursSource
)
) / 3600.0 AS Total_Crane_Hours,
ROUND(
COUNT(f.ContainerNumber) * 1.0 /
NULLIF(
SUM(
DATEDIFF(
SECOND,
CAST('00:00:00' AS TIME),
f.QCWorkingHoursSource
)
) / 3600.0,
0
),
1
) AS Moves_Per_Hour
FROM FactContainerOperations f
JOIN DimEquipment de
ON f.DimEquipment_EquipmentKey = de.EquipmentKey
WHERE de.EquipmentNumber IS NOT NULL
AND de.EquipmentNumber <> ''
AND MONTH(f.InDateOnly) IN (1,2,3)
GROUP BY FORMAT(f.InDateOnly, 'yyyy-MM')
ORDER BY YearMonth;
```

This exploration was valuable because it revealed an important grain issue: QC working-hour values can repeat across multiple container-operation rows. Direct summation at container grain can therefore overstate crane hours and distort productivity.

The project subsequently addressed this issue by creating FactQCPerformance at the appropriate vessel-call and equipment grain and validating QC-hour consistency before using the records in governed productivity measures.

QC productivity grain

Quay-crane productivity must be calculated using validated QC working hours at the appropriate vessel-call and equipment grain. Repeated source-hour values should not be summed directly across container-operation records. SQL queries using QCWorkingHoursSource at container grain are therefore treated as exploratory analysis that helped identify and resolve this modeling issue.

6.17. Dwell-Time Analysis

SQL analysis examined average StackDays for records with a valid recorded exit, including monthly and mode-specific comparisons.

Listing 6.11: Monthly dwell-time analysis

```
SELECT
DATEPART(YEAR, f.InDateOnly) AS Year,
DATEPART(MONTH, f.InDateOnly) AS Month,
dtp.Mode,
AVG(f.StackDays) AS Avg_Stack_Days
FROM FactContainerOperations f
JOIN DimTransportProfile dtp
ON f.DimTransportProfile_TransportProfileKey =
dtp.TransportProfileKey
WHERE f.OutDateOnly IS NOT NULL
AND MONTH(f.InDateOnly) IN (1,2,3)
GROUP BY
dtp.Mode,
DATEPART(YEAR, f.InDateOnly),
DATEPART(MONTH, f.InDateOnly)
ORDER BY Year, Month;
```

A year-over-year query further compared monthly dwell by operational mode, allowing the percentage change between 2025 and 2026 to be examined.

The results demonstrated that import and export dwell behavior did not necessarily move in the same direction. This distinction is important because an overall average can conceal differences between operational flows.

Observed changes in dwell time are treated as performance patterns. Explanations involving customs processes, shipping delays, congestion, or external disruptions require additional operational evidence before they can be established as causal findings.

6.18. Combined Monthly Activity Analysis

A combined SQL query brought together container-voyage counts, voyages, vessel calls, and average dwell time by month and operational mode.

Listing 6.12: Monthly activity and dwell analysis

```
SELECT
YEAR(f.InDateOnly) AS Year,
MONTH(f.InDateOnly) AS Month,
dtp.Mode,
COUNT(
DISTINCT CONCAT(
f.ContainerNumber,
```



```

    '- ',
    dvc.UserVoyage
  )
) AS Containers,
COUNT(DISTINCT dvc.UserVoyage) AS Voyages,
COUNT(
  DISTINCT CONCAT(
    dvc.VesselCode,
    '- ',
    dvc.CallingSequence
  )
) AS Vessel_Calls,
ROUND(AVG(f.StackDays), 2) AS Avg_Dwell_Time
FROM FactContainerOperations f
JOIN DimVesselCall dvc
ON f.VesselCallKey = dvc.VesselCallKey
JOIN DimTransportProfile dtp
ON f.DimTransportProfile_TransportProfileKey =
dtp.TransportProfileKey
WHERE YEAR(f.InDateOnly) IN (2025, 2026)
AND MONTH(f.InDateOnly) IN (1,2,3)
GROUP BY
YEAR(f.InDateOnly),
MONTH(f.InDateOnly),
dtp.Mode
ORDER BY Year, Month, dtp.Mode;

```

This analysis helps interpret dwell performance in the context of changing traffic levels rather than examining time-based KPIs independently of operational volume.

6.19. Vessel Turnaround Analysis

The detailed vessel-call fact table was used to calculate vessel turnaround directly at vessel-call grain.

Listing 6.13: Monthly vessel-turnaround analysis

```

SELECT
  DATEPART(YEAR, ATB) AS Year,
  DATEPART(MONTH, ATB) AS Month,
  AVG(VesselTurnaroundHours) AS Avg_Turnaround_Hours,
  COUNT(*) AS Vessel_Calls
FROM FactVesselPortCalls
GROUP BY
  DATEPART(YEAR, ATB),
  DATEPART(MONTH, ATB)
ORDER BY Year, Month;

```

This query preserves the appropriate vessel-call grain and avoids counting the same vessel turnaround repeatedly through container-level records.

6.20. Grain Risk Identified During Vessel Analysis

An early exploratory vessel-turnaround-by-mode query linked FactVesselPortCalls to FactContainerOperations. Because one vessel call can be associated with many container-operation records, this created a many-row expansion of the vessel-call observations.

The result demonstrates an important dimensional-modeling principle: two fact tables should not be joined directly unless their grains are made compatible first.

A vessel-level measure repeated across container rows can produce an apparently reasonable numerical result while still being structurally incorrect.

The appropriate resolution is to:

- aggregate each fact table to a common analytical grain before joining;
- use shared dimensions to coordinate filtering between facts; or
- calculate vessel measures directly from the vessel-call fact when container-level detail is not required.

This finding contributed directly to the refinement of the final multi-fact analytical architecture.

6.21. Import and Export Share

A window-function query was used to calculate the relative share of import and export activity within each year.

Listing 6.14: Year-level import/export share

```
SELECT
  dvc.CallingYear AS Year,
  dtp.Mode,
  COUNT(*) AS Total_Containers,
  ROUND(
    COUNT(*) * 100.0 /
    SUM(COUNT(*))
  OVER (PARTITION BY dvc.CallingYear),
  1
  ) AS Pct_Of_Year
FROM FactContainerOperations f
JOIN DimVesselCall dvc
ON f.VesselCallKey = dvc.VesselCallKey
JOIN DimTransportProfile dtp
ON f.DimTransportProfile_TransportProfileKey =
  dtp.TransportProfileKey
WHERE MONTH(f.InDateOnly) IN (1,2,3)
GROUP BY
  dvc.CallingYear,
  dtp.Mode
ORDER BY Year, dtp.Mode;
```

The analysis demonstrates a substantial change in the composition of import and export activity between the Q1 periods. This trade-mix shift is consistent with the broader comparative analysis and provides important context for interpreting changes in dwell time, yard workload, and terminal activity.

6.22. Country-Level Distribution

Country-level discharge analysis was performed using a Common Table Expression (CTE) followed by a window function.

Listing 6.15: Country share by calling year

```
WITH country_totals AS (
  SELECT
    dvc.CallingYear AS Year,
    dpod.POD_Master_Country AS Country,
```

```

COUNT(*) AS Total_Containers
FROM FactContainerOperations f
JOIN DimVesselCall dvc
ON f.VesselCallKey = dvc.VesselCallKey
JOIN DimPortPOD dpod
ON f.PODKey = dpod.PODKey
GROUP BY
dvc.CallingYear,
dpod.POD_Master_Country
)
SELECT
Year,
Country,
Total_Containers,
ROUND(
Total_Containers * 100.0 /
SUM(Total_Containers)
OVER (PARTITION BY Year),
1
) AS Pct_Of_Year
FROM country_totals
ORDER BY
Year,
Total_Containers DESC;

```

The CTE first aggregates activity by calling year and discharge country. The outer query then calculates each country's share of the corresponding annual total.

This technique provides a useful example of how SQL window functions can calculate relative contribution without losing the underlying grouped values.

6.23. Representative Analytical Techniques

The SQL workstream demonstrates several important analytical techniques used throughout the project:

- metadata inspection using `INFORMATION_SCHEMA`;
- primary- and foreign-key configuration;
- data-type validation and alignment;
- unique and referential constraints;
- joins across fact and dimension structures;
- `COUNT(DISTINCT ...)` for business-level counting;
- aggregation using `GROUP BY`;
- conditional filtering by year and month;
- Common Table Expressions;
- window functions using `OVER(PARTITION BY ...)`;
- percentage-of-total calculations;
- year-over-year comparison;
- operational ranking;
- validation of analytical grain; and
- identification of data-quality and modeling risks.

These techniques demonstrate that the SQL workstream supported both database structure and analytical reasoning rather than functioning only as a query-output layer.

6.24. Analytical Validation and Interpretation

The SQL analysis identified several important operational patterns, including changes in overall activity, vessel frequency, operator workload, cargo composition, port connectivity, capacity utilization, dwell behavior, crane productivity, vessel turnaround, and import/export balance.

However, interpretation was deliberately separated from causality. The operational data can demonstrate that two measures changed at the same time, but it cannot independently establish why those changes occurred.

For example, changes in throughput, dwell time, vessel turnaround, or port connectivity could potentially be influenced by factors such as:

- shipping-line schedule changes,
- berth availability,
- equipment availability,
- yard congestion,
- customs processes,
- maintenance activities,
- weather or navigation conditions,
- vessel cancellations or delays, or
- broader supply-chain disruptions.

These factors represent possible areas for further investigation rather than verified explanations unless supported by additional operational or external evidence.

Interpretation discipline

SQL results establish patterns, differences, associations, and operational distributions within the available data. They do not by themselves establish causal relationships. Explanations involving external events, maintenance, congestion, customs performance, shipping disruptions, or geopolitical conditions should therefore be treated as analytical hypotheses unless independently supported by additional evidence.

6.25. Contribution of the SQL Workstream

The SQL Server workstream strengthened the project in three principal ways. First, it provided structural validation through keys, data types, constraints, and relationship testing. Second, it enabled detailed exploration of the operational data across multiple business dimensions. Third, it helped identify important analytical risks, particularly those associated with fact-table grain, repeated values, and inappropriate aggregation.

The SQL results were reviewed alongside the wider project analysis so that the final conclusions remained aligned with common business definitions and validation principles. In this way, SQL contributed both technical assurance and analytical depth to the complete container-terminal business intelligence solution.

7. Python Data Cleaning and Workbook Preparation

7.1. Purpose of the Python workstream

Python was used to clean, inspect, document, and re-export the multi-sheet Excel workbook that contained the terminal data. The work was performed in a notebook environment using `pandas` for tabular processing and `openpyxl` for reading and writing modern Excel workbooks.

The objective was not simply to open the workbook and save another copy. The Python process was designed to make the workbook easier to trust and easier to use by:

- confirming that the source workbook could be found and opened;
- identifying every worksheet available in the workbook;
- standardizing column names and text values;
- removing completely blank rows and columns;
- protecting important key fields from missing and duplicate values;
- identifying and removing exact duplicate records where applicable;
- converting likely date and time fields into appropriate datetime values;
- standardizing selected commodity descriptions;
- preserving numeric and date fields while treating missing text values consistently;
- recording the effect of cleaning on every worksheet;
- producing a technical data dictionary;
- previewing the cleaned tables; and
- writing the results to a separate cleaned Excel workbook.

The process was therefore both a cleaning exercise and a quality-assurance exercise. Python applied the same controlled logic across multiple worksheets, recorded what happened during execution, and produced a reusable workbook rather than modifying the original source file directly.

7.2. Preparing the notebook environment

The notebook began by preparing the Python environment and loading the Excel-processing libraries required for the work. The `openpyxl` package was installed so that `pandas` could read and write `.xlsx` files reliably. The source and output paths were then defined so that the notebook could distinguish between the original workbook and the cleaned copy.

Figure 7.1 shows the initial notebook stage in which the required package was installed and the source workbook was made available in the working environment.



```
[1]
✓ 7m !pip -q install openpyxl

import pandas as pd
import numpy as np
import re
from google.colab import files
from IPython.display import display

print("Choose file...")
uploaded = files.upload()

file_name = next(iter(uploaded))
print(f"\n file loaded : {file_name}")

... Choose file...
Choose Files Dataset.xlsx
Dataset.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 53174899 bytes, last modified: 8/15/2026 - 100% done
Saving Dataset.xlsx to Dataset.xlsx

file loaded : Dataset.xlsx
```

Figure 7.1.: Preparing the Python environment, installing the Excel-processing dependency, and loading the source workbook.

Before reading any data, the script checked whether the input path existed and whether the file had an .xlsx extension. These checks prevented the remainder of the notebook from running against a missing file or an unsupported format. This was important because a failed workbook read later in the process would otherwise produce less informative errors after several processing steps had already started.

After the file check, the workbook was opened with `pd.ExcelFile`. Rather than assuming that a fixed set of worksheets would always be present, the notebook retrieved the worksheet names directly from the workbook. This allowed the analyst to see the actual structure of the submitted file before cleaning it.

Figure 7.2 shows the workbook inspection and the resulting worksheet inventory.



```
[4]
✓ 1s excel_file = pd.ExcelFile(file_name)
sheet_names = excel_file.sheet_names

print(f"\n Number of sheets: {len(sheet_names)}")
print("sheets:")
for i, sheet in enumerate(sheet_names, 1):
    print(f"{i}. {sheet}")

...
Number of sheets: 15
sheets:
1. StgContainerOperations
2. DimVesselCall
3. DimOperator
4. DimContainerSpecification
5. DimEquipment
6. DimCargo
7. PortMaster
8. DimPortPOL
9. POD_Master
10. DimPortPOD
11. DimTransportProfile
12. DimDamageCondition
13. CheckQCWorkingHours
14. FactQCPerformance
15. FactContainerOperations
```

Figure 7.2.: Inspecting the source workbook and listing the worksheets available for processing.

Each worksheet was then read into a separate pandas DataFrame. The DataFrames were stored in a

dictionary keyed by worksheet name. This structure made it possible to process all worksheets through the same loop while retaining each table as a separate object.

The workbook contained operational, dimensional, reference, and quality-control tables. Examples included container-operation records, vessel-call data, operator codes, cargo classifications, equipment identifiers, port reference tables, transport profiles, damage categories, and quay-crane validation tables. Loading every worksheet at the beginning provided a complete view of the workbook and ensured that cleaning was applied consistently rather than only to the most visible tables.

7.3. Standardizing worksheet structure

7.3.1. Cleaning column names

Column-name consistency was addressed before individual data values were changed. Spreadsheet exports can contain leading or trailing spaces, repeated spaces, inconsistent punctuation, unusual Unicode characters, and other invisible differences. Such differences can prevent a field from being found even when the visible header appears correct.

A reusable `clean_column_name` function was created to convert each header to text, trim surrounding spaces, normalize unwanted characters, and produce a more stable field name. Figure 7.3 shows the helper function used for this purpose.

```
[3]
✓ Os
def clean_column_name(col):
    col = str(col).strip()
    col = re.sub(r'\s+', '_', col)
    col = re.sub(r'^\w\u0600-\u06FF', '_', col)
    col = re.sub(r'+', '', col)
    col = col.strip('_')
    return col
```

Figure 7.3.: Reusable Python function for normalizing worksheet column names.

The function was applied to the headers of every worksheet. Standardizing headers early in the process made all subsequent references to fields more dependable. It also reduced the risk that two visually identical fields would be treated as different because one contained an extra space or nonstandard character.

Column cleaning can create a secondary issue. Two original headers that differ only by punctuation or spacing may become identical after normalization. The notebook therefore checked the cleaned column names for duplication. When a duplicate header was found, a numeric suffix was added so that every column remained uniquely addressable. This avoided silently overwriting one column with another.

7.3.2. Case-insensitive matching of important fields

The notebook did not assume that worksheet and column capitalization would always remain unchanged. When a specific field was required, the code compared names after trimming spaces and converting them to lowercase.

This approach was particularly important for the `UserVoyage` field in `DimVesselCall`. The script searched for the column case-insensitively before applying key-field cleaning. As a result, a variation such as `uservoyage`, `UserVoyage`, or a version containing surrounding spaces could still be found correctly.

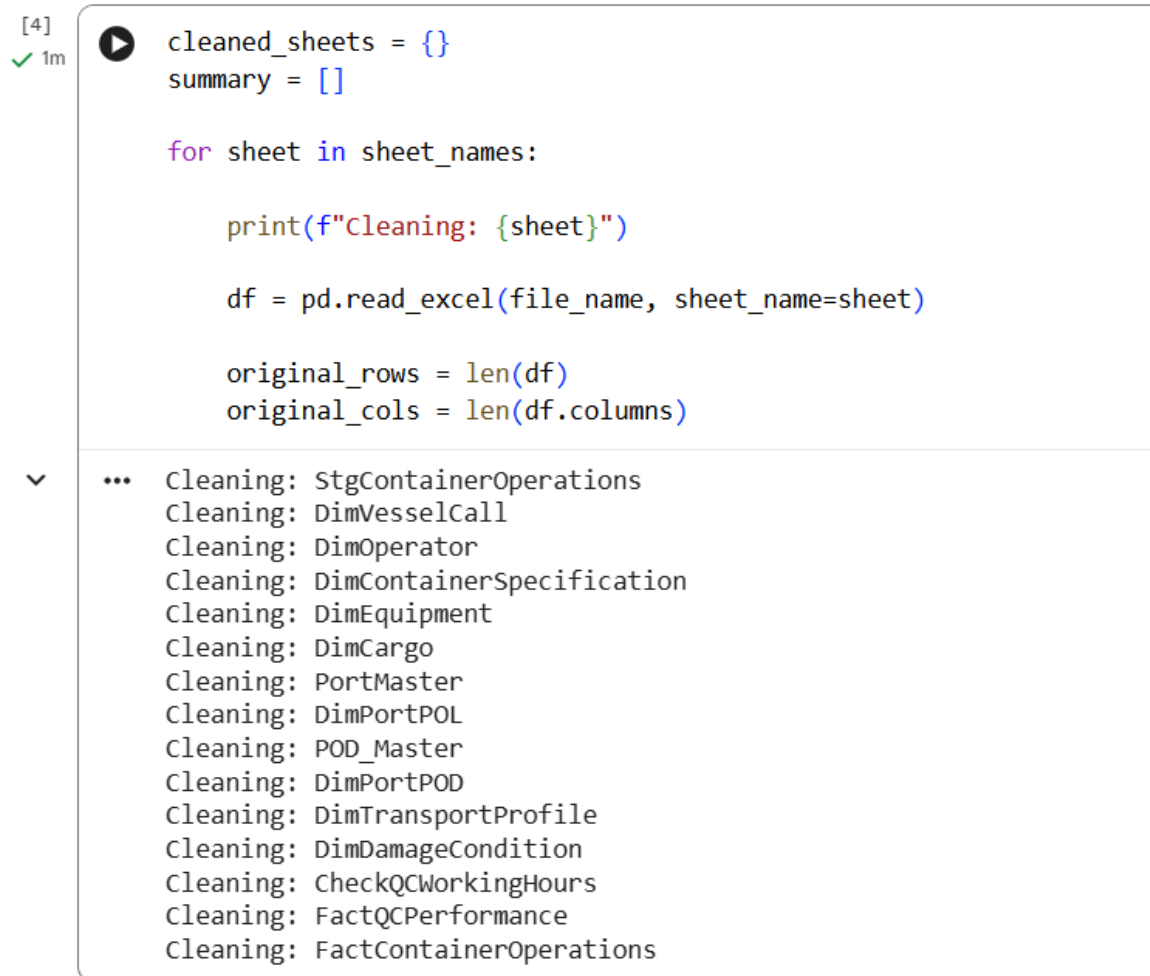
7.4. Processing the workbook sheet by sheet

After the source structure had been inspected, the notebook entered its main cleaning loop. Two principal Python objects were maintained during this stage:

- `cleaned_sheets`, which stored the cleaned DataFrame for every worksheet; and
- `summary`, which accumulated before-and-after quality statistics for each worksheet.

At the beginning of each loop iteration, the worksheet was read and its original number of rows and columns was recorded. These baseline values allowed the cleaning results to be measured rather than described only informally.

Figure 7.4 shows the beginning of the loop and the progress messages used to indicate which worksheet was being processed.



```
[4] ✓ 1m ▶ cleaned_sheets = {}
summary = []

for sheet in sheet_names:

    print(f"Cleaning: {sheet}")

    df = pd.read_excel(file_name, sheet_name=sheet)

    original_rows = len(df)
    original_cols = len(df.columns)

... Cleaning: StgContainerOperations
Cleaning: DimVesselCall
Cleaning: DimOperator
Cleaning: DimContainersSpecification
Cleaning: DimEquipment
Cleaning: DimCargo
Cleaning: PortMaster
Cleaning: DimPortPOL
Cleaning: POD_Master
Cleaning: DimPortPOD
Cleaning: DimTransportProfile
Cleaning: DimDamageCondition
Cleaning: CheckQCWorkingHours
Cleaning: FactQCPerformance
Cleaning: FactContainerOperations
```

Figure 7.4.: Beginning of the workbook-wide cleaning loop and initialization of the cleaned-sheet and summary objects.

The progress output was useful for two reasons. First, it confirmed that every worksheet was reached by the loop. Second, if an error occurred, the most recently printed worksheet name indicated where the process had stopped.

7.5. Removing structural noise

7.5.1. Completely blank rows and columns

Excel workbooks frequently contain unused formatted ranges, placeholder rows, or columns that are technically part of the worksheet but contain no values. These areas increase worksheet size and can interfere with row counts, previews, and later processing.

For each DataFrame, Python removed:

- rows in which every field was blank; and
- columns in which every field was blank.

Only completely empty structures were removed. A row containing a valid container number but a missing optional field was not discarded. This distinction preserved incomplete but potentially useful operational records while removing rows and columns that contained no information at all.

After blank rows were removed, the DataFrame index was reset. Resetting the index produced a clean sequential index and prevented gaps left by removed rows from appearing in later previews or exports.

7.5.2. Text normalization

Text cleaning was applied to object or string columns. Each text value was converted to a consistent string representation and stripped of leading and trailing spaces. This prevented values such as MSC and MSC_ from becoming separate categories.

Several common textual representations of missing data were also normalized. Empty strings and text values such as nan, NaN, and None were treated as missing values rather than legitimate categories.

Figure 7.5 shows the portion of the cleaning loop that handled headers, duplicated column names, blank structures, and text normalization.

```

cleaned_sheets = {}
summary = []

for sheet in sheet_names:

    print(f"Cleaning: {sheet}")

    df = pd.read_excel(file_name, sheet_name=sheet)

    original_rows = len(df)
    original_cols = len(df.columns)

    # Clean column names
    df.columns = [clean_column_name(c) for c in df.columns]

    # Handle duplicate column names
    new_columns = []
    used_columns = {}

    for col in df.columns:

        if col not in used_columns:
            used_columns[col] = 0
            new_columns.append(col)

        else:
            used_columns[col] += 1
            new_columns.append(
                f"{col}_{used_columns[col]}"
            )

    df.columns = new_columns

    # Remove completely empty columns
    df = df.dropna(axis=1, how='all')

    # Remove completely empty rows
    df = df.dropna(axis=0, how='all')

    # Clean text values
    for col in df.select_dtypes(include=['object']).columns:

        df[col] = (
            df[col]
            .astype("string")
            .str.strip()
        )

        df[col] = df[col].replace({
            "": pd.NA,
            "nan": pd.NA,
            "None": pd.NA,
            "NaN": pd.NA
        })

```

<https://colab.research.google.com/drive/1YhqX0H4IZSLUD09rqDqxZPhuQEPL-dmu#scrollTo=d-n0IKusdiji>

Figure 7.5.: Standardizing headers, resolving duplicate column names, removing completely empty structures, and cleaning text values.

These operations were especially valuable for categorical fields such as vessel names, operator codes, cargo types, commodities, port names, equipment numbers, and status labels. Small differences in white-space would otherwise fragment one real-world category into multiple analytical values.

7.6. Protecting key and categorical fields

7.6.1. Cleaning UserVoyage

The `DimVesselCall` worksheet received an additional targeted check because `UserVoyage` was treated as an important identifying field.

Python first located the field using case-insensitive matching. Rows with a missing `UserVoyage` value were removed because they could not support a reliable vessel-voyage reference. The remaining records were checked for duplicate `UserVoyage` values. When duplicates were found, the first occurrence was retained and the later occurrences were removed.

The script recorded:

- the number of rows removed because the field was null; and
- the number of duplicate key values removed.

This made the key-field cleaning transparent. The notebook did not remove records silently; it reported how many records were affected.

7.6.2. Standardizing commodity descriptions

Two commodity descriptions were standardized because they represented inconsistent names for the same intended categories:

- `Danger Goods` was changed to `Dangerous Goods`; and
- `Electrical Appliances` was changed to `Electronic Devices`.

The replacements were applied to the `Commodity` field in both `DimCargo` and `StgContainerOperations`. Applying the same replacement in the dimensional and staging tables prevented the old and new descriptions from coexisting in different parts of the workbook.

7.6.3. Trimming selected business fields

A worksheet-specific trimming map was used to identify the text columns that required explicit whitespace removal. The mapping covered fields from the operational, dimensional, reference, and quality-control tables.

Examples included:

- vessel codes, vessel names, vessel business keys, and vessel-call labels;
- operator and equipment codes;
- cargo types, commodities, and cargo labels;
- container size, type, height, width, and TEU mapping labels;
- POL and POD codes, names, countries, regions, and labels;
- transport status, mode, delivery, dispatch, and full/empty fields;
- damage-condition descriptions;
- container numbers, weight groups, match-status fields, and age buckets; and
- quay-crane working-hour status fields.

The actual column name was identified case-insensitively before trimming. Trimming was applied only when the located column had a text-compatible data type.

Preserving numeric and date types

Missing-value replacement was limited to text columns. Numeric fields and date fields were not converted to text merely to display the word `Unknown`. This preserved the ability to calculate totals, averages, durations, and date differences after cleaning.

7.7. Duplicate detection and date preparation

7.7.1. Exact duplicate rows

The workbook-wide cleaning loop counted exact duplicate rows before removing them. An exact duplicate was a row whose complete set of values matched another row in the same worksheet.

The duplicate count was retained in the cleaning summary so that the analyst could distinguish between:

- a worksheet that originally contained no duplicates; and
- a worksheet from which duplicated records had been removed.

This was preferable to calling `drop_duplicates()` without recording its effect. The resulting audit information made the transformation measurable.

7.7.2. Date and time conversion

Operational workbooks often contain date values stored as text. The notebook examined column names for indicators such as `date`, `time`, `datetime`, `created`, and `updated`. Columns whose names suggested temporal content were tested with `pd.to_datetime`.

The conversion used `errors="coerce"`, which changed unparseable values to missing datetime values rather than stopping the whole notebook. A field was accepted as a date column only when a sufficient share of its values could be parsed successfully. The notebook used a 70% valid-value threshold.

This threshold reduced the risk of converting a non-date field merely because its name contained a date-related word. It also allowed genuinely temporal columns with a limited number of missing or malformed values to be converted successfully.

Figure 7.6 shows the duplicate check, date-conversion logic, and the construction of the per-sheet summary record.

```

2:55 - 2026/8/20
Untitled6.ipynb - Colab

# Count duplicates
duplicates = int(df.duplicated().sum())

# Remove duplicate rows
df = df.drop_duplicates()

# Try to convert date columns
for col in df.columns:
    col_name = str(col).lower()

    if any(
        x in col_name
        for x in [
            'date',
            'time',
            'datetime',
            'created',
            'updated'
        ]
    ):
        try:
            converted = pd.to_datetime(
                df[col],
                errors='coerce'
            )

            valid_ratio = (
                converted.notna().mean()
                if len(df) > 0
                else 0
            )

            if valid_ratio >= 0.70:
                df[col] = converted

        except Exception:
            pass

    cleaned_sheets[sheet] = df

summary.append({
    "Sheet": sheet,
    "Original Rows": original_rows,
    "Clean Rows": len(df),
    "Original Columns": original_cols,
    "Clean Columns": len(df.columns),
    "Duplicates Removed": duplicates,
    "Missing Values": int(df.isna().sum().sum())
})

print(
    f"Completed: {original_rows:,} rows -> "
    f"{len(df):,} rows | "
    f"{original_cols:,} columns -> "

```

<https://colab.research.google.com/drive/1YhqX0H4IZSLUD09rqDqxZPhuQEPL-dmu#scrollTo=d-n0IKusdiji> 2/4

Figure 7.6.: Counting duplicate rows, converting likely temporal fields, and recording worksheet-level cleaning statistics.

7.8. Recording the result of each cleaning pass

After a worksheet had been cleaned, its DataFrame was placed in `cleaned_sheets`. A summary record was then created containing:

- worksheet name;
- original number of rows;
- cleaned number of rows;
- original number of columns;

- cleaned number of columns;
- number of duplicate rows removed; and
- total number of missing values remaining after cleaning.

The console printed a short before-and-after message for every worksheet. This provided immediate evidence that the workbook was being processed and showed whether the number of rows or columns changed.

Figure 7.7 shows the completion messages produced while the loop moved through the workbook.

```

2:55 - 2026/8/20
Untitled6.ipynb - Colab

f"{len(df.columns)} columns"
)
print()
print("All sheets have been cleaned successfully")

Cleaning: StgContainerOperations
/twp/ipykernel_2031/2793637908.py:79: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'. To ensure parsing is
converted = pd.to_datetime(
Completed: 98,052 rows -> 98,052 rows | 37 columns -> 37 columns
Cleaning: DimVesselCall
Completed: 201 rows -> 201 rows | 7 columns -> 7 columns
Cleaning: DimOperator
Completed: 33 rows -> 33 rows | 2 columns -> 2 columns
Cleaning: DimContainerSpecification
Completed: 76 rows -> 76 rows | 8 columns -> 8 columns
Cleaning: DimEquipment
Completed: 9 rows -> 9 rows | 2 columns -> 2 columns
Cleaning: DimCargo
Completed: 240 rows -> 240 rows | 5 columns -> 5 columns
Cleaning: PortMaster
Completed: 71 rows -> 71 rows | 4 columns -> 4 columns
Cleaning: DimPortPOL
Completed: 71 rows -> 71 rows | 7 columns -> 7 columns
Cleaning: POD_Master
Completed: 51 rows -> 51 rows | 4 columns -> 4 columns
Cleaning: DimPortFOD
Completed: 52 rows -> 52 rows | 6 columns -> 6 columns
Cleaning: DimTransportProfile
Completed: 269 rows -> 269 rows | 12 columns -> 12 columns
Cleaning: DimDamageCondition
Completed: 4 rows -> 4 rows | 2 columns -> 2 columns
Cleaning: CheckQCWorkingHours
Completed: 1,134 rows -> 1,134 rows | 11 columns -> 11 columns
Cleaning: FactQCPerformance
Completed: 1,134 rows -> 1,134 rows | 13 columns -> 13 columns
Cleaning: FactContainerOperations
Completed: 98,052 rows -> 98,052 rows | 33 columns -> 33 columns

All sheets have been cleaned successfully
/twp/ipykernel_2031/2793637908.py:79: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'. To ensure parsing is
converted = pd.to_datetime(

```

<https://colab.research.google.com/drive/1YhqX0H4IZSLUD09rqDqxZPhuQEPL-dmu#scrollTo=d-nOIKudjji> 3/4

Figure 7.7.: Console output confirming the progress and completion of worksheet cleaning.

The console output also displayed a date-parsing warning during automatic conversion. Such a warning indicates that Python was inferring date formats instead of using one explicitly supplied format. The notebook continued because invalid values were handled through coercion and the conversion acceptance threshold.

After all worksheets had been processed, the list of summary records was converted into a DataFrame named `summary_df`. Figure 7.8 shows the final completion message and the creation of this summary table.

```

[18]: print()
print("All sheets have been cleaned successfully")

Cleaning: StgContainerOperations
/tmp/ipykernel_2031/3657802595.py:98: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'. To ensure parsing is consistent and as-expected, please specify a format.
  converted = pd.to_datetime(
Completed: 98,052 rows -> 98,052 rows | 37 columns -> 37 columns
Cleaning: DimVesselCall
Completed: 201 rows -> 201 rows | 7 columns -> 7 columns
Cleaning: DimOperator
Completed: 33 rows -> 33 rows | 2 columns -> 2 columns
Cleaning: DimContainerSpecification
Completed: 76 rows -> 76 rows | 8 columns -> 8 columns
Cleaning: DimEquipment
Completed: 9 rows -> 9 rows | 2 columns -> 2 columns
Cleaning: DimCargo
Completed: 240 rows -> 240 rows | 5 columns -> 5 columns
Cleaning: PortMaster
Completed: 71 rows -> 71 rows | 4 columns -> 4 columns
Cleaning: DimPortPOL
Completed: 71 rows -> 71 rows | 7 columns -> 7 columns
Cleaning: POD_Master
Completed: 51 rows -> 51 rows | 4 columns -> 4 columns
Cleaning: DimPortPOD
Completed: 52 rows -> 52 rows | 6 columns -> 6 columns
Cleaning: DimTransportProfile
Completed: 269 rows -> 269 rows | 12 columns -> 12 columns
Cleaning: DimDamageCondition
Completed: 4 rows -> 4 rows | 2 columns -> 2 columns
Cleaning: CheckQCWorkingHours
Completed: 1,134 rows -> 1,134 rows | 11 columns -> 11 columns
Cleaning: FactQCPerformance
Completed: 1,134 rows -> 1,134 rows | 13 columns -> 13 columns
Cleaning: FactContainerOperations
Completed: 98,052 rows -> 98,052 rows | 33 columns -> 33 columns

All sheets have been cleaned successfully
/tmp/ipykernel_2031/3657802595.py:98: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'. To ensure parsing is consistent and as-expected, please specify a format.
  converted = pd.to_datetime(

[19]: summary_df = pd.DataFrame(summary)

```

Figure 7.8.: Completion of the workbook-wide cleaning loop and construction of the cleaning-summary DataFrame.

7.9. Reviewing the cleaning summary

The cleaning summary converted the notebook run into an auditable result. Instead of relying on a general statement that the data had been cleaned, the table showed the effect on each worksheet.

Figure 7.9 presents the rendered cleaning summary and the confirmation that the cleaned workbook was written successfully.

```

=====
.. CLEANING SUMMARY
=====
/tmp/ipykernel_822/2407386915.py:87: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'.
  converted = pd.to_datetime(

```

	Sheet	Original Rows	Clean Rows	Original Columns	Clean Columns	Duplicates Removed	Missing Values
0	StgContainerOperations	98052	98052	37	37	0	238669
1	DimVesselCall	201	201	7	7	0	15
2	DimOperator	33	33	2	2	0	0
3	DimContainerSpecification	76	76	8	8	0	1
4	DimEquipment	9	9	2	2	0	0
5	DimCargo	240	240	5	5	0	2
6	PortMaster	71	71	4	4	0	0
7	DimPortPOL	71	71	7	7	0	0
8	POD_Master	51	51	4	4	0	0
9	DimPortPOD	52	52	6	6	0	3
10	DimTransportProfile	269	269	12	12	0	0
11	DimDamageCondition	4	4	2	2	0	0
12	CheckQCWorkingHours	1134	1134	11	11	0	51
13	FactQCPerformance	1134	1134	13	13	0	254
14	FactContainerOperations	98052	98052	33	33	0	98494

```

=====
file are cleaned successfully
result PowerBI_Extract_Cleaned.xlsx
=====

```

Figure 7.9.: Worksheet-level cleaning summary and confirmation that the cleaned workbook copy was created.

The summary showed that the large operational tables retained their main row populations while still containing measurable missing values in optional or conditionally populated fields. This was an important outcome: the cleaning process did not treat every blank cell as a reason to remove the entire record.

Small dimension and reference tables could also be reviewed quickly for duplicate removal and missing-value counts. Tables with no duplicates and no missing values required no additional row-level

intervention, while tables with remaining blanks could be examined using the technical data dictionary produced in the next stage.

The summary therefore answered several practical questions:

- Did a worksheet lose rows during cleaning?
- Were completely empty columns removed?
- Did exact duplicate rows exist?
- How many missing values remained?
- Was the cleaned output created successfully?

7.10. Creating a technical data dictionary

Once the data had been cleaned, Python was used to describe the contents of the cleaned workbook at column level. The notebook iterated through every column of every cleaned DataFrame and recorded:

- worksheet name;
- column name;
- Python/pandas data type;
- worksheet row count;
- number of missing values;
- percentage of missing values; and
- number of distinct nonblank values.

The accumulated records were converted into `dictionary_df`. Figure 7.10 shows the code used to build the dictionary and part of the resulting table.

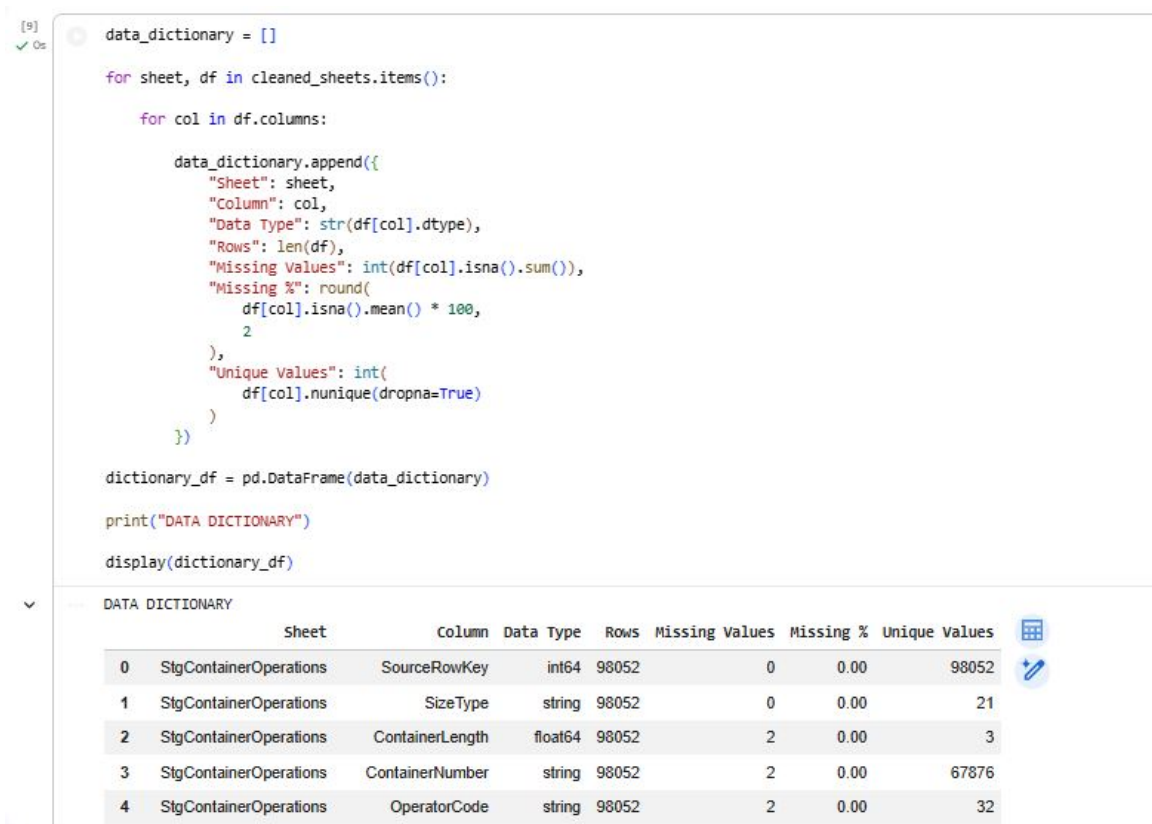


Figure 7.10.: Construction and display of the column-level technical data dictionary.

The data dictionary supported more detailed quality review than the worksheet summary. A worksheet

might have thousands of missing values, but the dictionary identified exactly which fields contained them and what proportion of each field was affected.

The distinct-value count also helped reveal the likely function of a column. For example:

- a row key should have a very high distinct count;
- an operator field should contain a limited set of codes;
- a status field should contain a small set of repeated categories;
- a container-number field should contain many distinct values; and
- a date field should appear with a temporal data type after conversion.

The dictionary therefore served as a technical inventory of the cleaned workbook and provided a concise basis for future inspection.

7.11. Writing the cleaned Excel workbook

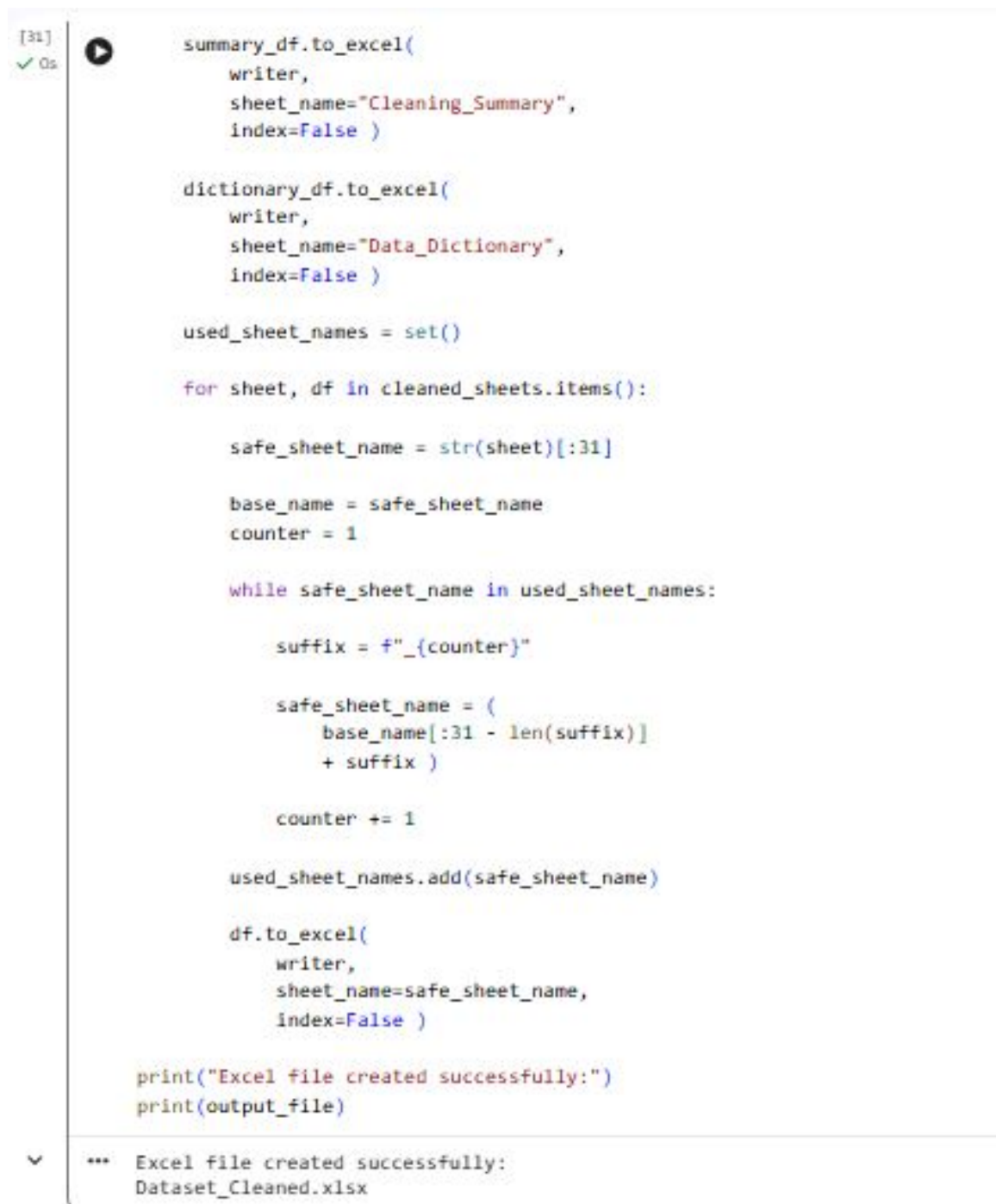
The cleaned DataFrames, cleaning summary, and technical data dictionary were written to a new Excel workbook. The original workbook was not overwritten.

The export process created dedicated worksheets for:

- the cleaning summary;
- the technical data dictionary; and
- every cleaned source worksheet.

Excel limits worksheet names to 31 characters. The Python code therefore generated safe worksheet names by truncating long names where necessary. It also checked for collisions after truncation. If two resulting worksheet names would have been identical, a numeric suffix was added to keep them unique.

Figure 7.11 shows the export process and the safe worksheet-name handling.



```
[31] summary_df.to_excel(
      writer,
      sheet_name="Cleaning_Summary",
      index=False )

dictionary_df.to_excel(
      writer,
      sheet_name="Data_Dictionary",
      index=False )

used_sheet_names = set()

for sheet, df in cleaned_sheets.items():

    safe_sheet_name = str(sheet)[:31]

    base_name = safe_sheet_name
    counter = 1

    while safe_sheet_name in used_sheet_names:

        suffix = f"_{counter}"

        safe_sheet_name = (
            base_name[:31 - len(suffix)]
            + suffix )

        counter += 1

    used_sheet_names.add(safe_sheet_name)

    df.to_excel(
        writer,
        sheet_name=safe_sheet_name,
        index=False )

print("Excel file created successfully:")
print(output_file)
```

*** Excel file created successfully:
Dataset_Cleaned.xlsx

Figure 7.11.: Writing the cleaning summary, data dictionary, and cleaned worksheets to a new Excel workbook.

This export design produced a self-contained deliverable. A reviewer opening the cleaned workbook could first inspect the summary and dictionary, then navigate to the individual cleaned tables.

7.12. Inspecting the cleaned worksheets

The final notebook stage previewed the first rows of every cleaned worksheet. The preview loop printed a divider and worksheet name before displaying the first ten records. This provided a final visual inspection before the file was downloaded.

The previews were not intended to calculate new operational KPIs. Their purpose was to confirm that

the cleaned tables remained readable, that headers were stable, that representative values were preserved, and that the main identifiers and categories appeared in the expected format.

7.12.1. Staging operations preview

Figure 7.12 shows the first portion of the preview output for `StgContainerOperations`. The displayed records confirm that container identifiers, operator codes, size types, and temporal values remained available after cleaning.

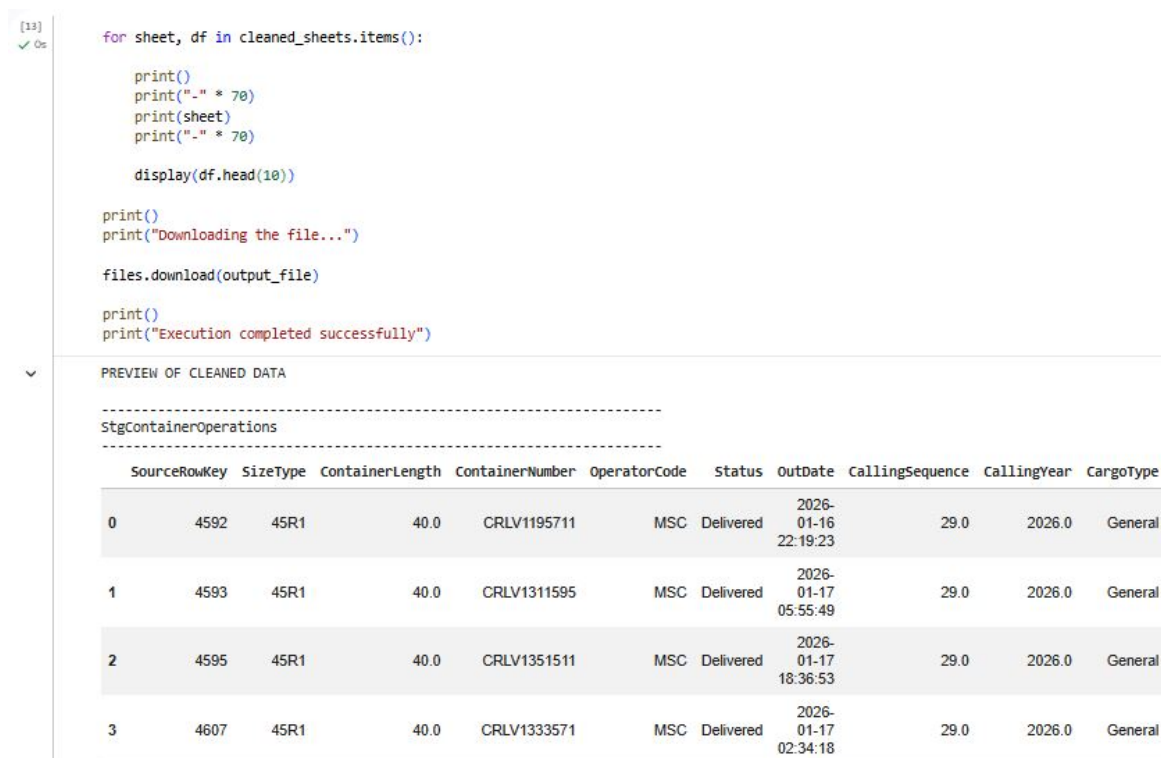


Figure 7.12.: Preview of the cleaned `StgContainerOperations` worksheet.

The preview allowed the analyst to check that:

- rows had not shifted during processing;
- container numbers remained text values;
- categorical codes were still populated;
- dates were displayed consistently; and
- the worksheet retained its operational detail.

7.12.2. Vessel-call and operator reference data

Figure 7.13 displays the previews of `DimVesselCall` and `DimOperator`.

DimVesselCall							
	VesselCallKey	CallingSequence	CallingYear	VesselCode	VesselName	UserVoyage	VesselCallBusinessID
0	1	502.0	2025.0	LANG	LANGENESS	94390.0	2025 502 LANG 94390
1	5	510.0	2025.0	XHAI	XIN XIN HAI 1	94283.0	2025 510 XHAI 94283
2	18	496.0	2025.0	LIAN	XIN LIAN YUN GANG	94547.0	2025 496 LIAN 94547
3	36	489.0	2025.0	KOPE	OOCL KOBE	94245.0	2025 489 KOPE 94245
4	53	514.0	2025.0	EXSP	EXPRESS SPAIN	94705.0	2025 514 EXSP 94705
5	63	485.0	2025.0	SHOU	SC HOUSTON	94423.0	2025 485 SHOU 94423
6	77	519.0	2025.0	LUCY	LUCY BORCHARD	94430.0	2025 519 LUCY 94430
7	79	526.0	2025.0	JUBO	JUDITH BORCHARD	94402.0	2025 526 JUBO 94402
8	90	79.0	2025.0	ANTW	COSCO ANTWERP	89525.0	2025 79 ANTW 89525
9	91	65.0	2025.0	CHAT	CHATTANOOGA	89314.0	2025 65 CHAT 89314

DimOperator		
	OperatorKey	OperatorCode
0	1	WSI
1	2	NNS
2	3	ONE
3	4	LID
4	5	RDL
5	6	ESA

Figure 7.13.: Preview of the cleaned vessel-call and operator reference worksheets.

The vessel-call preview was important because the Python process had specifically checked the `UserVoyage` field for blanks and duplicates. The displayed result allowed the retained vessel codes, names, call references, and voyage identifiers to be reviewed together.

The operator preview confirmed that the operator codes remained compact text values and that whitespace cleaning had not altered the codes themselves.

7.12.3. Container specifications and equipment

Figure 7.14 shows the cleaned `DimContainerSpecification` and `DimEquipment` worksheets.

DimContainerspecification								
ContainerSpecificationKey	SizeType	ContainerSpecificationLabelCustom	ContainerType	ContainerHeightwidth	ContainerLength	TEUFactor	TEUMappingStatus	
0	1	22G1	20 H: 8'6 W: 2.348 General Dry	General Dry	H: 8'6 W: 2.348	20	1.0	Mapped
1	2	45G1	40 H: 9'6 W: 2.348 General Dry	General Dry	H: 9'6 W: 2.348	40	2.0	Mapped
2	3	22T1	20 H: 8'6 W: 2.348 Tank	Tank	H: 8'6 W: 2.348	20	1.0	Mapped
3	4	45R1	40 H: 9'6 W: 2.348 Refrigerated	Refrigerated	H: 9'6 W: 2.348	40	2.0	Mapped
4	5	22P1	20 H: 8'6 W: 2.348 Platform-Fixed	Platform-Fixed	H: 8'6 W: 2.348	20	1.0	Mapped
5	6	42P1	40 H: 8'6 W: 2.348 Platform-Fixed	Platform-Fixed	H: 8'6 W: 2.348	40	2.0	Mapped
6	7	42G1	40 H: 8'6 W: 2.348 General Dry	General Dry	H: 8'6 W: 2.348	40	2.0	Mapped
7	8	45V1	40 H: 9'6 W: 2.348 General Dry	General Dry	H: 9'6 W: 2.348	40	2.0	Mapped
8	9	22T1	20 H: 8'6 W: 2.348 General Dry	General Dry	H: 8'6 W: 2.348	20	1.0	Mapped
9	10	22G1	20 H: 8'6 W: 2.348 Tank	Tank	H: 8'6 W: 2.348	20	1.0	Mapped

DimEquipment		
EquipmentKey	EquipmentNumber	
0	1	QC02
1	2	QC01
2	3	QC08
3	4	QC10
4	5	QC03
5	6	QC09

Figure 7.14.: Preview of the cleaned container-specification and equipment worksheets.

The container-specification preview contains size and type descriptors used to distinguish different equipment configurations. It also provides a quick check that fields such as container length, height/width description, type, and mapping status remained correctly aligned.

The equipment preview confirms the cleaned equipment identifiers. Because equipment codes are short categorical values, removing leading and trailing spaces was especially important. A visually identical crane code containing one hidden trailing space would otherwise be treated as a separate equipment category.

7.12.4. Cargo and port master data

Figure 7.15 shows DimCargo and PortMaster after cleaning.

DimCargo					
	CargoKey	CargoType	Commodity	CargoLabel	IsUnknownCargo
0	1	General	Home Furniture	General Home Furniture	False
1	2	Dangerous	Home Furniture	Dangerous Home Furniture	False
2	3	Dangerous	Dangerous Goods	Dangerous Dangerous Goods	False
3	4	Reefer	Orange	Reefer Orange	False
4	5	Over Dimension	Home Furniture	Over Dimension Home Furniture	False
5	6	General	Carpets and Home Furnishings	General Carpets and Home Furnishings	False
6	7	Over Dimension	Heavy Equipment	Over Dimension Heavy Equipment	False
7	8	General	Accessories	General Accessories	False
8	9	General	Wood	General Wood	False
9	10	Dangerous	Accessories	Dangerous Accessories	False

PortMaster				
	PortCode	PortName	Country	Region
0	APT	APAPA TERMINAL	NIGERIA	WEST AFRICA
1	BEANR	ANTWERP	BELGIUM	NORTH EUROPE
2	CNNGB	NINGBO	CHINA	EAST ASIA
3	CNNSA	NANSHA	CHINA	SOUTH CHINA
4	CNNSS	NANTONG	CHINA	EAST CHINA
5	CNRZH	RIZHAO	CHINA	NORTH CHINA

Figure 7.15.: Preview of the cleaned cargo dimension and port master reference data.

The cargo preview allowed the standardized commodity descriptions to be reviewed. The replacements for `Danger Goods` and `Electrical Appliances` prevented those descriptions from remaining as separate variants.

The port master preview confirmed that port codes, names, countries, and regions were still aligned. These fields are sensitive to whitespace and spelling variation because they are used repeatedly as geographic classifications.

7.12.5. Port-of-loading and discharge master data

Figure 7.16 presents the cleaned `DimPortPOL` and `POD_Master` worksheets.

DimPortPOL							
POLKey	POLCode	IsUnknownPOL	PortMaster_PortName	PortMaster_Country	PortMaster_Region	POLLabel	
0	1	EGDAM	False	DAMIETTA	EGYPT	EASTERN MEDITERRANEAN	EGDAM DAMIETTA
1	60	APT	False	APAPA TERMINAL	NIGERIA	WEST AFRICA	APT APAPA TERMINAL
2	2	CNNGB	False	NINGBO	CHINA	EAST ASIA	CNNGB NINGBO
3	13	BEANR	False	ANTWERP	BELGIUM	NORTH EUROPE	BEANR ANTWERP
4	3	CNNSA	False	NANSHA	CHINA	SOUTH CHINA	CNNSA NANSHA
5	4	CNSHA	False	SHANGHAI	CHINA	EAST CHINA	CNSHA SHANGHAI
6	5	ESVLC	False	VALENCIA	SPAIN	WESTERN MEDITERRANEAN	ESVLC VALENCIA
7	61	CNNSS	False	NANTONG	CHINA	EAST CHINA	CNNSS NANTONG
8	6	CNTAO	False	QINGDAO	CHINA	NORTH CHINA	CNTAO QINGDAO
9	9	CNRZH	False	RIZHAO	CHINA	NORTH CHINA	CNRZH RIZHAO

POD_Master			
PODCode	PortName	Country	Region
0	BEANR	Antwerp	Belgium
1	CNNGB	Ningbo	China
2	CNNSA	Nansha	China
3	CNSHA	Shanghai	China
4	CNTAO	Qingdao	China
5	CNTXG	Tianjin Xingang	China

Figure 7.16.: Preview of the cleaned port-of-loading dimension and discharge-port master worksheet.

The preview demonstrates that loading-port codes remained associated with their descriptive port, country, and region fields. It also shows the corresponding discharge-port master values.

Python's contribution at this stage was mainly consistency. Codes and geographic descriptions were preserved while surrounding spaces and blank textual values were standardized.

7.12.6. Port-of-discharge and transport-profile data

Figure 7.17 shows DimPortPOD and DimTransportProfile.

DimPortPOD						
	PODKey	PODCode	POD_Master_PortName	POD_Master_Country	Region	PODLabel
0	1	ESVLC	Valencia	Spain	Western Mediterranean	ESVLC Valencia
1	10	BEANR	Antwerp	Belgium	North Europe	BEANR Antwerp
2	2	EGDAM	Damietta	Egypt	Eastern Mediterranean	EGDAM Damietta
3	9	CNNGB	Ningbo	China	East China	CNNGB Ningbo
4	3	ILASH	Ashdod	Israel	Eastern Mediterranean	ILASH Ashdod
5	51	CNNSA	Nansha	China	South China	CNNSA Nansha
6	4	ILHFA	Haifa	Israel	Eastern Mediterranean	ILHFA Haifa
7	26	CNSHA	Shanghai	China	East China	CNSHA Shanghai
8	5	TRIZT	İzmit	Türkiye	Sea of Marmara	TRIZT İzmit
9	27	CNTAO	Qingdao	China	North China	CNTAO Qingdao

DimTransportProfile									
	TransportProfileKey	TransportProfileLabel	Status	Delivery	DispatchModeIn	FullEmptyStatus	LCLFCL	Mode	SA
0	1	DELIVERED E TRUCK VESSEL	DELIVERED	UNKNOWN	EXPORT	E	UNKNOWN	EXPORT	GLOG
1	8	DELIVERED F TRUCK VESSEL	DELIVERED	UNKNOWN	EXPORT	F	UNKNOWN	EXPORT	KDMR
2	10	DELIVERED F TRUCK VESSEL	DELIVERED	UNKNOWN	EXPORT	F	UNKNOWN	EXPORT	SOBK

Figure 7.17.: Preview of the cleaned port-of-discharge and transport-profile worksheets.

The discharge-port preview confirmed the consistency of port codes and geographic descriptions. The transport-profile preview displayed combined operational classifications such as status, delivery, full/empty condition, mode, and inbound or outbound transport attributes.

Because transport-profile labels are assembled from several categorical fields, trimming each component reduced the possibility that logically identical profiles would differ only because of hidden spaces.

7.12.7. Damage-condition and QC consistency data

Figure 7.18 shows DimDamageCondition and CheckQCWorkingHours.

DimDamageCondition		
DamageConditionKey	DamageCondition	
0	1	UNKNOWN
1	2	SOUND
2	3	HEAVY
3	4	MINOR

CheckQCWorkingHours								
CallingYear	CallingSequence	VesselCode	UserVoyage	EquipmentNumber	ContainerRowCount	MinimumQCWorkingHours	MaximumQCWorkingHours	
0	2025.0	65.0	CHAT	89314.0	QC09	10	0.04	0.04
1	2025.0	145.0	LEAA	89913.0	QC09	116	0.04	0.04
2	2025.0	6.0	ANTW	88830.0	QC09	72	0.04	0.04
3	2025.0	117.0	XTNG	89862.0	QC09	47	0.04	0.04
4	2025.0	60.0	HNRO	89060.0	QC09	16	0.04	0.04
5	2025.0	22.0	BALA	89034.0	QC09	29	0.04	0.04
6	2025.0	92.0	MOND	89460.0	QC09	27	0.04	0.04

Figure 7.18.: Preview of the cleaned damage-condition lookup and quay-crane working-hour check worksheet.

The damage-condition table is a small categorical lookup. Its preview confirmed that the condition labels remained separate and readable after text normalization.

The QC check worksheet contains vessel, equipment, and working-hour consistency information. Trimming the vessel code, equipment number, and status fields ensured that the same QC group was not fragmented by text-formatting differences.

7.12.8. QC performance and container-operation facts

Figure 7.19 shows the preview of FactQCPerformance and the beginning of FactContainerOperations.

FactQCPerformance							
QCPerformanceKey	ContainerRowCount	MinimumQCWorkingHours	MaximumQCWorkingHours	DistinctQCWorkingHourCount	BlankQCWorkingHourRows	QCWorkingHoursStatus	
0	1	25	0.05	0.05	1	0	Consistent Repeated Value
1	11	11	0.05	0.05	1	0	Consistent Repeated Value
2	21	7	0.05	0.05	1	0	Consistent Repeated Value
3	25	7	0.05	0.05	1	0	Consistent Repeated Value
4	33	4	0.05	0.05	1	0	Consistent Repeated Value
5	35	7	0.05	0.05	1	0	Consistent Repeated Value
6	42	13	0.05	0.05	1	0	Consistent Repeated Value
7	45	99	0.05	0.05	1	0	Consistent Repeated Value
8	51	58	0.05	0.05	1	0	Consistent Repeated Value
9	54	107	0.05	0.05	1	0	Consistent Repeated Value

FactContainerOperations							
-------------------------	--	--	--	--	--	--	--

Figure 7.19.: Preview of the cleaned QC-performance fact and the beginning of the container-operations fact.

The QC-performance preview made it possible to review grouped equipment-performance fields and

QC working-hour status values after cleaning. The beginning of the container-operations fact confirmed that the central operational table was still present in the exported workbook with its detailed records.

The final preview stage therefore acted as a practical visual check. The analyst could inspect the actual tables rather than relying only on summary statistics.

7.13. Downloading and retaining the cleaned output

After the previews were completed, the notebook downloaded the cleaned workbook from the Colab environment to the local computer. A final execution message confirmed that the notebook had reached the end of the process.

The output workbook was a separate file from the source. This preserved the raw extract and provided a controlled cleaned copy containing:

- normalized worksheet data;
- the cleaning summary;
- the technical data dictionary; and
- the original worksheet separation.

Saving a separate copy supported traceability. The original workbook remained available for comparison, while the cleaned copy could be used for review, further preparation, or controlled downstream work.

7.14. How Python improved data preparation

Python contributed to the data-cleaning process in several important ways.

7.14.1. Consistency across worksheets

The same cleaning logic was applied programmatically to all worksheets. This reduced the risk of cleaning one table differently from another or forgetting to apply a correction to a related table.

For example, commodity replacements were applied to both the cargo dimension and the staging operations table. Similarly, worksheet-specific text mappings ensured that vessel, operator, cargo, equipment, port, transport, and quality-status fields were treated consistently.

7.14.2. Protection of data types

The script distinguished between text, numeric, and temporal data. Text blanks could be represented consistently without turning numeric or datetime columns into strings.

This preserved the analytical usefulness of the workbook. Numeric fields remained suitable for aggregation, and date fields remained suitable for sorting and duration calculation.

7.14.3. Measured rather than silent cleaning

The cleaning process recorded the original and final worksheet shapes, duplicate counts, and remaining missing values. This allowed the effect of cleaning to be reviewed after execution.

Rows were not simply removed without explanation. Key-field deletions, duplicate removal, and worksheet-level changes were reported through console messages and the summary table.

7.14.4. Improved transparency

The technical data dictionary documented each field's type, completeness, and cardinality. The preview sequence then showed representative records from the cleaned worksheets.

Together, the summary, dictionary, and previews made the process easier to audit. A reviewer could see both high-level quality statistics and actual cleaned rows.

7.14.5. Repeatability

The notebook converted a sequence of manual spreadsheet-cleaning tasks into a repeatable procedure. The same script could be executed again when a refreshed workbook was supplied, subject to the same worksheet and field expectations.

This repeatability was one of the most important benefits of Python. Manual editing can be difficult to reproduce and may leave no record of exactly what was changed. The notebook preserved the cleaning logic in executable form.

7.15. Final outcome

The Python workstream transformed a multi-sheet Excel workbook into a cleaner and more documented workbook copy. The process did not depend on a single cleaning command. It combined structural checks, text normalization, key-field protection, duplicate handling, date preparation, category standardization, missing-value treatment, quality reporting, documentation, export, and final visual inspection.

The resulting workbook retained the individual operational and reference tables while adding the information needed to understand the cleaning process itself. Python therefore helped prepare the data in three complementary ways:

1. it improved the consistency of the worksheet contents;
2. it documented the quality and structure of the cleaned data; and
3. it produced a separate, reusable Excel deliverable without overwriting the original source.

The nineteen notebook figures included in this chapter record the complete progression from environment setup and workbook discovery through cleaning, summary generation, data-dictionary creation, export, and final worksheet inspection.

8. Power BI Semantic Model, DAX, and Validation

8.1. Purpose and Scope

The Power BI semantic model transforms the prepared Power Query outputs into a structured analytical environment for container-terminal performance analysis. The model is designed not simply to store transformed tables, but to define the business meaning of each analytical entity, preserve the correct grain of operational processes, establish controlled relationships, and provide a governed foundation for DAX measures and dashboard reporting.

The final design uses multiple fact tables because container movements, quay-crane performance, and vessel-port-call activity occur at different operational grains. These facts are connected through reusable dimensions that provide consistent filtering and business context. The resulting structure supports container flow, TEU, dwell time, vessel performance, equipment workload, cargo analysis, port connectivity, current-state monitoring, and data-quality validation without forcing fundamentally different operational processes into a single wide table.

The principal model-design objectives were to:

- preserve the correct grain of each operational process;
- separate descriptive attributes from measurable operational events;
- create consistent technical and business keys;
- prevent repeated values from producing double counting;
- support reusable filtering through dimensions;
- distinguish entry, exit, and current-state analytical logic;
- centralize governed DAX calculations;
- maintain explicit data-quality controls; and
- provide a traceable foundation for the final Power BI dashboards.

Semantic-model design principle

Each KPI is evaluated at an appropriate and documented analytical grain. Container activity remains at container-operation grain, QC performance is validated at vessel-call–equipment grain, and vessel turnaround is retained at vessel-port-call grain. Dimensions provide shared descriptive context without changing the natural grain of the underlying facts.

8.2. Overall Semantic-Model Architecture

The final analytical architecture is best described as a multi-fact semantic model. It contains three principal fact tables:

- FactContainerOperations,
- FactQCPerformance, and
- FactVesselPortCalls.

These facts are supported by dimensions representing vessel calls, vessels, operators, container specifications, cargo, equipment, loading ports, discharge ports, transport profiles, damage conditions, and calendar dates.

The model separates operational events according to their natural business meaning rather than attempting to combine them into a single structure. This is particularly important because one vessel call may be associated with many container-operation records and several equipment groups. Maintaining separate fact tables prevents vessel-level and equipment-level values from being repeated unnecessarily across container rows.

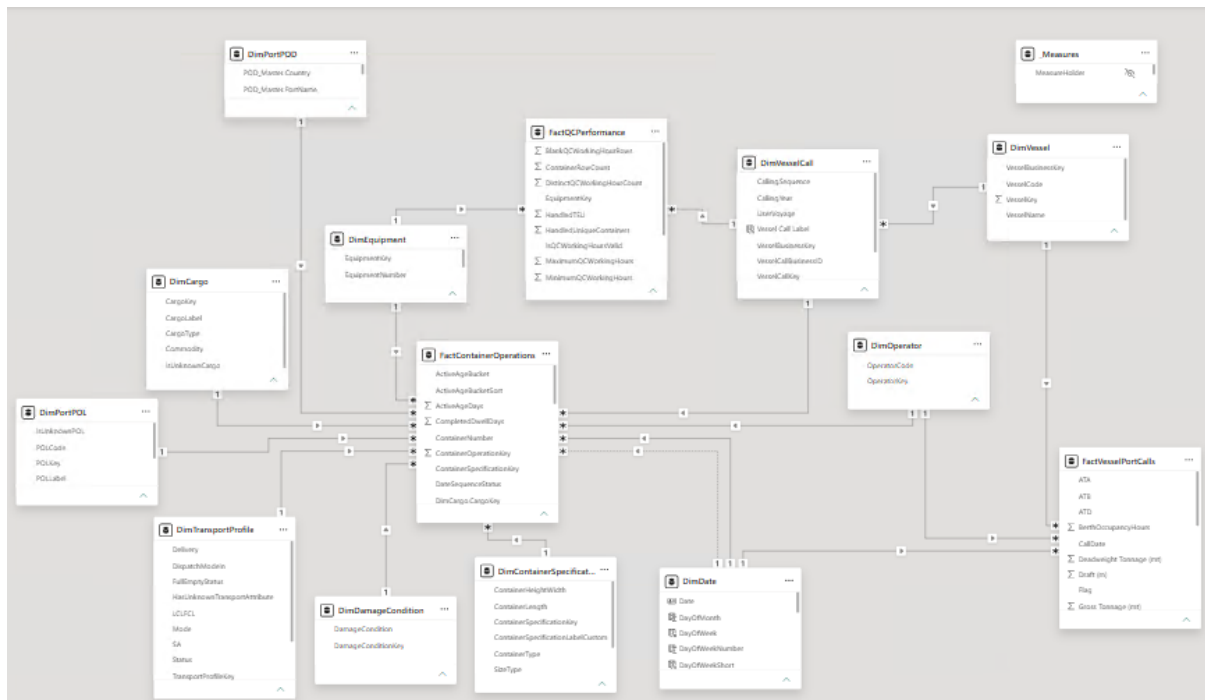


Figure 8.1: Power BI model view showing the multi-fact semantic architecture, supporting dimensions, date relationships, and centralized measure table.

8.3. Declared Fact-Table Grains

A fundamental requirement of the model is that the meaning of one row in every fact table is clearly understood. Table 8.1 summarizes the grain and analytical responsibility of the three implemented facts.

Table 8.1.: Fact-table grains and analytical responsibilities

Fact table	Declared grain	Principal analytical responsibility
FactContainerOperations	One imported container-operation record	Container activity, entry and exit flow, dwell, weight, cargo, ports, transport characteristics, equipment association, TEU through container specification, and current-state controls.
FactQCPerformance	One vessel-call × equipment group	Validated quay-crane working hours, handled source records, unique handled containers, handled TEU, QC consistency status, and productivity analysis.
FactVesselPortCalls	One supplied vessel-port-call record	ATA, ATB, ATD, vessel turnaround, pre-berth waiting, berth occupancy, vessel characteristics, dimensions, draft, deadweight, gross tonnage, and call-level analysis.

FactContainerOperations is the principal container-activity fact. Its generated ContainerOperationKey identifies individual imported operational rows and provides technical traceability.

FactQCPerformance exists because QC working hours in the original source may be repeated across multiple container rows belonging to the same vessel-call and equipment combination. The QC fact restructures this information at the appropriate grain before productivity calculations are performed.

FactVesselPortCalls preserves vessel timing and physical characteristics independently from container volume. This prevents vessel turnaround or berth-duration values from being multiplied when a vessel call is associated with many containers.

Three Fact Tables at Declared Grains

Queries [19] < ✓ f₁ = Table.TransformColumnTypes(#"Added Custom12",{"IsCompleted", type

	ContainerOperationKey	ContainerNumber	VesselCallKey	Open
1	1	AAV9111711		1
2	2	AAV9111993		1
3	3	AAV9111835		1
4	4	AAV9111590		1
5	5	AAV9111927		1
6	80	AAV9979551		1
7	98	AAV5557917		1
8	109	ADMV9111511		1
9	110	ADMV9111773		1
10	115	ADMV9111339		1
11	35	AAV5151131		1
12	41	AAV5151791		1
13	55	AAV5155557		1
14	62	AAV517311		1
15	64	AAV5159539		1
16	65	AAV5159735		1
17	79	AAV9111153		1
18	107	AAV9351551		1
19	121	ADMV531799		1
20	44	AAV5151173		1
21	45	AAV5151111		1

Activate Windows

FactContainerOperations - container-operation grain

Queries [19] < ✓ f₁ = Table.RemoveColumns(#"Reordered Columns",{"CallingYear", "Call

	QCPerformanceKey	ContainerRowCount	MinimumQCWorkingHours
1	1	25	0.05
2	2	152	0.06
3	3	99	0.06
4	4	24	0.06
5	5	15	0.06
6	6	2	0.06
7	7	205	0.05
8	8	34	0.05
9	9	242	0.02
10	10	213	0.04
11	11	12	0.05
12	12	100	0.04
13	13	133	0.02
14	14	368	0.05
15	15	245	0.06
16	16	148	0.06
17	17	123	0.02
18	18	7	0.05
19	19	174	0.02
20	20	274	0.04

FactQCPerformance - vessel-call x equipment grain

Queries [19] < ✓ f₁ = Table.AddColumn(#"Added Vessel Time Reconciliation Variance", "callDate", each Date.From([ATA]))

	VesselOperationKey	Operator	Vessel Name	Vessel Code	Flag	Year Built
1	MITO	MITO	MITO	MITO	PORTUGAL	2006
2	DARI	MSC DARIEN	MSC DARIEN	DARI	PORTUGAL	2003
3	YMEV	YMEV	YMEV	YMEV	TAIWAN	2014
4	MOND	MOND	MOND	MOND	PANAMA	2007
5	LUCY	LUCY	LUCY	LUCY	ANTIGUA	2006
6	MORL	MORL	MORL	MORL	MALTA	2007
7	MITO	MITO	MITO	MITO	PORTUGAL	2006
8	LOBO	LOBO	LOBO	LOBO	ANTIGUA	2006
9	SANU	SANU	SANU	SANU	PANAMA	2003
10	CHAT	CHAT	CHAT	CHAT	SINGAPORE	2012
11	ANTW	ANTW	ANTW	ANTW	UNITED KINGDOM	2002
12	ASTR	ASTR	ASTR	ASTR	ANTIGUA	2006
13	LUCY	LUCY	LUCY	LUCY	ANTIGUA	2006
14	SHOU	SHOU	SHOU	SHOU	MARSHALL ISLANDS	2010
15	ZYNP	ZYNP	ZYNP	ZYNP	TUVALU	2003
16	SHAI	SHAI	SHAI	SHAI	UNITED KINGDOM	2002
17	KOPE	KOPE	KOPE	KOPE	HONG KONG	2007
18	LOBO	LOBO	LOBO	LOBO	ANTIGUA	2006
19	WITT	WITT	WITT	WITT	ITALY	2024
20	DEGO	DEGO	DEGO	DEGO	UNITED KINGDOM	2017

FactVesselPortCalls - vessel-call grain

Figure 8.2.: Power Query previews of the three implemented fact tables, each representing a distinct operational grain.

8.4. FactContainerOperations: Central Container-Activity Fact

FactContainerOperations is derived from StgContainerOperations. The fact retains the operational record grain established during staging while replacing repeated descriptive attributes with technical dimension keys where the corresponding dimensions have been implemented.

The staging field SourceRowKey is renamed to ContainerOperationKey, emphasizing that the generated identifier represents one imported operational record rather than a globally unique physical container.

The fact contains, among other fields:

- ContainerOperationKey,
- ContainerNumber,
- VesselCallKey,
- OperatorKey,
- ContainerSpecificationKey,
- cargo, port, equipment, damage-condition, and transport-profile keys,
- InDate and OutDate,
- InDateOnly and OutDateOnly,
- StackDays,
- CompletedDwellDays,
- QCWorkingHoursSource,
- IMDG,
- WeightGroup,
- WeightValue,
- UnitCount,
- lifecycle and date-quality controls,
- current-state controls, and
- weight-quality controls.

The fact is enriched through a sequence of left-outer joins to dimensions. Once a descriptive attribute has been represented by a dimension key, repeated descriptive fields can be removed from the fact where appropriate. This produces a more compact model and centralizes categorical definitions within their dimensions.

8.5. Container Lifecycle and Current-State Controls

Several generated fields support lifecycle interpretation and validation.

8.5.1. Completed Lifecycle

The fact contains a Boolean lifecycle control that identifies records with a valid entry and exit sequence. A record is considered completed only when both timestamps exist and OutDate is not earlier than InDate.

The numeric CompletedDwellDays field is calculated only for valid completed records:

Listing 8.1: Completed dwell calculation

```
if [InDate] = null or [OutDate] = null then
    null
else if [OutDate] < [InDate] then
    null
else
```

```
Duration.TotalDays([OutDate] - [InDate])
```

Incomplete and invalid records remain blank rather than being assigned a value of zero. This distinction is important because zero would imply a valid completed lifecycle with no elapsed dwell time.

8.5.2. Date-Sequence Status

The staging workflow also classifies source records according to date completeness and chronological validity:

Listing 8.2: Date-sequence validation logic

```
if [InDate] = null then
    "Missing In Date"
else if [OutDate] = null then
    "Active or Missing Out Date"
else if [OutDate] < [InDate] then
    "Invalid: Out Before In"
else
    "Valid Completed Record"
```

This classification allows incomplete or invalid records to remain visible for data-quality analysis while preventing them from entering calculations that require a valid completed lifecycle.

8.5.3. Current-State Identification

A separate control identifies records with a recorded entry but no recorded exit:

Listing 8.3: Current-state identification

```
if [InDate] <> null and [OutDate] = null then
    true
else
    false
```

The resulting `IsCurrentlyInPort` field represents the latest known state within the available extract. It should therefore be interpreted as a source-record status rather than independent confirmation that the physical container remains in the terminal after the reporting cutoff.

8.5.4. Active-Age Monitoring

For records with an entry date and no exit date, `ActiveAgeDays` calculates elapsed age. The implemented workflow groups active records into ordered bands:

- 0–3 days,
- 4–7 days,
- 8–14 days,
- 15–30 days, and
- 31+ days.

A corresponding numeric sort field maintains the intended bucket order in Power BI visuals.

The current Power Query implementation derives active age using the refresh-time date. For a production environment requiring strict reproducibility, the same logic can be anchored to a controlled extract or reporting timestamp so that identical source extracts always reproduce identical age values.

8.6. Weight Controls

The source weight value is retained in `WeightValue`. Because the physical unit is not independently confirmed within the source implementation, the semantic model preserves the value without converting it to kilograms, tonnes, or another assumed unit.

A dedicated `WeightDataStatus` classifies each row as:

- Missing Weight,
- Invalid Negative Weight,
- Zero Weight, or
- Valid Positive Weight.

The Boolean `HasValidWeight` provides an additional analytical control for measures that should use only positive weight values.

This approach distinguishes data-quality status from business interpretation and avoids attaching an unsupported physical unit to the source values.

8.7. Dimension Inventory

The final model uses dimensions to provide reusable descriptive context for the facts. Most dimensions are generated in Power Query from the governed staging layer, while `DimDate` is created in DAX for calendar-based analysis.

Table 8.3 summarizes the principal dimension structures and their analytical roles.

Table 8.3.: Implemented dimensions and their analytical roles

Dimension	Key / grain	Principal attributes	Analytical role
DimVesselCall	One distinct vessel-call combination; VesselCallKey	Calling sequence, calling year, vessel code/name, user voyage, vessel business key, vessel-call business ID	Connects container and QC activity to vessel-call context.
DimVessel	One cleaned vessel identity; VesselKey	Vessel business key, vessel code, vessel name	Provides a reusable vessel identity across vessel calls and detailed vessel-port-call records.
DimOperator	One cleaned operator code; OperatorKey	Operator code	Supports operator-level workload and performance analysis.
DimContainerSpecification	One distinct container specification; ContainerSpecificationKey	Size type, container length, height/width, container type, specification label, TEU factor, mapping status	Supports container-profile analysis and governed TEU conversion.
DimCargo	One cargo-type-commodity combination; CargoKey	Cargo type, commodity, combined cargo label, unknown indicator	Supports cargo-type and commodity analysis.
DimEquipment	One equipment identifier; EquipmentKey	Equipment number	Supports equipment workload and validated QC analysis.
DimPortPOL	One loading-port code; POLKey	POL code, loading-port label, port name, country, region, unknown control	Provides the loading-port role and origin-side connectivity analysis.
DimPortPOD	One discharge-port code; PODKey	POD code, discharge-port label, port name, country, region	Provides the discharge-port role and destination-side connectivity analysis.
DimTransportProfile	One unique combination of operational classifications; TransportProfileKey	Status, delivery, dispatch mode, full/empty, LCL/FCL, mode, S/A, inbound and outbound transport types	Provides a consolidated dimension for multiple low-cardinality operational classifications.
DimDamageCondition	One standardized damage-condition category; DamageConditionKey	Damage condition	Supports condition-based filtering and segmentation.
DimDate	One calendar date	Year, quarter, month, week, weekday, sorting attributes, and other calendar fields	Supports entry-date, exit-date, time-intelligence, and comparative analysis.

8.8. Vessel and Vessel-Call Dimensions

8.8.1. DimVesselCall

DimVesselCall creates the operational call context shared by container activity and QC performance. A composite VesselCallBusinessID is generated from:

- calling year,
- calling sequence,
- vessel code, and
- user voyage.

Where one of these values is missing, an explicit placeholder is used when constructing the business identifier. The dimension then assigns a generated integer VesselCallKey.

A separate VesselBusinessKey is constructed from cleaned vessel code and vessel name:

Listing 8.4: Vessel business-key construction

```
Text.Upper(  
  Text.Trim(  
    Text.Clean([VesselCode])  
  )  
)  
& "|" &  
Text.Upper(  
  Text.Trim(  
    Text.Clean([VesselName])  
  )  
)
```

The vessel-call dimension therefore answers the question “which operational call?” rather than merely “which vessel?”.

Queries [19] <

StgContainerOperations

DimVesselCall

FactContainerOperations

DimOperator

DimContainerSpecification

DimCargo

DimEquipment

DimPortPOL

PortMaster

DimPortPOD

POD_Master

DimTransportProfile

DimDamageCondition

_Measures

CheckQCWorkingHours

FactQCPerformance

Append2_Vessel

FactVesselPortCalls

DimVessel

fx = Table.Distinct(#"Removed Other Columns", {"CallingSequence",

	1 ² 3 VesselCallKey	A ^B _C VesselBusinessKey	A ^B _C CallingSequence	A ^B _C CallingYear
1		1	LANG LANGENESS	502
2		5	BZCH BZ CHONGFU	86
3		6	NITA ANITA A	93
4		38	REMA REMAH	114
5		108	XHAI XIN XIN HAI 1	510
6		129	REMA REMAH	9
7		130	ZYNP ZEYNEP	78
8		131	LIPE LIDER PERIHAN	75
9		132	BRNS BARENTS	91
10		133	KUZN S.KUZNETSOV	31
11		134	HUID HUI DA 9	17
12		136	KRCA KR CELEBES	72
13		137	VESP VERSA SPIRIT	73
14		139	JEON MSC JEONGMIN	1
15		140	LANG LANGENESS	64
16		141	VALC VALENCE	7
17		142	LIAN XIN LIAN YUN GANG	496
18		145	XHAI XIN XIN HAI 1	102
19		156	LANG LANGENESS	21
20		159	DLNS DELPHINUS	50
21		160	DLNS DELPHINUS	123
22		161	DLNS DELPHINUS	77

Activate Windows
Go to Settings to activate Windows

Figure 8.3.: Power Query preview of DimVesselCall, including the generated vessel-call key and vessel business identity.

8.8.2. DimVessel

DimVessel represents the reusable vessel identity. It combines vessel information available from DimVesselCall with vessel identities found in FactVesselPortCalls, filters invalid business keys, removes duplicate vessel identities, and generates a surrogate VesselKey.

The final dimension contains:

- VesselKey,
- VesselBusinessKey,
- VesselCode, and
- VesselName.

This dimension separates enduring vessel identity from individual vessel calls. A vessel can therefore appear in multiple calls while retaining one common identity for cross-analysis.

Queries [19] < fx = Table.ReorderColumns(

	¹ ₂ VesselKey	^A _B VesselBusinessKey	^A _B VesselCode	^A _C VesselName
1	1	LANG LANGENESS	LANG	LANGENESS
2	2	BZCH BZ CHONGFU	BZCH	BZ CHONGFU
3	3	NITA ANITA A	NITA	ANITA A
4	4	REMA REMAH	REMA	REMAH
5	5	XHAI XIN XIN HAI 1	XHAI	XIN XIN HAI 1
6	6	ZYNP ZEYNEP	ZYNP	ZEYNEP
7	7	LIPE LIDER PERIHAN	LIPE	LIDER PERIHAN
8	8	BRNS BARENTS	BRNS	BARENTS
9	9	KUZN S.KUZNETSOV	KUZN	S.KUZNETSOV
10	10	HUID HUI DA 9	HUID	HUI DA 9
11	11	KRCA KR CELEBES	KRCA	KR CELEBES
12	12	VESP VERSA SPIRIT	VESP	VERSA SPIRIT
13	13	JEON MSC JEONGMIN	JEON	MSC JEONGMIN
14	14	VALC VALENCE	VALC	VALENCE
15	15	LIAN XIN LIAN YUN GANG	LIAN	XIN LIAN YUN GANG
16	16	DLNS DELPHINUS	DLNS	DELPHINUS
17	17	CETS CETUS	CETS	CETUS
18	18	AURE AURELIA	AURE	AURELIA
19	19	XMEN XIN XIA MEN	XMEN	XIN XIA MEN
20	20	ERAS ERASMUS NINJA	ERAS	ERASMUS NINJA
21	21	ZLNG ZHONGCHENGCHANGLING	ZLNG	ZHONGCHENGCHANGLING
22	22	ELMA MSC ELMA	ELMA	MSC ELMA

Figure 8.4.: Power Query preview of DimVessel, showing the surrogate vessel key and cleaned vessel business key.

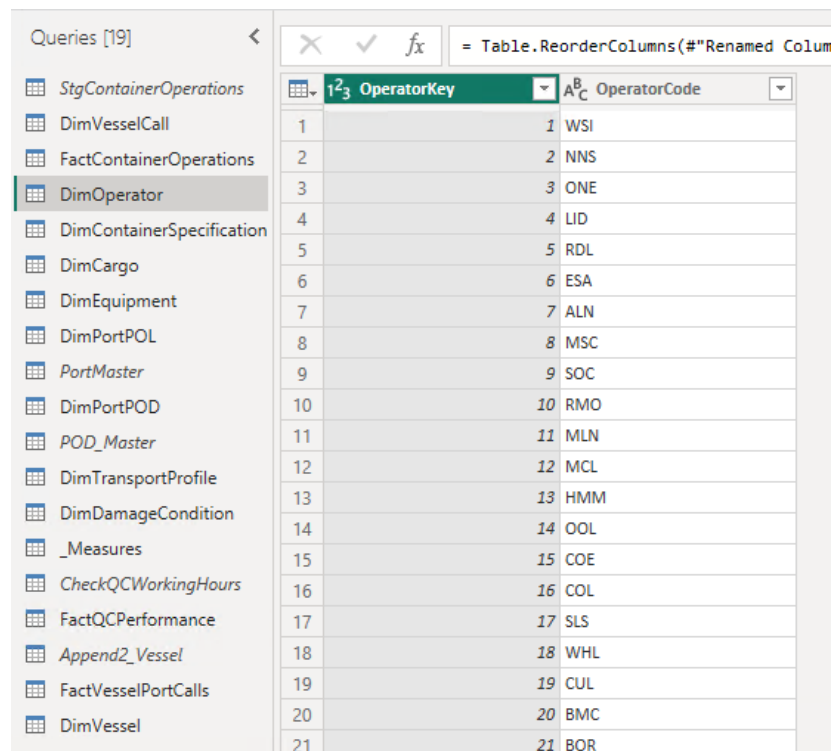
Why both vessel dimensions are required

DimVessel represents the vessel itself, whereas DimVesselCall represents a specific operational visit. This distinction allows repeated calls by the same vessel to be analyzed without duplicating vessel identity and supports both vessel-level and call-level reporting.

8.9. Operator Dimension

DimOperator contains one standardized operator code per row. Source values are trimmed and cleaned, while values such as NULL, N/A, and actual null values are converted to an explicit unknown category before duplicate removal and key assignment.

The generated OperatorKey provides a compact technical key for model relationships.



The image shows a Power Query preview of the 'DimOperator' table. On the left, a list of queries is shown, with 'DimOperator' selected. The main area displays a table with two columns: 'OperatorKey' and 'OperatorCode'. The 'OperatorKey' column contains integers from 1 to 21, and the 'OperatorCode' column contains corresponding three-letter codes. The formula bar at the top shows the DAX formula: `= Table.ReorderColumns(#"Renamed Column")`.

OperatorKey	OperatorCode
1	WSI
2	NNS
3	ONE
4	LID
5	RDL
6	ESA
7	ALN
8	MSC
9	SOC
10	RMO
11	MLN
12	MCL
13	HMM
14	OOL
15	COE
16	COL
17	SLS
18	WHL
19	CUL
20	BMC
21	BOR

Figure 8.5.: Power Query preview of DimOperator, containing one generated key per standardized operator code.

The source field OPR is retained under the neutral business term *operator*. The model does not assign a more specific contractual or commercial meaning without formal business confirmation.

8.10. Container-Specification Dimension and TEU Mapping

DimContainerSpecification centralizes container physical and type attributes:

- SizeType,
- ContainerLength,
- ContainerHeightWidth, and
- ContainerType.

Duplicate combinations are removed and a generated ContainerSpecificationKey is assigned.

A readable specification label combines the principal physical attributes and supports report-level interpretation.

The dimension also contains the governed TEUFactor. The final implemented conversion is:

- 20 feet = 1.00 TEU,
- 40 feet = 2.00 TEU,
- 45 feet = 2.25 TEU.

All other values remain blank rather than being assigned an assumed conversion.

A TEUMappingStatus field classifies specifications as:

- Mapped,
- Unknown Container Length, or
- Unmapped Container Length.

This separation makes the TEU conversion auditable and allows unsupported source values to remain visible as data-quality exceptions.

Queries [19] <

StgContainerOperations
DimVesselCall
FactContainerOperations
DimOperator
DimContainerSpecification
DimCargo
DimEquipment
DimPortPOL
PortMaster
DimPortPOD
POD_Master
DimTransportProfile
DimDamageCondition
_Measures
CheckQCWorkingHours
FactQCPerformance
Append2_Vessel
FactVesselPortCalls
DimVessel

fx = Table.AddColumn(#"Removed Columns1", "TEUMappingStatus", each if

	1-3 ContainerSpecificationKey	A ^B _C SizeType	ABC 123 ContainerSpecificationLabelCustom
1	1	22G1	20 H: 8'6 W: 2,348 General Dry
2	2	45G1	40 H: 9'6 W: 2,348 General Dry
3	3	22T1	20 H: 8'6 W: 2,348 Tank
4	4	45R1	40 H: 9'6 W: 2,348 Refrigerated
5	5	22P1	20 H: 8'6 W: 2,348 Platform-Fixed
6	6	42P1	40 H: 8'6 W: 2,348 Platform-Fixed
7	7	42G1	40 H: 8'6 W: 2,348 General Dry
8	8	45V1	40 H: 9'6 W: 2,348 General Dry
9	9	22T1	20 H: 8'6 W: 2,348 General Dry
10	10	22G1	20 H: 8'6 W: 2,348 Tank
11	11	45G1	40 H: 9'6 W: 2,348 Tank
12	12	45R1	40 H: 9'6 W: 2,348 General Dry
13	13	45G1	40 H: 9'6 W: 2,348 Refrigerated
14	14	4EG1	40 H: 9'6 W: >2,438 =<2,5 Refrigerated
15	15	4EG1	40 H: 9'6 W: >2,438 =<2,5 General
16	16	22G1	20 H: 8'6 W: 2,348 Refrigerated
17	17	44R1	40 H: 9' W: 2,348 Refrigerated
18	18	42P1	40 H: 8'6 W: 2,348 General Dry
19	19	21G1	20 H: 8' W: 2,348 Refrigerated
20	20	45G1	40 H: 9'6 W: 2,348 Platform-Fixed
21	21	45U1	40 H: 9'6 W: 2,348 General Dry
22	22	45P1	40 H: 9'6 W: 2,348 General Dry

Figure 8.6.: Power Query preview of DimContainerSpecification, containing the container profile and governed TEU mapping.

8.11. Cargo Dimension

DimCargo is generated from the combination of CargoType and Commodity. Text values are typed, trimmed, and cleaned before distinct combinations are retained.

The dimension contains:

- CargoKey,
- CargoType,
- Commodity,
- CargoLabel, and
- IsUnknownCargo.

CargoLabel combines the broad cargo type and commodity description into a readable analytical label.

IsUnknownCargo provides a quality indicator for combinations containing unknown cargo-type or commodity information.

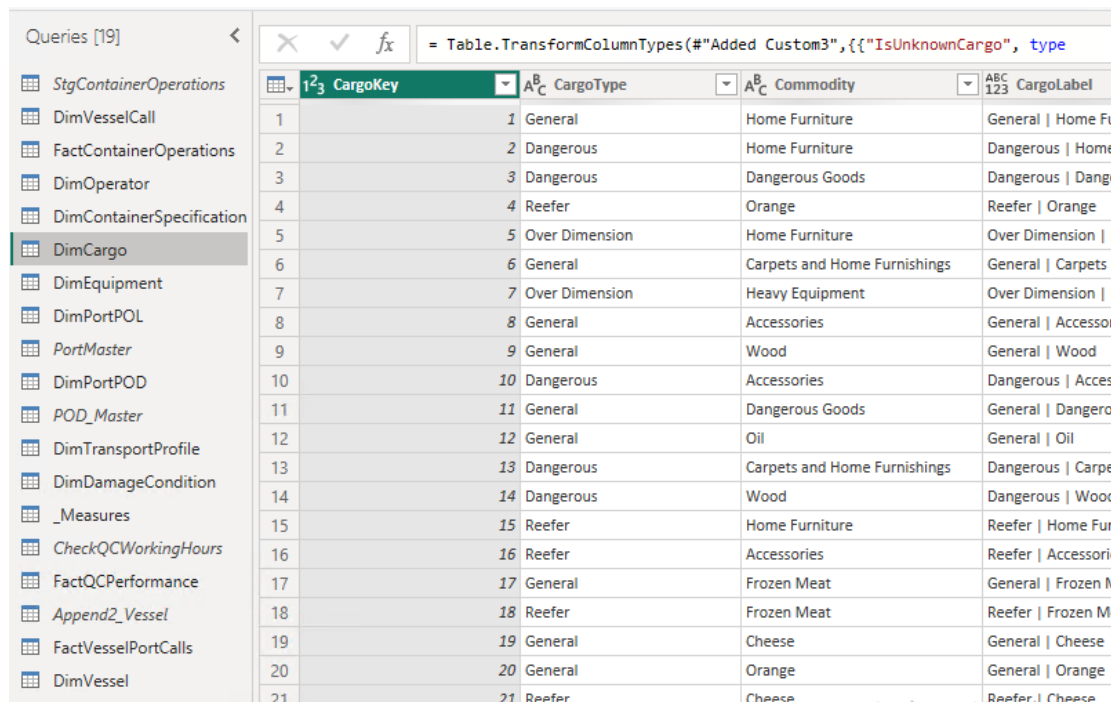


Figure 8.7 shows a Power Query preview of the DimCargo table. The table has 21 rows and 4 columns: CargoKey (integer), CargoType (text), Commodity (text), and CargoLabel (text). The CargoLabel column is a concatenation of CargoType and Commodity, separated by a vertical bar. The data is as follows:

CargoKey	CargoType	Commodity	CargoLabel
1	General	Home Furniture	General Home Furniture
2	Dangerous	Home Furniture	Dangerous Home Furniture
3	Dangerous	Dangerous Goods	Dangerous Dangerous Goods
4	Reefer	Orange	Reefer Orange
5	Over Dimension	Home Furniture	Over Dimension Home Furniture
6	General	Carpets and Home Furnishings	General Carpets and Home Furnishings
7	Over Dimension	Heavy Equipment	Over Dimension Heavy Equipment
8	General	Accessories	General Accessories
9	General	Wood	General Wood
10	Dangerous	Accessories	Dangerous Accessories
11	General	Dangerous Goods	General Dangerous Goods
12	General	Oil	General Oil
13	Dangerous	Carpets and Home Furnishings	Dangerous Carpets and Home Furnishings
14	Dangerous	Wood	Dangerous Wood
15	Reefer	Home Furniture	Reefer Home Furniture
16	Reefer	Accessories	Reefer Accessories
17	General	Frozen Meat	General Frozen Meat
18	Reefer	Frozen Meat	Reefer Frozen Meat
19	General	Cheese	General Cheese
20	General	Orange	General Orange
21	Reefer	Cheese	Reefer Cheese

Figure 8.7.: Power Query preview of DimCargo, showing the generated cargo key and combined cargo label.

8.12. Equipment Dimension

DimEquipment contains one standardized equipment identifier per row. Equipment values are converted to text, trimmed, cleaned, uppercased, deduplicated, and assigned an integer EquipmentKey.

The dimension intentionally contains equipment identity rather than QC working hours. Working hours vary according to vessel-call and equipment context and therefore belong in the QC fact rather than in a descriptive equipment dimension.

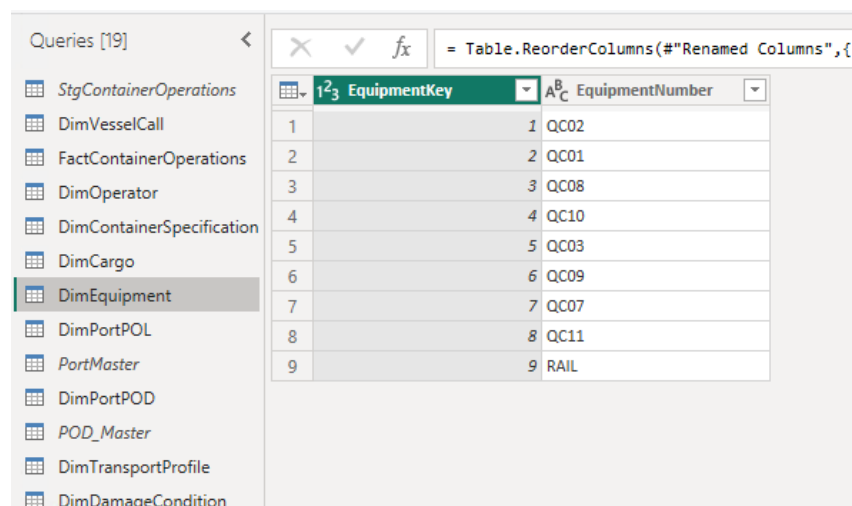


Figure 8.8 shows a Power Query preview of the DimEquipment table. The table has 9 rows and 2 columns: EquipmentKey (integer) and EquipmentNumber (text). The data is as follows:

EquipmentKey	EquipmentNumber
1	QC02
2	QC01
3	QC08
4	QC10
5	QC03
6	QC09
7	QC07
8	QC11
9	RAIL

Figure 8.8.: Power Query preview of DimEquipment, containing one key per distinct standardized equipment identifier.

8.13. Role-Specific Port Dimensions

The model maintains separate loading-port and discharge-port dimensions. Although both describe ports, POL and POD represent different analytical roles and must remain semantically distinct.

8.13.1. DimPortPOL

DimPortPOL standardizes POLCode, converts missing codes to UNKNOWN, removes duplicates, and creates POLKey. The dimension is then enriched through PortMaster, which provides port name, country, and region.

A readable loading-port label combines the source port code and port name where available.

	POLKey	POLCode	IsUnknown
1		EGDAM	
2		58 APT	
3		2 CNNGB	
4		12 BEANR	
5		3 CNNSA	
6		4 CNSHA	
7		5 ESVLC	
8		59 CNSS	
9		6 CNTAO	
10		7 ESBCN	
11		8 SAJED	
12		25 CNSHK	
13		9 RUNVS	
14		46 CNTAG	
15		10 SADMM	
16		11 DZALG	
17		26 CYLMS	
18		13 GRPIR	
19		35 DEBRV	
20		14 RULED	
21		15 DEHAM	
22		43 DETCT	

Figure 8.9.: Power Query preview of the loading-port dimension and its reference-data enrichment.

8.13.2. DimPortPOD

DimPortPOD performs the corresponding discharge-port role. The source PODCode is standardized, missing values are represented explicitly, duplicate codes are removed, and a PODKey is generated.

The dimension is enriched from POD_Master with port name, country, and region.

The image shows a Power Query editor window. On the left, a list of queries is shown, with 'DimPortPOD' selected. The main area displays a table with columns: 'PODKey', 'PODCode', 'POD_Master.PortName', and 'POD_Master.Country'. The table contains 17 rows of data, including ports like ES VLC, BE ANR, EG DAM, etc. The formula bar at the top shows the DAX formula: `= Table.RenameColumns("#Added Custom",{{"POD_Master.Region", "Region"}})`.

PODKey	PODCode	POD_Master.PortName	POD_Master.Country
1	ES VLC	Valencia	Spain
2	BE ANR	Antwerp	Belgium
3	EG DAM	Damietta	Egypt
4	CN NGB	Ningbo	China
5	ILASH	Ashdod	Israel
6	CN NSA	Nansha	China
7	ILHFA	Haifa	Israel
8	CN SHA	Shanghai	China
9	TRIZT	Izmit	Türkiye
10	CN TAO	Qingdao	China
11	TRALI	Aliaga	Türkiye
12	CN TXG	Tianjin Xingang	China
13	RU LED	St. Petersburg	Russia
14	CY LMS	Limassol	Cyprus
15	RU NVS	Novorossiysk	Russia
16	DE BRV	Bremerhaven	Germany
17	DE HAM	Hamburg	Germany

Figure 8.10.: Power Query preview of the discharge-port dimension and its reference-data enrichment.

8.13.3. Supporting Port Reference Tables

The project deliberately uses two similar port-master structures in different roles:

- PortMaster enriches DimPortPOL;
- POD_Master enriches DimPortPOD.

This role separation prevents loading-port and discharge-port descriptions from being confused simply because the two reference structures contain similar attributes.

The image shows two Power Query editor windows. Window (a) shows the 'PortMaster' table with columns: 'PortCode', 'PortName', 'Country', and 'Region'. Window (b) shows the 'POD_Master' table with columns: 'PODCode', 'PortName', 'Country', and 'Region'. Both tables contain 17 rows of data, including ports like APT, BE ANR, EG DAM, etc. The formula bar at the top of both windows shows the DAX formula: `= Table.TransformColumns("#Cleaned Text",{{"PortCode", Text.Upper, type text}})`.

(a) PortMaster: loading-port enrichment.

(b) POD_Master: discharge-port enrichment.

Figure 8.11.: Reference tables supporting the two role-specific port dimensions.

8.14. Transport-Profile Dimension

DimTransportProfile consolidates several operational classifications into one reusable dimension rather than creating a separate small dimension for every low-cardinality field.

The dimension contains:

- Status,
- Delivery,
- DispatchModeIn,
- FullEmptyStatus,
- LCLFCL,

- Mode,
- SA,
- TransportTypeIn, and
- TransportTypeOut.

Each component is standardized through trimming, cleaning, uppercasing, and explicit UNKNOWN handling. Duplicate combinations are removed and a generated TransportProfileKey is assigned.

The dimension also includes:

- a readable TransportProfileLabel, and
- HasUnknownTransportAttribute.

This structure functions as a controlled combined or junk dimension, reducing the number of small independent lookup tables while keeping recurring transport classifications out of the central fact.

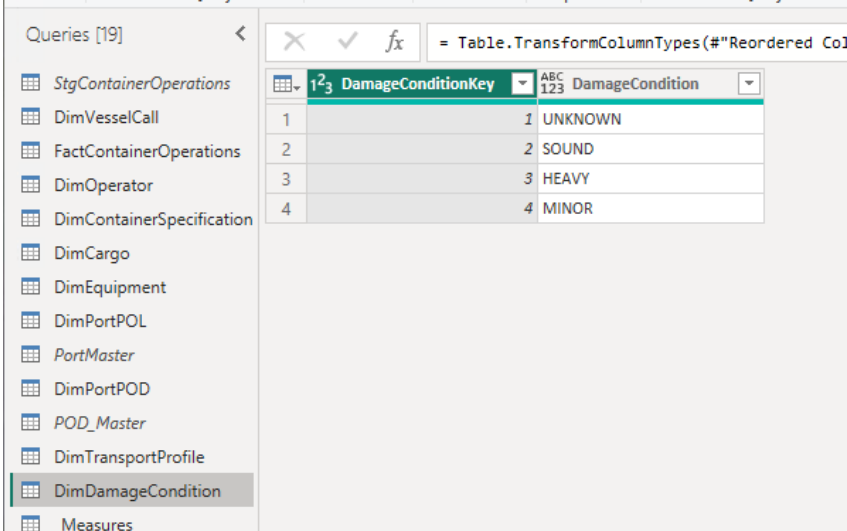
	1 ² 3	TransportProfileKey	TransportProfileLabel	ABC 123	Status	ABC 123	Deliv
1		1	DELIVERED E TRUCK VESSEL		DELIVERED		UNKNOV
2		2	DELIVERED F VESSEL MODE CHANGE		DELIVERED		TRANSS
3		3	DELIVERED F VESSEL VESSEL		DELIVERED		TRANSS
4		4	DELIVERED E VESSEL MODE CHANGE		DELIVERED		UNKNOV
5		5	DELIVERED F VESSEL VESSEL		DELIVERED		TRANSS
6		6	DELIVERED F VESSEL TRUCK		DELIVERED		UNKNOV
7		7	DELIVERED F TRUCK VESSEL		DELIVERED		UNKNOV
8		8	DELIVERED E VESSEL MODE CHANGE		DELIVERED		UNKNOV
9		9	DELIVERED F TRUCK VESSEL		DELIVERED		UNKNOV
10		10	DELIVERED F VESSEL MODE CHANGE		DELIVERED		TRANSS
11		11	DELIVERED E VESSEL MODE CHANGE		DELIVERED		TRANSS
12		12	DELIVERED E VESSEL VESSEL		DELIVERED		TRANSS
13		13	DELIVERED F TRUCK VESSEL		DELIVERED		UNKNOV
14		14	DELIVERED F VESSEL TRUCK		DELIVERED		UNKNOV
15		15	DELIVERED F VESSEL TRUCK		DELIVERED		UNKNOV
16		16	DELIVERED E TRUCK VESSEL		DELIVERED		UNKNOV
17		17	DELIVERED E TRUCK VESSEL		DELIVERED		UNKNOV
18		18	DELIVERED F VESSEL CFS		DELIVERED		UNKNOV
19		19	DELIVERED F VESSEL TRUCK		DELIVERED		UNKNOV
20		20	DELIVERED F TRUCK VESSEL		DELIVERED		UNKNOV

Figure 8.12.: Power Query preview of the transport-profile dimension, showing the generated key and combined operational classification structure.

8.15. Damage-Condition Dimension

DimDamageCondition standardizes the recorded damage-condition values by converting missing or blank values to UNKNOWN, applying text cleaning and uppercasing, removing duplicates, and generating DamageConditionKey.

The dimension supports filtering and comparison using the recorded source condition categories without introducing an unsupported binary damaged/not-damaged business rule.



	1-3 DamageConditionKey	ABC 123 DamageCondition
1		UNKNOWN
2		SOUND
3		HEAVY
4		MINOR

Figure 8.13.: Power Query preview of the standardized damage-condition dimension.

8.16. Fields Retained in the Container Fact

Not every source classification was promoted into a separate dimension.

IMDG remains in `FactContainerOperations`. The source semantics require further business clarification before a dedicated dangerous-goods dimension would represent a governed classification.

`WeightGroup` also remains in the fact. The source provides the categories, but an approved range definition and formal analytical hierarchy were not required for the implemented model.

These decisions preserve the source information while avoiding unnecessary dimensions whose business definitions have not been fully established.

8.17. QC Working-Hours Validation

QC working hours require special treatment because the source value appears on container-operation rows even though the measurement belongs at a higher operational grain.

Directly summing `QCWorkingHoursSource` across container rows could repeatedly count the same working-hour value. The project therefore introduced `CheckQCWorkingHours` as an explicit validation query.

The source is grouped by:

- calling year,
- calling sequence,
- vessel code,
- user voyage, and
- equipment number.

For each vessel-call-equipment group, the query calculates:

- `ContainerRowCount`,
- minimum QC working hours,
- maximum QC working hours,
- `DistinctQCWorkingHourCount`,
- number of blank QC-hour rows, and
- `QCWorkingHoursStatus`.

The status is classified as:

Listing 8.5: QC working-hour consistency classification

```
if [DistinctQCWorkingHourCount] = 0 then
    "Missing QC Working Hours"
else if [DistinctQCWorkingHourCount] = 1 then
    "Consistent Repeated Value"
else
    "Multiple QC Working Hour Values"
```

This validation step distinguishes a repeated but internally consistent higher-grain value from a group containing conflicting source values.

8.18. FactQCPerformance

FactQCPerformance is created from CheckQCWorkingHours. A QC-hour value is accepted for productivity reporting only when exactly one distinct nonblank QC-hour value exists within the vessel-call–equipment group.

Listing 8.6: Validated QC working-hours rule

```
if [DistinctQCWorkingHourCount] = 1 then
    [MaximumQCWorkingHours]
else
    null
```

The fact therefore contains:

- QCPerformanceKey,
- VesselCallKey,
- EquipmentKey,
- ContainerRowCount,
- MinimumQCWorkingHours,
- MaximumQCWorkingHours,
- DistinctQCWorkingHourCount,
- BlankQCWorkingHourRows,
- QCWorkingHoursStatus,
- ValidatedQCWorkingHours,
- IsQCWorkingHoursValid,
- HandledUniqueContainers, and
- HandledTEU.

Handled TEU is calculated from container length using the same project mapping applied elsewhere:

- 20 feet = 1.00 TEU,
- 40 feet = 2.00 TEU,
- 45 feet = 2.25 TEU.

The QC fact therefore allows equipment workload and productivity to be calculated without directly summing repeated source QC-hour values from the container fact.

QC productivity

Quay-crane productivity should use ValidatedQCWorkingHours from FactQCPerformance. Groups containing missing or multiple conflicting working-hour values remain available for quality reporting but are not treated as valid productivity denominators.

8.19. Detailed Vessel-Port-Call Fact

FactVesselPortCalls extends the model with detailed vessel-port-call records.

The source includes vessel identification, operator, owner, vessel characteristics, voyage information, and the principal operational timestamps:

- ATA — Actual Time of Arrival,
- ATB — Actual Time of Berthing, and
- ATD — Actual Time of Departure.

The query generates the same standardized VesselBusinessKey used for vessel identity and derives several vessel-performance fields.

8.19.1. Vessel Turnaround

Listing 8.7: Vessel turnaround calculation

```
if [ATA] = null or [ATD] = null then
    null
else if [ATD] < [ATA] then
    null
else
    Duration.TotalHours([ATD] - [ATA])
```

8.19.2. Berth Occupancy

Listing 8.8: Berth occupancy calculation

```
if [ATB] = null or [ATD] = null then
    null
else if [ATD] < [ATB] then
    null
else
    Duration.TotalHours([ATD] - [ATB])
```

8.19.3. Pre-Berth Waiting

Listing 8.9: Pre-berth waiting calculation

```
if [ATA] = null or [ATB] = null then
    null
else if [ATB] < [ATA] then
    null
else
    Duration.TotalHours([ATB] - [ATA])
```

8.19.4. Vessel Timestamp Validation

A `VesselTimeSequenceStatus` field identifies missing or invalid timestamp sequences. A reconciliation field then checks the relationship:

$$\text{Vessel Turnaround} = \text{Pre-Berth Waiting} + \text{Berth Occupancy}.$$

The resulting `VesselTimeReconciliationVariance` supports validation of the vessel timing calculations.

A `CallDate` is also derived from ATA to provide a date-level vessel-call field for analytical use.

8.20. Model Relationships and Filter Propagation

The semantic model uses dimension-to-fact relationships to propagate analytical filters while preserving the grain of each fact.

The verified relationship structure includes the following principal paths:

Table 8.4.: Principal semantic-model relationship paths

Dimension	Key	Related analytical structure	Purpose
DimVesselCall	VesselCallKey	FactContainerOperations	Filters container activity by vessel call, calling year, sequence, vessel, and voyage.
DimVesselCall	VesselCallKey	FactQCPerformance	Associates validated equipment/QC groups with vessel-call context.
DimVessel	VesselBusinessKey	DimVesselCall and FactVesselPortCalls	Provides a common vessel identity across operational calls and detailed vessel-call records.
DimOperator	OperatorKey	FactContainerOperations	Supports operator-level container, TEU, flow, and dwell analysis.
DimContainerSpecification	ContainerSpecificationKey	FactContainerOperations	Provides container characteristics and TEU factor.
DimCargo	CargoKey	FactContainerOperations	Supports cargo-type and commodity analysis.
DimEquipment	EquipmentKey	FactContainerOperations and FactQCPerformance	Provides common equipment context for workload and validated QC analysis.
DimPortPOL	POLKey	FactContainerOperations	Provides the loading-port analytical role.
DimPortPOD	PODKey	FactContainerOperations	Provides the discharge-port analytical role.
DimTransportProfile	TransportProfileKey	FactContainerOperations	Provides operational status, mode, full/empty, and transport classifications.
DimDamageCondition	DamageConditionKey	FactContainerOperations	Provides damage-condition filtering.
DimDate	Date	FactContainerOperations[InDateOnly]	Provides the default entry-date analytical path.
DimDate	Date	FactContainerOperations[OutDateOnly]	Provides the alternate exit-date analytical path for measures that explicitly activate it.

The design avoids direct analytical dependence between facts where that would produce grain multiplication. When measures from different facts are compared, common dimensions provide the appropriate filter context.

Fact-grain protection

A vessel-call record must not be multiplied by the number of containers associated with that vessel call, and a QC-hour value must not be multiplied by the number of container rows carrying the same source value. Separate facts and conformed dimensions are used specifically to prevent these forms of fan-out.

8.21. DimDate and Role-Playing Date Analysis

DimDate provides a governed calendar structure for date-based analysis. Unlike the Power Query dimensions, it is implemented as a DAX calculated table.

Its date range is derived dynamically from the available entry and exit dates in FactContainerOperations.

Listing 8.10: Dynamic DimDate boundary logic

```
DimDate =
VAR MinimumInDate =
MINX (
FactContainerOperations,
FactContainerOperations[InDateOnly]
)
VAR MaximumInDate =
MAXX (
FactContainerOperations,
FactContainerOperations[InDateOnly]
)
VAR MaximumOutDate =
MAXX (
FactContainerOperations,
FactContainerOperations[OutDateOnly]
)
VAR MinimumModelDate =
DATE ( YEAR ( MinimumInDate ), 1, 1 )
VAR MaximumAvailableDate =
MAX ( MaximumInDate, MaximumOutDate )
VAR MaximumModelDate =
DATE ( YEAR ( MaximumAvailableDate ), 12, 31 )
RETURN
CALENDAR (
MinimumModelDate,
MaximumModelDate
)
```

The date dimension supports calendar attributes such as year, quarter, month, month number, year-month, week, weekday, sorting fields, and other time-intelligence structures required by the dashboards.

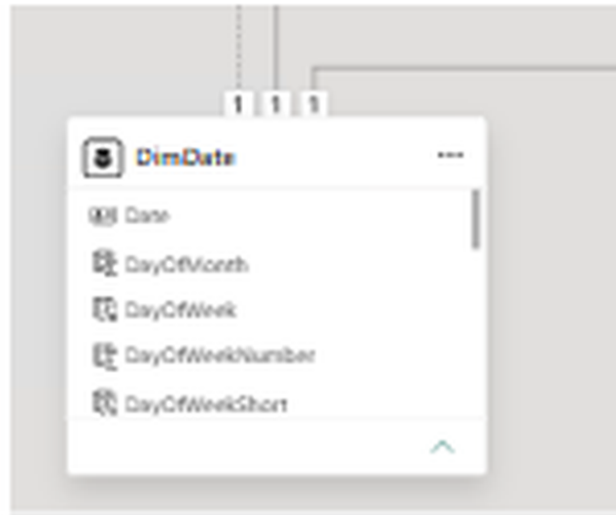


Figure 8.14.: Power BI model view of DimDate and representative calendar attributes.

8.21.1. Entry-Date Role

The principal date relationship uses:

`DimDate[Date] → FactContainerOperations[InDateOnly]`

This relationship provides the default date context for container-entry analysis.

8.21.2. Exit-Date Role

A second relationship connects `DimDate[Date]` to `OutDateOnly`. Because two date relationships between the same tables cannot both serve as the default active filter path, exit-based DAX measures explicitly activate the exit relationship when required.

This allows entry and exit activity to be analyzed against the same governed calendar without creating separate calendar tables for the two lifecycle roles.

8.22. Centralized DAX Measure Layer

DAX measures are organized within the dedicated `_Measures` table rather than being distributed across individual fact and dimension tables.

This provides a centralized location for:

- container volume measures,
- TEU measures,
- entry and exit measures,
- current-state measures,
- dwell measures,
- weight measures,
- vessel measures,
- equipment and QC measures,
- connectivity measures,
- comparison measures,

- data-quality measures, and
- technical validation helpers.

The measure table is an organizational structure rather than a business dimension and therefore does not participate in analytical relationships.

8.23. Container Entry and Exit Measures

Container entries use the normal entry-date relationship:

Listing 8.11: Container-entry measure

```
Container Entries =  
CALCULATE (  
    COUNTROWS ( FactContainerOperations ),  
    NOT ISBLANK (  
        FactContainerOperations[InDateOnly]  
    )  
)
```

Container exits activate the exit-date relationship:

Listing 8.12: Container-exit measure

```
Container Exits =  
CALCULATE (  
    COUNTROWS ( FactContainerOperations ),  
    USERELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    ),  
    NOT ISBLANK (  
        FactContainerOperations[OutDateOnly]  
    )  
)
```

A reporting-window difference can then be calculated as:

Listing 8.13: Net container flow

```
Net Container Flow =  
[Container Entries] - [Container Exits]
```

This value represents the difference between entries and exits in the selected reporting context. It should not automatically be interpreted as the number of containers currently inside the terminal.

8.24. Current Inventory Measures

Current-state measures use records with a valid entry and no recorded exit.

Two conceptually different current-state counts are useful:

- current operational records inside, and
- current unique container identifiers inside.

The distinction is important because `ContainerNumber` is not treated as a universally unique operation-level identifier.

Current-state KPIs represent the latest available extract condition. They are not historical inventory snapshots. Reconstructing inventory as of an earlier date would require an appropriate historical snapshot

design or equivalent event-state reconstruction.

8.25. Dwell-Time Measures

Dwell measures use the governed numeric CompletedDwellDays field created during Power Query preparation.

Because invalid or incomplete lifecycle sequences return blank, standard DAX aggregation naturally excludes those rows.

Representative measures include:

Listing 8.14: Average completed dwell

```
Average Completed Dwell Days =  
CALCULATE (  
    AVERAGE (  
        FactContainerOperations[CompletedDwellDays]  
    ),  
    USERELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    )  
)
```

Listing 8.15: Median completed dwell

```
Median Completed Dwell Days =  
CALCULATE (  
    MEDIAN (  
        FactContainerOperations[CompletedDwellDays]  
    ),  
    USERELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    )  
)
```

Listing 8.16: P90 completed dwell

```
P90 Completed Dwell Days =  
CALCULATE (  
    PERCENTILEX.INC (  
        FILTER (  
            FactContainerOperations,  
            NOT ISBLANK (  
                FactContainerOperations[CompletedDwellDays]  
            )  
        ),  
        FactContainerOperations[CompletedDwellDays],  
        0.90  
    ),  
    USERELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    )  
)
```

Using multiple dwell statistics is preferable to relying on a single average because terminal dwell distributions can contain long-duration records that materially affect the mean.

8.26. TEU Measures

The core TEU calculation retrieves the governed factor from DimContainerSpecification for each container-operation row:

Listing 8.17: Container-operation TEU calculation

```
Total TEU =
SUMX (
    FactContainerOperations,
    RELATED (
        DimContainerSpecification[TEUFactor]
    )
)
```

The project distinguishes lifecycle TEU measures according to whether the associated container entered or exited the terminal.

Listing 8.18: TEU entries

```
TEU Entries =
CALCULATE (
    [Total TEU],
    NOT ISBLANK (
        FactContainerOperations[InDateOnly]
    )
)
```

Listing 8.19: TEU exits

```
TEU Exits =
CALCULATE (
    [Total TEU],
    USERELATIONSHIP (
        DimDate[Date],
        FactContainerOperations[OutDateOnly]
    ),
    NOT ISBLANK (
        FactContainerOperations[OutDateOnly]
    )
)
```

For reporting where “Total TEU” is intended to represent total handled TEU activity rather than the TEU represented by the container rows themselves, the project definition should remain explicit:

Listing 8.20: Total handled TEU

```
Total Handled TEU =
[TEU Entries] + [TEU Exits]
```

This distinction prevents the label *Total TEU* from being used ambiguously for two different business concepts.

TEU terminology

The relationship-based SUMX calculation returns the TEU equivalent represented by container-operation rows in the current filter context. Entry TEU and exit TEU apply lifecycle logic. When the required KPI is total handled TEU activity, it is defined as TEU entries plus TEU exits and should be labeled accordingly.

8.27. QC Performance Measures

QC productivity measures should be calculated from `FactQCPerformance` rather than by summing `QCWorkingHoursSource` directly from the container fact.

Representative components include:

- validated QC working hours,
- handled unique containers,
- handled TEU,
- valid QC group count,
- invalid QC group count, and
- moves per validated QC hour.

Conceptually:

$$\text{QC Productivity} = \frac{\text{Handled Containers or Moves}}{\text{Validated QC Working Hours}}.$$

Only groups satisfying the QC consistency rule should contribute validated hours to the productivity denominator.

This approach provides a controlled productivity KPI while retaining inconsistent groups for data-quality analysis.

8.28. Vessel-Performance Measures

Vessel-performance calculations originate from `FactVesselPortCalls` rather than from repeated container-operation rows.

The fact supports measures such as:

- vessel calls,
- unique vessels,
- average vessel turnaround hours,
- average pre-berth waiting hours,
- average berth occupancy hours,
- average LOA,
- average draft,
- average gross tonnage, and
- average deadweight tonnage.

The three principal vessel timing components are defined as:

$$\text{Turnaround} = \text{ATD} - \text{ATA},$$

$$\text{Pre-Berth Waiting} = \text{ATB} - \text{ATA},$$

and

$$\text{Berth Occupancy} = \text{ATD} - \text{ATB}.$$

For valid timestamp sequences:

$$\text{Turnaround} = \text{Pre-Berth Waiting} + \text{Berth Occupancy}.$$

The reconciliation variance generated in Power Query provides a direct validation of this relationship.

8.29. Connectivity Measures

Port-connectivity analysis uses the two independent port dimensions.

DimPortPOL supports loading-port, loading-country, and loading-region analysis, while DimPortPOD supports discharge-port, discharge-country, and discharge-region analysis.

This structure allows the model to answer questions such as:

- which loading ports contribute the greatest operational volume;
- which discharge ports receive the greatest operational volume;
- which countries and regions dominate the recorded port network;
- how TEU or container volume varies by loading or discharge role; and
- how port connectivity changes across reporting periods.

The roles remain separate throughout analysis so that a port's loading activity is not confused with its discharge activity.

8.30. Validation Framework

Validation is embedded into the analytical architecture rather than performed only after dashboard construction. Table 8.5 summarizes the principal implemented controls.

Table 8.5.: Principal semantic-model validation controls

Control	Implementation	Analytical risk addressed
Technical row identity	Generated ContainerOperati onKey	Prevents ambiguity between physi- cal container identity and imported operational-row identity.
Vessel-call matching	VesselCallMatchStatus	Identifies container records that do not resolve to the vessel-call dimen- sion.
Operator matching	OperatorMatchStatus	Identifies source operators that do not resolve to the operator dimen- sion.
Date sequence	DateSequenceStatus and blank invalid dwell	Prevents negative or unsupported completed-dwell calculations.
Lifecycle completion	IsCompleted	Distinguishes valid completed oper- ations from open or invalid records.
Current state	IsCurrentlyInPort, Active AgeDays, and age buckets	Separates latest known active records from completed lifecycle records.
Stack-days comparison	StackDaysVariance	Allows source-provided stack days to be compared with timestamp- derived completed dwell.
TEU mapping	TEUMappingStatus	Prevents unsupported container lengths from receiving an assumed TEU factor.
Weight validity	WeightDataStatus and HasVal idWeight	Distinguishes valid positive source weights from missing, zero, or neg- ative values.
QC-hour consistency	DistinctQCWorkingHourCou nt, QCWorkingHoursStatus, and IsQCWorkingHoursValid	Prevents repeated or conflicting QC hours from distorting equipment pro- ductivity.
Vessel timestamp se- quence	VesselTimeSequenceStatus	Prevents invalid ATA–ATB–ATD sequences from entering duration measures.
Vessel time reconcilia- tion	VesselTimeReconciliation Variance	Checks that turnaround reconciles with waiting plus berth occupancy.
Fact grain	Separate container, QC, and vessel facts	Prevents fan-out, duplicated dura- tions, and inappropriate aggregation across operational grains.
Role-specific ports	Separate DimPortPOL and DimP ortPOD	Prevents loading and discharge roles from being mixed.
Role-specific dates	Entry and exit date relationships	Prevents entry-period and exit- period measures from being interpreted as the same lifecycle event.

8.31. Semantic and Business-Definition Guardrails

A technically valid field or relationship does not automatically establish a final business definition. Several source attributes are therefore retained conservatively.

The model does not assign unsupported meanings to:

- the precise commercial role represented by OPR;
- the unresolved source abbreviation S/A;
- detailed business semantics of the source IMDG field;
- the physical unit represented by WGT; or
- any binary damaged/not-damaged rule beyond the recorded source condition.

These fields can be analyzed as recorded, but further business interpretation requires confirmation from the responsible terminal or data owner.

Business-definition boundary

Data modeling can standardize, relate, and validate source values, but it should not create business meanings that are not supported by the source or formally approved. Unresolved source semantics remain documented rather than being silently converted into governed business rules.

8.32. Why the Multi-Fact Design Matters

The final architecture resolves several important analytical risks.

First, container operations remain at their original operational grain, allowing container counts, flow, cargo, ports, dwell, weight, and TEU to be analyzed without losing row-level traceability.

Second, QC working hours are moved into a dedicated grouped structure before productivity is calculated. This prevents the same crane-hour value from being counted repeatedly simply because several containers were handled during the same vessel-call–equipment activity.

Third, vessel timing remains at vessel-call grain. Turnaround, waiting, and berth occupancy are therefore not duplicated according to the number of containers associated with a vessel.

Fourth, reusable dimensions allow multiple analytical processes to share consistent descriptive context. Vessel identity, equipment, operator, date, and other dimensions can therefore support several report pages without embedding repeated descriptive values within every fact.

Finally, the dedicated DAX layer separates business measures from raw fields and provides a controlled location for KPI definitions, comparison logic, lifecycle rules, and validation calculations.

8.33. Chapter Conclusion

The Power BI semantic model provides the governed analytical foundation for the final reporting solution. Its strength lies in maintaining the correct relationship between business process, analytical grain, dimensional context, and KPI logic.

FactContainerOperations provides the principal container-operation view. FactQCPerformance protects quay-crane productivity from repeated or conflicting source-hour values. FactVesselPortCalls preserves detailed vessel timing and physical characteristics at vessel-call grain.

The supporting dimensions organize vessel identity, vessel-call context, operators, container specifications, cargo, equipment, loading ports, discharge ports, transport classifications, damage conditions, and calendar dates. The governed TEU mapping, date-role logic, current-state controls, QC validation, weight checks, and vessel-time reconciliation provide additional analytical assurance.

Together, the fact tables, dimensions, relationships, DAX measures, and validation controls form a coherent semantic layer capable of supporting traceable analysis of container flow, TEU activity, dwell performance, current inventory, vessel operations, equipment workload, cargo mix, port connectivity, and data quality. This model forms the direct analytical foundation for the Power BI dashboards documented in the following chapter.

9. Power BI Dashboard Suite and Visualization Design

9.1. Dashboard Suite Overview

Following the data preparation, semantic modeling, DAX development, and validation processes described in the preceding chapters, the final stage of the Power BI implementation focused on converting the governed analytical model into an interactive and management-oriented dashboard suite.

The completed Power BI report consists of **nine dashboard pages**. Each page addresses a distinct analytical area while remaining connected to the same underlying semantic model. Together, the dashboards provide views of terminal throughput, port connectivity, operational workload, quay-crane activity, container flow and dwell, cargo and commodity composition, weight profiles, vessel-port-call performance, and detailed container lifecycle records.

The dashboard suite was designed to support different levels of analysis. Management users can begin with high-level operational indicators and year-over-year comparisons, while operational and analytical users can progressively move into equipment, cargo, vessel, port, and individual container details.

Table 9.1 summarizes the final report structure.

Table 9.1.: Final Power BI dashboard suite

No.	Report page	Displayed title	Primary analytical purpose
01	Executive Overview	Terminal Executive Performance Overview	High-level terminal throughput, dwell, vessel-call, and operator performance with Q1 comparison.
02	Outbound Connectivity	Port Connectivity and Trade Lanes	Outbound network structure, destination regions and countries, discharge ports, trade lanes, TEU, and route-level dwell.
03	Workload Analysis	Vessel, Operator, and Equipment Workload Analysis	Combined analysis of vessel calls, operators, equipment, container characteristics, and workload concentration.
04	Equipment Dashboard	Equipment Workload and Productivity Analysis	Quay-crane workload, handled containers and TEU, validated QC activity, and productivity analysis.
05	Container Flow and Dwell	Container Flow, Inventory, and Dwell Performance	Container and TEU entry/exit flow, current inventory, completed dwell, and quarterly comparison.
06	Cargo and Commodity	Cargo and Commodity Performance Analysis	Cargo mix, container types, commodities, TEU contribution, container volume, and commodity dwell.
07	Weight Profile	Weight Profile and Workload Analysis	Weight distribution, completeness, operator workload, cargo characteristics, and weight-versus-dwell exploration.
08	Vessel Port Call	Vessel Port Call and Berth Performance	Vessel turnaround, pre-berth waiting, berth occupancy, operator performance, and vessel characteristics.
09	DT Container Detail	Container Detail and Lifecycle	Record-level container lifecycle, cargo, weight, operator, and port traceability.

9.2. Dashboard Design and Navigation

A consistent visual design was applied across the nine report pages to create a unified analytical experience. The dashboards use a 16:9 report canvas with a dark navy outer frame, white visual containers, pale-blue KPI cards, prominent page titles, consistent typography, and persistent page navigation.

KPI cards are positioned prominently to provide immediate visibility into the principal measures associated with each analytical topic. Detailed charts are then used to explain the composition, trend, distribution, or relationship behind those headline values.

Slicers are incorporated where analytical segmentation is required and include dimensions such as operational date, operator, cargo type, commodity, container type, vessel, equipment, region, and container length. This enables the same dashboard to be explored from multiple operational perspectives without changing the underlying KPI definitions.

The page sequence also reflects a deliberate analytical progression:

1. executive performance overview;
2. port connectivity and trade lanes;
3. combined operational workload;
4. detailed equipment and quay-crane performance;
5. container lifecycle, flow, and dwell;
6. cargo and commodity composition;
7. weight and workload characteristics;
8. vessel-port-call and berth performance; and
9. individual container detail and traceability.

This structure allows users to move from aggregate management indicators toward increasingly detailed operational analysis.

9.3. Page 01: Executive Overview

Page identity

Report page: 01ExecutiveOverview

Displayed title: Terminal Executive Performance Overview

Primary focus: Overall terminal performance and Q1 comparison

9.3.1. Purpose

The Executive Overview serves as the principal management entry point to the Power BI report. It consolidates the most important terminal performance indicators into one page and provides a direct comparison between Q1 2025 and Q1 2026.

The dashboard is designed to answer several high-level management questions:

- How much container and TEU activity was handled?
- Did throughput increase or decrease?
- How did completed dwell change?
- How many vessel calls are represented in the selected executive reporting context?
- Which operators account for the largest TEU contribution?

9.3.2. Key Performance Indicators

The principal KPI group reports:

- **95,325 container entries,**
- **89,221 container exits,**
- **168,345 TEU entered,**
- **158,090 TEU exited,**
- **179 vessel calls,** and
- **6.60 days average completed dwell.**

The dashboard also presents direct Q1 comparison cards:

- Average dwell increased from approximately **6.4 days** in Q1 2025 to **6.7 days** in Q1 2026, representing a displayed increase of **4.4%**.
- TEU entered increased by approximately **4.5%**.
- Container entries increased by approximately **3.1%**.

9.3.3. Visual Analysis

The *Quarterly Container Entries and Exits* visual compares container movement between Q1 2025 and Q1 2026 and provides an immediate view of changes in terminal flow.

The *Completed Dwell by Exit Quarter* visual presents average, median, and P90 completed dwell together. The displayed values indicate:

- average dwell increasing from approximately 6.45 to 6.74 days;
- median dwell increasing from 4 to 5 days; and
- P90 remaining approximately 13.2 days.

Displaying these measures together is important because average dwell alone does not describe the full distribution of container residence time.

The *Top 10 Operators by TEU* visual shows the concentration of handled TEU across operator codes, with MSC leading the displayed ranking, followed by COE and ONE.

9.3.4. Management Interpretation

The executive dashboard shows that Q1 2026 experienced higher container-entry and TEU-entry activity than Q1 2025. However, completed dwell also increased. The operational interpretation is therefore more nuanced than a simple volume-growth story: increased throughput was accompanied by additional time spent within the terminal for completed container records.

The page also establishes an important distinction between **flow** and **inventory**. The difference between entries and exits over a reporting period should not be interpreted as the number of containers currently inside the terminal. Current inventory is addressed separately on Page 9.7 using the latest lifecycle status available in the model.

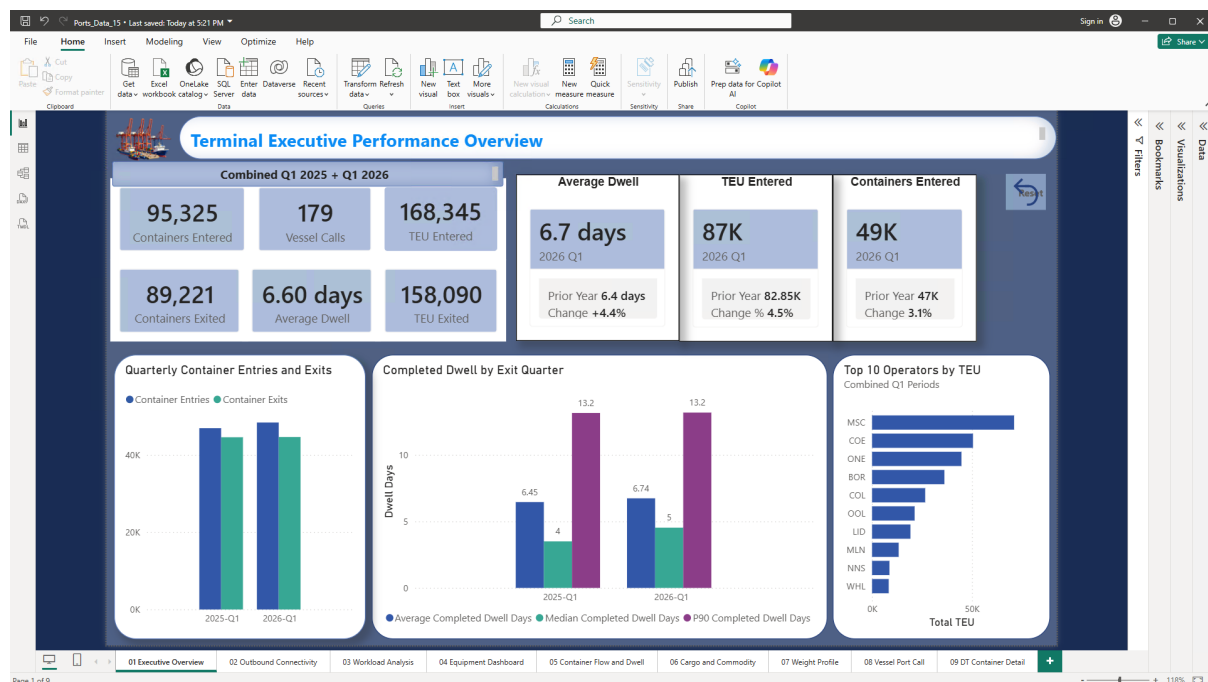


Figure 9.1.: Power BI Page 01, Executive Overview: Terminal Executive Performance Overview.

9.4. Page 02: Outbound Connectivity

Page identity

Report page: 020outboundConnectivity

Displayed title: Port Connectivity and Trade Lanes

Primary focus: Outbound network structure and destination analysis

9.4.1. Purpose

The Outbound Connectivity dashboard analyzes the geographic structure of terminal traffic. It uses the role-specific loading-port and discharge-port dimensions developed in the semantic model to examine destination regions, countries, discharge ports, and POL–POD trade lanes.

The page is intended to answer:

- How geographically diverse is the terminal's outbound network?
- Which discharge ports receive the greatest activity?
- Which trade lanes contribute the largest TEU?
- Which destination ports experience the highest completed dwell?

9.4.2. Key Performance Indicators

The KPI cards report:

- **15 destination regions,**
- **21 destination countries,**
- **50 discharge ports,**
- **118 active trade lanes,** and
- approximately **158.09K outbound TEU.**

Available slicers include operational date, operator, and region, enabling connectivity patterns to be examined under different analytical contexts.

9.4.3. Trade-Lane Analysis

The *Top 10 Outbound Trade Lanes by TEU* visual identifies the highest-volume POL–POD combinations.

The leading visible routes include:

- BEANR → EGDAM,
- RUNVS → EGDAM,
- EGDAM → GRPIR,
- EGDAM → RUNVS,
- GRPIR → EGDAM,
- DEHAM → EGDAM,
- EGDAM → BEANR,
- NLRTM → EGDAM,
- SAJED → EGDAM, and
- EGDAM → DEHAM.

This ranking identifies the routes that contribute the greatest TEU volume within the selected report context.

9.4.4. Destination-Port Analysis

The discharge-port ranking highlights major destination ports such as Piraeus, Novorossiysk, Antwerp, Hamburg, Rotterdam, Ashdod, Mersin, and Salerno.

A separate visual ranks destination ports by average completed dwell, subject to the minimum-record threshold applied to the visual. Ports such as Trieste and Hamburg appear among the higher-dwell destinations in the displayed analysis.

9.4.5. Operational Interpretation

The dashboard demonstrates that network reach and operational concentration are different concepts. Although the terminal connects to numerous countries, regions, and discharge ports, a smaller group of routes accounts for a substantial share of outbound TEU.

The page also shows that a high-volume discharge port is not necessarily a high-dwell port. Route-volume management and dwell-exception management should therefore be treated as related but separate analytical tasks.

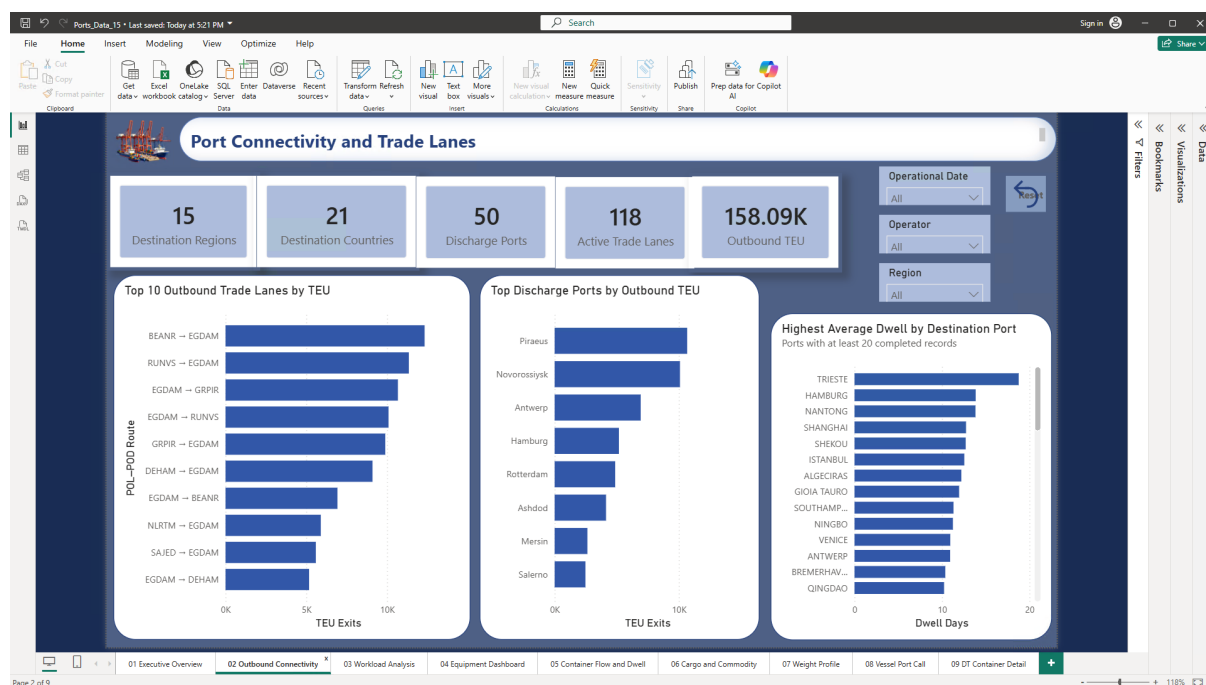


Figure 9.2.: Power BI Page 02, Outbound Connectivity: Port Connectivity and Trade Lanes.

9.5. Page 03: Workload Analysis

Page identity

Report page: 03WorkloadAnalysis

Displayed title: Vessel, Operator, and Equipment Workload Analysis

Primary focus: Combined workload concentration across operational entities

9.5.1. Purpose

The Workload Analysis dashboard provides a consolidated view of how terminal workload is distributed across vessels, vessel calls, operators, equipment, and container specifications.

It acts as a bridge between the high-level Executive Overview and the more specialized Equipment and Vessel Port Call dashboards.

9.5.2. Key Performance Indicators

The KPI cards display approximately:

- **191 vessel calls**,
- **68K unique containers**,
- **96K container-operation records**, and
- **338.8K TEU activity** within the measure context used on this page.

A container-length selector provides the principal values 20, 40, and 45, while slicers support filtering by operational date, container type, operator, vessel, and equipment.

9.5.3. Equipment Workload

The *Equipment Workload by Container Records* visual shows how container-operation records are distributed across equipment identifiers.

QC10 carries the largest visible workload, followed by QC01, QC08, QC02, and QC07.

A separate validated-equipment visual introduces the quality-controlled QC working-hour information developed in Chapter 8. This allows equipment workload to be interpreted together with validated operational-hour information rather than relying on raw repeated source values.

9.5.4. Vessel-Call and Operator Workload

The *Top Vessel Calls by TEU* ranking identifies vessel calls associated with the largest TEU activity in the selected context.

The *Top 10 Operators by TEU* chart provides the corresponding operator-level concentration, with MSC, COE, and ONE among the leading displayed operators.

9.5.5. Operational Interpretation

The dashboard shows that terminal workload is concentrated across several dimensions simultaneously. A small group of operators, equipment units, and vessel calls contributes a substantial proportion of recorded activity.

This page is particularly useful for identifying where further operational analysis should be directed. It does not, however, imply that the equipment handling the greatest volume is necessarily the most or least productive. Productivity requires the validated working-hour context provided on the dedicated equipment dashboard.

The vessel-call value shown on this page also reflects its own analytical context and should not be treated as interchangeable with the detailed vessel-call population analyzed on Page 9.10.

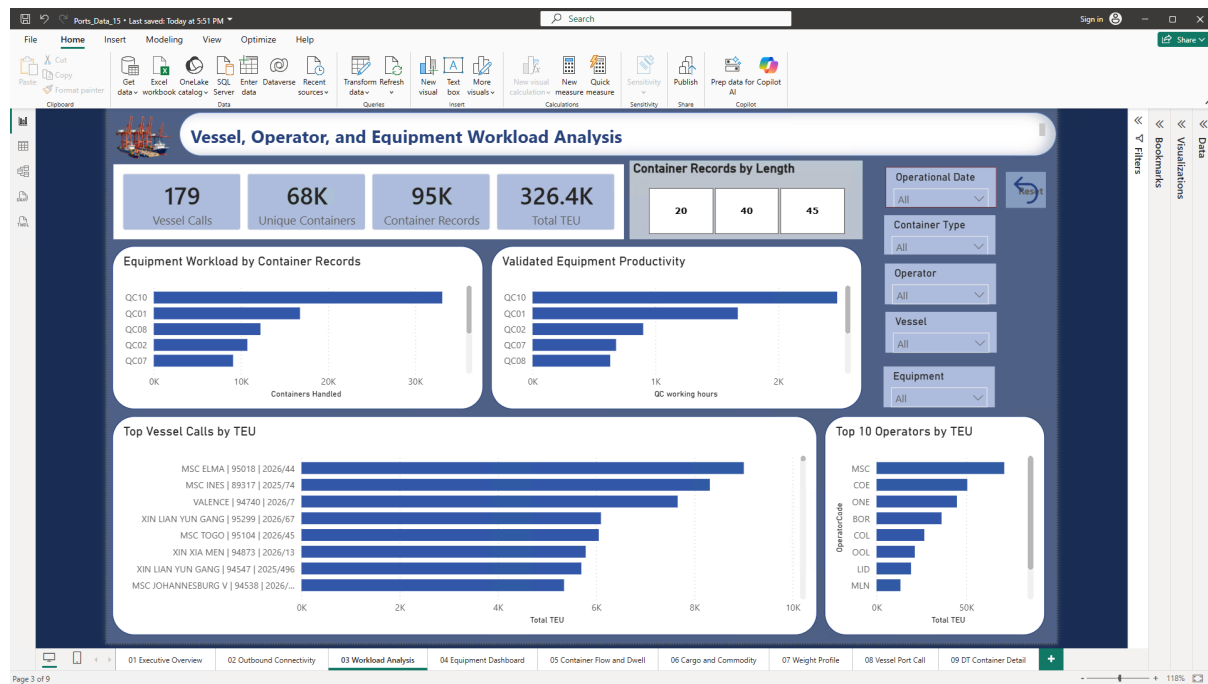


Figure 9.3.: Power BI Page 03, Workload Analysis: Vessel, Operator, and Equipment Workload Analysis.

9.6. Page 04: Equipment Dashboard

Page identity

Report page: 04EquipmentDashboard

Displayed title: Equipment Workload and Productivity Analysis

Primary focus: Quay-crane workload and validated productivity

9.6.1. Purpose

The Equipment Dashboard provides a focused analysis of quay-crane workload and validated productivity. It combines handled container records, TEU activity, validated QC workload, and TEU-per-working-hour measures so that equipment performance can be interpreted from both workload and efficiency perspectives.

The dashboard uses the QC-validation structure described in Chapter 8, ensuring that productivity measures are based on QC working-hour values validated at the appropriate vessel-call–equipment grain.

9.6.2. Key Performance Indicators

The dashboard reports approximately:

- **95.3K containers handled,**
- **326.4K TEU handled,**
- **9 active quay cranes,** and
- approximately **51.9K validated QC handled records** in the displayed validation context.

The page can be filtered by operational date, cargo type, operator, and commodity.

9.6.3. Workload Analysis

The *Containers Handled by Quay Crane* visual shows substantial equipment workload concentration. QC10 carries approximately 24.0K container records, followed by QC01, QC08, QC07, QC02, QC09, QC03, and QC11 in the visible ranking.

The *TEU Handled by Quay Crane* visual provides the corresponding TEU workload. QC10 again leads the visible ranking at approximately 66K TEU.

These visuals identify workload concentration but should not independently be interpreted as productivity measures.

9.6.4. Validated QC Analysis

The *Productivity by Quay Crane* visual ranks equipment using valid QC handled records. QC01 leads the visible validated-record ranking, followed by QC10 and QC02.

The *TEU Productivity by Quay Crane* visual evaluates TEU relative to validated QC working hours. QC10 leads the visible TEU-per-QC-hour ranking, followed by QC01, QC02, QC07, and QC09.

9.6.5. Operational Interpretation

The dashboard illustrates why workload and productivity must be evaluated separately. QC10 handles the greatest visible container and TEU workload, while QC01 leads the displayed valid-QC-record ranking. TEU-per-QC-hour provides another perspective on equipment utilization.

Accordingly, equipment decisions should not be based solely on handled volume. A more appropriate review sequence is to:

1. confirm QC working-hour validity;
2. evaluate handled container and TEU workload;
3. compare validated productivity;
4. examine vessel-call and cargo context; and
5. investigate persistent workload or productivity differences.

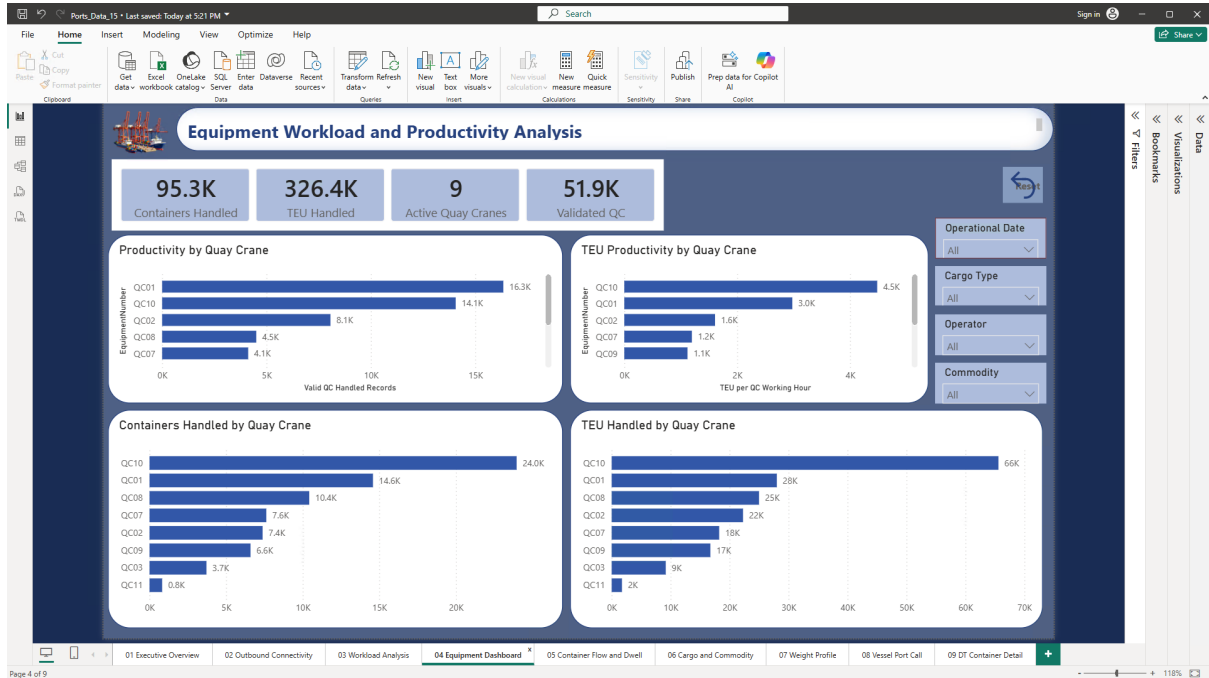


Figure 9.4.: Power BI Page 04, Equipment Dashboard: Equipment Workload and Productivity Analysis.

9.7. Page 05: Container Flow and Dwell

Page identity

Report page: 05ContainerFlowandDwell

Displayed title: Container Flow, Inventory, and Dwell Performance

Primary focus: Container lifecycle, current state, and dwell efficiency

9.7.1. Purpose

The Container Flow and Dwell dashboard is the principal lifecycle-performance page in the report. It integrates entry activity, exit activity, TEU movement, current active records, and completed-dwell statistics.

The page separates three operational concepts that should not be treated as equivalent:

- movement during a reporting period;
- current active inventory; and
- elapsed dwell for completed container lifecycles.

9.7.2. Key Performance Indicators

The dashboard reports:

- **95,325 container entries,**
- **89,221 container exits,**
- **168,345 TEU entered,**
- **158,090 TEU exited,**
- **19 current terminal inventory records,**
- **6.6 average completed dwell days,**
- **4 median completed dwell days, and**
- **13.2 P90 completed dwell days.**

The dashboard provides a cargo-type selector and slicers for operational date, operator, and container length.

9.7.3. Container and TEU Flow

The *Monthly Container Flow – Entries vs Exits* visual compares entry and exit activity across the Q1 months represented in 2025 and 2026.

The *Monthly TEU Flow – Entered vs Exited* visual provides the equivalent capacity-based view.

The *Quarterly Container Flow* visual summarizes entry, exit, and net-flow behavior for the two Q1 periods.

These visuals collectively enable users to distinguish changes in container count from changes in TEU-equivalent activity.

9.7.4. Dwell Distribution

The dashboard deliberately reports average, median, and P90 dwell together:

- median = 4 days,
- average = 6.6 days,
- P90 = 13.2 days.

The difference between these statistics indicates a right-skewed completed-dwell distribution. Most completed records have shorter dwell than the average, while a smaller group of long-stay containers increases the mean.

9.7.5. Inventory Interpretation

The reporting-period difference between container entries and exits is:

$$95,325 - 89,221 = 6,104.$$

This value is a **reporting-window flow difference**; it is not the current terminal inventory.

The current-state KPI reports 19 records under the implemented logic requiring a valid entry and no recorded exit. This measure represents the latest state available in the source extract.

Historical inventory should not be reconstructed simply by applying historical date filters to this current-state measure. A dedicated snapshot or equivalent historical-state structure would be required for true inventory-over-time analysis.

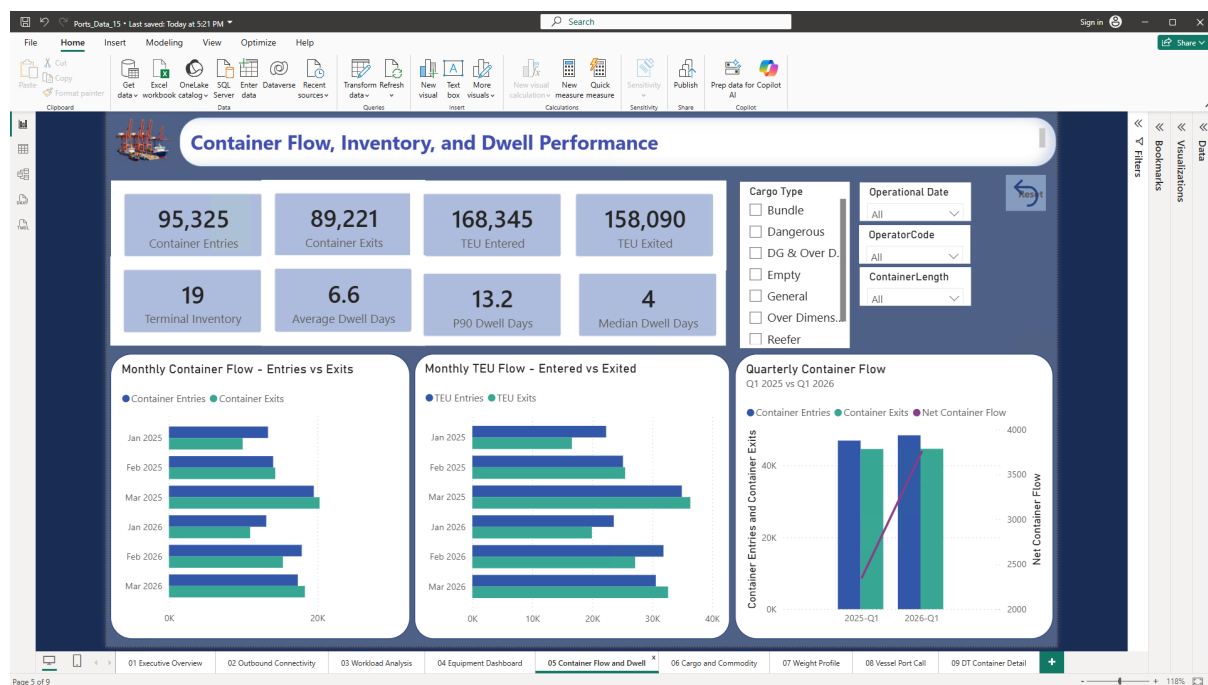


Figure 9.5.: Power BI Page 05, Container Flow and Dwell: Container Flow, Inventory, and Dwell Performance.

9.8. Page 06: Cargo and Commodity

Page identity

Report page: 06CargoandCommodity

Displayed title: Cargo and Commodity Performance Analysis

Primary focus: Cargo composition, commodity workload, and dwell

9.8.1. Purpose

The Cargo and Commodity dashboard evaluates the composition of terminal activity by cargo type, commodity, and container type.

Its analytical objective is to distinguish **volume concentration** from **dwell concentration**. A commodity responsible for high TEU volume is not necessarily the commodity with the highest completed dwell.

9.8.2. Key Performance Indicators

The KPI cards display approximately:

- **326K TEU**,
- **68K unique containers**,
- **9 cargo types**, and
- **52 commodities**.

The page can be filtered by operational date, cargo type, commodity, and container type.

9.8.3. Container-Type and Cargo-Type Mix

The *TEU Distribution by Container Type* visual shows General Dry as the dominant visible container category, followed by Refrigerated equipment. Open-top, Platform-Fixed, Tank, and Platform containers contribute smaller shares of the displayed TEU activity.

The *Cargo Type Mix by TEU* visual provides a broader cargo classification view, with General cargo representing the largest visible category, followed by Empty and Reefer.

The *Containers by CargoType* chart provides the corresponding container-count perspective.

9.8.4. Commodity Analysis

The *Top Commodities by TEU Handled* visual identifies major commodities including Orange, Wood, Accessories, Clothes, Vaccines, and Electrical Appliances among the leading displayed categories.

A separate *Commodities with Highest Average Dwell* visual ranks commodities according to completed dwell rather than volume. The visible ranking includes categories such as Reinforced Iron, Sugar, Potatoes, Sensitive Tools, Wood, Shrimp, Vaccines, and Special Chemicals.

9.8.5. Operational Interpretation

The dashboard demonstrates that high-volume cargo and high-dwell cargo require different management responses.

High-volume commodities are primarily relevant to capacity, equipment, storage, and planning decisions. High-dwell commodities may require investigation of operational handling, documentation, service requirements, cargo characteristics, or scheduling conditions.

The visual separation between volume and dwell therefore helps prevent operational priority from being assigned solely on the basis of throughput.

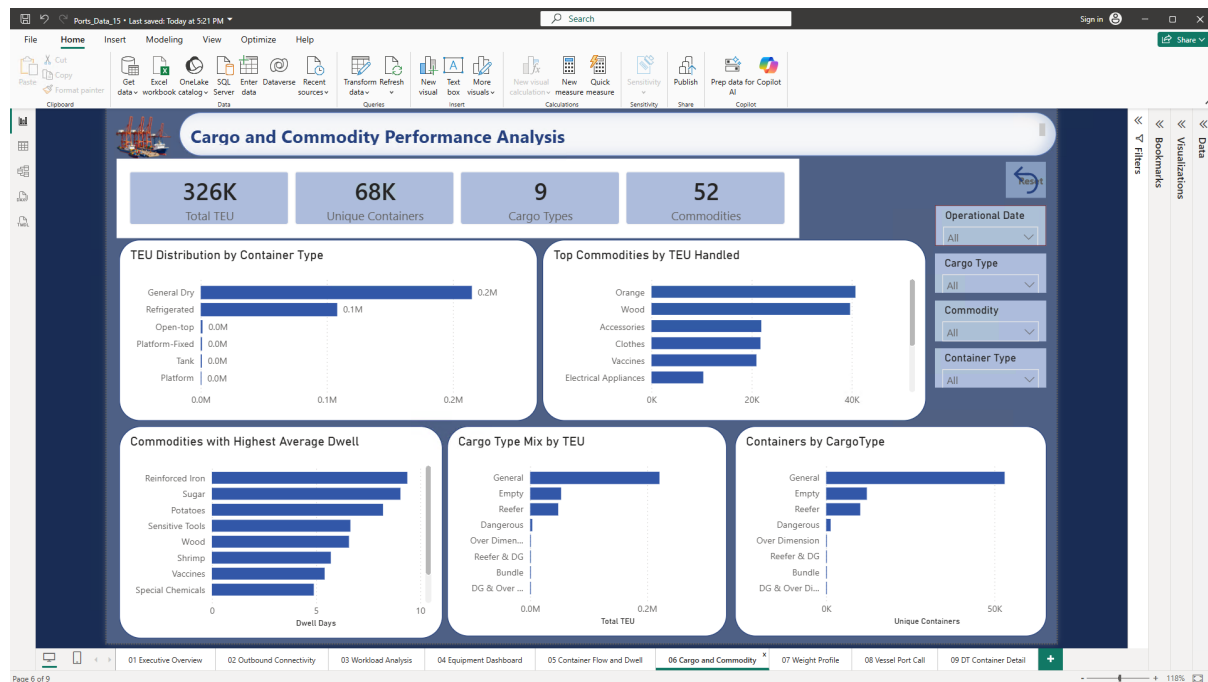


Figure 9.6.: Power BI Page 06, Cargo and Commodity: Cargo and Commodity Performance Analysis.

9.9. Page 07: Weight Profile

Page identity

Report page: 07WeightProfile

Displayed title: Weight Profile and Workload Analysis

Primary focus: Weight distribution, workload, and data readiness

9.9.1. Purpose

The Weight Profile dashboard examines the distribution and analytical completeness of the source weight values while relating weight to container type, cargo type, operator workload, and completed dwell.

The source field is analyzed in its recorded unit because the dataset does not independently confirm whether WGT represents kilograms, tonnes, gross container weight, cargo weight, verified gross mass, or another operational measure.

Accordingly, the dashboard supports relative comparison without imposing an unsupported physical-unit interpretation.

9.9.2. Key Performance Indicators

The dashboard reports approximately:

- **1.71 billion total handled source weight,**
- **21.50K median weight,**
- **95.33K valid positive-weight records,** and
- **17.94K average container weight.**

A year slicer supports temporal filtering.

9.9.3. Weight Distribution

The *Containers by Weight Group* visual shows the distribution of container records across the source-defined weight categories. Extra Heavy and Extra Light represent major visible groups, followed by Heavy and Light.

The *Average and Median Weight by Container Type* visual compares central weight tendencies across container types such as Tank, General Dry, and Open-top.

The *Average Container Weight by Cargo Type* visual provides the equivalent cargo-level comparison, with Reefer among the highest visible categories.

9.9.4. Operator Workload

The *Top Operators by Total Handled Weight* chart identifies operators associated with the largest aggregate source-weight values. COE leads the visible ranking, followed by ONE, MSC, and BOR.

9.9.5. Weight and Dwell Relationship

The *Weight vs. Dwell by Operator* scatter plot compares average source weight and average completed dwell across operators.

This chart is exploratory. It can identify clusters, unusual observations, or operators requiring further examination, but it does not demonstrate that heavier containers cause longer dwell.

9.9.6. Interpretation

The dashboard shows that weight information has sufficient coverage to support comparative profiling. However, all numerical values must remain associated with the **source weight unit** until the operational definition of WGT is formally confirmed.

The weight-group categories are likewise treated as source-defined business categories rather than reconstructed ranges unless their thresholds are formally documented.

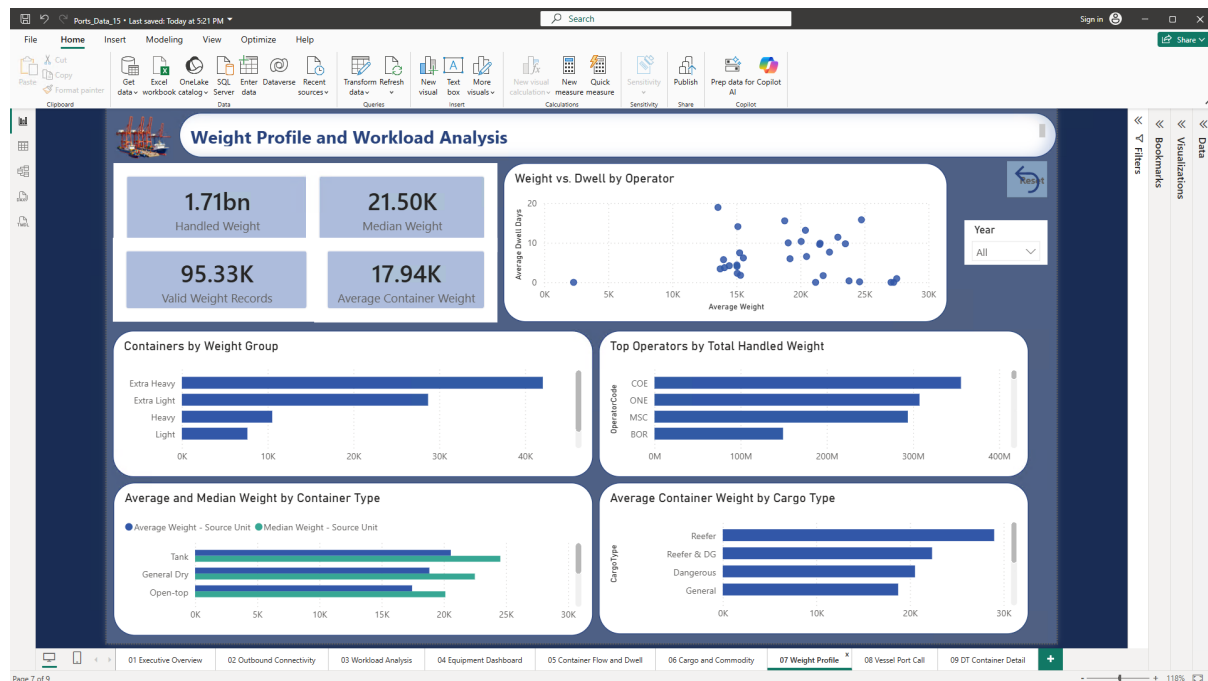


Figure 9.7.: Power BI Page 07, Weight Profile: Weight Profile and Workload Analysis.

9.10. Page 08: Vessel Port Call

Page identity

Report page: 08VesselPortCall

Displayed title: Vessel Port Call and Berth Performance

Primary focus: Vessel turnaround, waiting, berth occupancy, and vessel characteristics

9.10.1. Purpose

The Vessel Port Call dashboard analyzes the detailed vessel-call dataset introduced in the semantic model. Unlike container-level pages, this dashboard operates on the vessel-port-call fact, where one record represents one detailed vessel call.

The dashboard examines:

- total vessel calls,
- unique vessels,
- vessel turnaround,
- pre-berth waiting,
- berth occupancy,
- operator-level timing variation, and
- the relationship between vessel dimensions and turnaround.

9.10.2. Key Performance Indicators

The KPI cards report:

- **177 detailed vessel calls,**
- **82 vessels,**
- **46.4 hours average turnaround,**
- **23.9 hours average berth occupancy,** and
- **22.5 hours average pre-berth waiting.**

These measures are derived from the vessel timing sequence:

$$\text{Turnaround} = \text{ATD} - \text{ATA},$$

$$\text{Pre-Berth Waiting} = \text{ATB} - \text{ATA},$$

and

$$\text{Berth Occupancy} = \text{ATD} - \text{ATB}.$$

9.10.3. Vessel Turnaround Analysis

The *Top 10 Vessels by Average Turnaround* visual identifies vessels with the highest average recorded turnaround among the selected detailed calls. The visible ranking includes Penang Bridge, BZ Chongfu, Wan Hai 622, Lider Perihan, JY Bonito, MSC Lea II, Hui Da 9, Xin Xin Hai 1, MSC Ubert, and MSC Togo.

This ranking is useful for identifying calls or vessels that may warrant additional operational review, but high average turnaround alone does not establish the reason for the observed duration.

9.10.4. Operator-Level Performance

The *Top 10 Operators by Average Turnaround* chart compares turnaround performance across operator codes.

The *Average Waiting vs. Berth Time by Operator* visual further decomposes each operator's call time into:

- time before berthing; and
- time occupying the berth.

This decomposition is important because two operators may have similar total turnaround while experiencing different operational patterns before and after berthing.

9.10.5. Pre-Berth Waiting and Berth Occupancy

Average pre-berth waiting is nearly equal to average berth occupancy. The selected detailed calls report approximately 22.5 hours of waiting and 23.9 hours of berth occupancy within an average total turnaround of 46.4 hours.

This indicates that the period between vessel arrival and actual berthing represents a substantial portion of the overall port-call cycle. Pre-berth waiting is therefore an important candidate for operational investigation alongside berth-service time.

Potential factors requiring further investigation could include berth availability, overlapping vessel arrivals, arrival schedule variation, resource readiness, vessel sequencing, navigational requirements, or other operational conditions. The dashboard itself does not establish which factor caused the observed waiting time.

9.10.6. Vessel Size and Turnaround

The *Vessel Size vs. Turnaround Time* scatter plot compares Length Overall (LOA) with vessel turnaround hours.

The displayed observations do not form a single tight relationship in which increasing vessel length consistently produces proportionally longer turnaround. Vessel size alone therefore does not appear sufficient to explain the observed variation in vessel-call duration.

Other operational factors may also contribute, including cargo volume, crane allocation, equipment availability, service pattern, berth conditions, handling requirements, and call-specific operational circumstances.

The scatter plot should therefore be interpreted as an exploratory relationship rather than a causal model.

9.10.7. Scope of the Vessel-Call Population

The **177 vessel calls** shown on this dashboard represent the detailed records available in `FactVesselPortCalls` for this analysis.

This value should not be interpreted as a universal count of all vessel calls represented elsewhere in the project. Other pages can use different facts, grains, filters, or call definitions.

For example, container-level analysis can associate many container-operation records with one vessel call, while the detailed vessel fact contains one call-level observation per source record.

Accordingly, the vessel-call KPI on this page must always be interpreted within the grain and scope of `FactVesselPortCalls`.

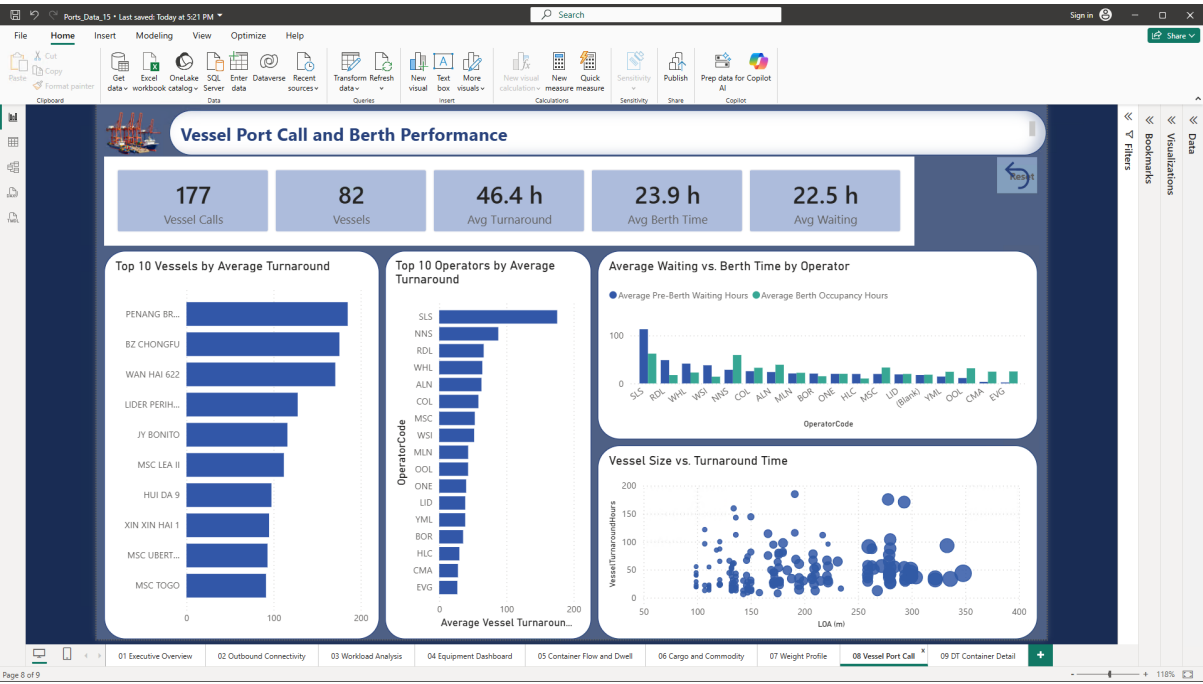


Figure 9.8.: Power BI Page 08, Vessel Port Call: Vessel Port Call and Berth Performance.

9.11. Page 09: DT Container Detail

Page identity

Report page: 09DTContainerDetail

Displayed title: Container Detail and Lifecycle

Primary focus: Record-level operational traceability

9.11.1. Purpose

The final dashboard page provides the detailed record-level view supporting the aggregated analytical pages.

Its purpose is not to introduce another summary KPI layer. Instead, it allows users to move from an aggregate finding to the individual container-operation records contributing to that result.

This page supports:

- operational verification;
- anomaly investigation;
- container-level follow-up;
- validation of dashboard selections; and
- traceability between aggregate analysis and source-level records.

9.11.2. Search and Filtering

Two prominent search controls allow users to locate records by:

- container number; and
- commodity.

These controls enable rapid investigation of individual containers or groups of containers associated with a specific commodity.

9.11.3. Container Detail Fields

The detailed table includes operational fields such as:

- container number,
- entry date and time,
- exit date and time,
- completed dwell days,
- source weight value,
- weight group,
- operator code,
- cargo type,
- commodity, and
- port-of-loading code.

Additional fields are available through horizontal navigation within the detail table.

9.11.4. Analytical Role

This page completes the dashboard hierarchy by providing an audit path from management-level indicators to individual operational records.

For example, a user can identify a high-dwell commodity on Page 9.8, apply the corresponding filter, and then inspect individual lifecycle records on the detailed page.

Similarly, operator, weight, cargo, or lifecycle anomalies identified elsewhere in the dashboard suite can be traced to the relevant operational rows.

ContainerNumber	InDate	OutDate	CompletedDwellDays	WeightValue	WeightGroup	OperatorCode	CargoType	Commodity	POLCode
TRHV5539177	1/1/2025 12:23:56 AM	1/18/2025 6:00:58 PM	17.73	28740	Extra Heavy	COE	General	Sugar	EGDAM
SZLV9533575	1/1/2025 12:25:39 AM	1/3/2025 9:27:38 PM	2.88	19380	Heavy	COE	Reefer	Shrimp	EGDAM
MSNV3393339	1/1/2025 12:30:18 AM	1/2/2025 8:56:45 PM	1.85	28460	Extra Heavy	MSC	General	Electronic Devices	EGDAM
ONEV1199333	1/1/2025 12:31:06 AM	1/4/2025 11:56:21 AM	3.48	28000	Extra Heavy	ONE	General	Marble and Granite	EGDAM
ONEV3313311	1/1/2025 12:33:16 AM	1/4/2025 1:13:23 PM	3.53	26230	Extra Heavy	ONE	General	Medicinal Herbs	EGDAM
ONEV3759373	1/1/2025 12:33:16 AM	1/4/2025 1:16:52 PM	3.53	26230	Extra Heavy	ONE	General	Medicinal Herbs	EGDAM
FSCV5777311	1/1/2025 12:33:41 AM	1/3/2025 7:34:08 PM	2.79	31000	Extra Heavy	MSC	Reefer	Orange	EGDAM
OTPV7519597	1/1/2025 12:35:13 AM	1/3/2025 7:33:50 PM	2.79	31060	Extra Heavy	MSC	Reefer	Frozen Fruits	EGDAM
TCNV5937151	1/1/2025 12:36:12 AM	1/4/2025 11:53:21 AM	3.47	28240	Extra Heavy	ONE	General	Clothes	EGDAM
TEMV9511515	1/1/2025 12:37:32 AM	1/3/2025 7:30:55 PM	2.79	31600	Extra Heavy	MSC	Reefer	Orange	EGDAM
MEDV5355939	1/1/2025 12:44:09 AM	1/2/2025 9:32:28 PM	1.87	29000	Extra Heavy	MSC	General	Accessories	EGDAM
MSDV2511597	1/1/2025 12:44:09 AM	1/2/2025 9:29:37 PM	1.86	29000	Extra Heavy	MSC	General	Accessories	EGDAM
OERV5191159	1/1/2025 12:45:59 AM	1/16/2025 11:27:28 PM	15.95	31960	Extra Heavy	COE	Reefer	Onions	EGDAM
MEDV3719959	1/1/2025 12:49:20 AM	1/2/2025 9:26:54 PM	1.86	29890	Extra Heavy	MSC	General	Dry Food	EGDAM
MEDV3799355	1/1/2025 12:49:20 AM	1/2/2025 9:29:36 PM	1.86	29890	Extra Heavy	MSC	General	Accessories	EGDAM
FBV1559931	1/1/2025 12:53:18 AM	1/2/2025 9:23:21 PM	1.85	25700	Extra Heavy	MSC	General	Ceramic and Porcelain	EGDAM
MEDV5137951	1/1/2025 12:53:18 AM	1/2/2025 9:26:53 PM	1.86	25700	Extra Heavy	MSC	General	Dangerous Goods	EGDAM
TEMV1737931	1/1/2025 12:56:42 AM	1/3/2025 6:28:42 PM	2.73	21740	Heavy	YML	General	Perfume Compound	EGDAM
MEDV5193737	1/1/2025 12:58:28 AM	1/2/2025 9:23:19 PM	1.85	27850	Extra Heavy	MSC	General	Dangerous Goods	EGDAM
MSMV1753971	1/1/2025 12:58:28 AM	1/2/2025 9:20:32 PM	1.85	27850	Extra Heavy	MSC	General	Accessories	EGDAM
MCLV5115351	1/1/2025 1:03:11 AM	1/4/2025 7:52:37 PM	3.78	8120	Light	BOR	General	Foodstuffs	EGDAM
GAOV7795197	1/1/2025 1:08:39 AM	1/4/2025 6:53:20 PM	3.74	32360	Extra Heavy	BOR	General	Accessories	EGDAM
KKPV9155991	1/1/2025 1:10:51 AM	1/4/2025 11:45:18 AM	3.44	28420	Extra Heavy	ONE	General	Accessories	EGDAM
TRHV5795371	1/1/2025 1:12:03 AM	1/4/2025 11:52:02 AM	3.44	28140	Extra Heavy	ONE	General	Paper Roll	EGDAM

Figure 9.9.: Power BI Page 09, DT Container Detail: Container Detail and Lifecycle.

9.12. Cross-Page Analytical Context

Although the nine dashboards are built from the same governed semantic model, they do not all operate at the same analytical grain or use identical measure contexts. Similar-looking KPIs must therefore be interpreted according to the page on which they appear.

Table 9.3 summarizes several important examples.

Table 9.3.: Cross-page analytical context

Analytical item	Dashboard context	Interpretation
Dashboard suite	Nine final pages	The Power BI report contains nine analytical pages covering executive performance through record-level detail.
Vessel calls	179 on Page 01; 191 on Page 03; 177 on Page 08	The values originate from different analytical contexts and grains and should not be treated as interchangeable counts.
Container flow	95,325 entries and 89,221 exits	These values describe lifecycle movements within the reporting context rather than current physical inventory.
Current inventory	19 on Page 05	Represents the latest records with a valid entry and no recorded exit under the implemented current-state logic.
TEU	Entry, exit, handled, and workload views appear on different pages	TEU measures must be interpreted according to their lifecycle and analytical definition rather than relying only on the word “TEU” in the visual label.
QC performance	Pages 03 and 04	Workload is distinguished from validated QC working-hour productivity to prevent repeated source hours from distorting interpretation.
Weight	Page 07	Weight values remain in the source unit because the physical unit has not been independently confirmed.
Vessel timing	Page 08	Turnaround, waiting, and berth occupancy are calculated at detailed vessel-port-call grain rather than container-operation grain.

Maintaining these distinctions is essential for reliable dashboard interpretation. The semantic model described in Chapter 8 provides the technical structure that enables each page to evaluate its measures at the appropriate grain.

9.13. Dashboard-to-Decision Analytical Flow

The nine-page suite supports a progressive analytical workflow rather than nine independent reports.

A typical management investigation can proceed as follows:

1. use Executive Overview to identify a major performance change;

2. use Outbound Connectivity, Workload Analysis, or Cargo and Commodity to determine where the change is concentrated;
3. use Equipment Dashboard, Container Flow and Dwell, Weight Profile, or Vessel Port Call to investigate the relevant operational dimension; and
4. use DT Container Detail to examine the underlying operational records.

For example, an increase in completed dwell identified on the Executive Overview can be explored by destination port, commodity, operator, container type, weight profile, or individual container lifecycle.

Similarly, a high equipment workload identified on the Workload Analysis page can be evaluated against validated QC productivity on the Equipment Dashboard before any operational conclusion is reached.

This progressive design strengthens analytical traceability and encourages users to move from summary indicators toward supporting operational detail.

9.14. Chapter Conclusion

The Power BI dashboard suite converts the governed semantic model into a structured decision-support environment for container-terminal operations.

The nine dashboards address complementary management and operational questions:

- Executive Overview summarizes overall terminal performance;
- Outbound Connectivity evaluates the port network and trade lanes;
- Workload Analysis identifies workload concentration;
- Equipment Dashboard examines quay-crane workload and validated productivity;
- Container Flow and Dwell evaluates container movement, current state, and dwell;
- Cargo and Commodity analyzes cargo composition and commodity performance;
- Weight Profile evaluates weight distribution and data readiness;
- Vessel Port Call analyzes vessel turnaround and berth performance; and
- DT Container Detail provides record-level operational traceability.

The value of the dashboard suite lies not only in presenting individual KPIs, but in connecting throughput, time efficiency, workload, cargo characteristics, connectivity, vessel performance, and data quality within one analytical framework.

The dashboard design also preserves the modeling principles established in Chapter 8: measures are interpreted at their appropriate grain, current-state values are distinguished from reporting-period flow, validated QC data is separated from raw repeated working-hour values, loading and discharge port roles remain distinct, and unresolved source definitions are not converted into unsupported business assumptions.

Together, the semantic model and the nine-page dashboard suite provide the final business intelligence environment through which the project's operational findings can be explored, validated, and communicated.

10. Results, Findings, and Operational Insights

10.1. Results Overview

The completed analytical project provides a consolidated view of container-terminal performance across throughput, container flow, dwell time, port connectivity, equipment workload, cargo composition, weight characteristics, vessel-port-call performance, and record-level traceability.

The results should be interpreted as a connected operational picture rather than as isolated KPI movements. In particular, higher throughput does not automatically indicate higher efficiency, high equipment workload does not automatically indicate higher productivity, and a large reporting-period difference between entries and exits does not represent current terminal inventory.

The final analysis highlights several recurring themes:

- container and TEU entry activity increased in the Q1 comparison;
- completed dwell also increased, indicating that throughput growth was accompanied by weaker time performance;
- container flow, current inventory, and completed dwell represent different operational concepts and must be monitored separately;
- outbound traffic reaches a broad international network but remains concentrated in a smaller group of ports and trade lanes;
- workload is concentrated among selected operators, quay cranes, vessel calls, cargo categories, and commodities;
- QC productivity requires validated working-hour data rather than raw repeated source values;
- high-volume commodities are not necessarily the commodities with the highest dwell;
- pre-berth waiting represents a substantial proportion of vessel turnaround;
- vessel size alone does not visually explain turnaround variation; and
- detailed container records provide a traceable path from aggregate indicators to individual operational observations.

Table 10.1 summarizes the principal findings associated with the nine final Power BI pages.

Table 10.1.: Principal analytical findings across the final Power BI dashboards

Page	Dashboard	Principal analytical finding
01	Executive Overview	Q1 2026 recorded higher container entries and TEU entries than Q1 2025, while average completed dwell also increased.
02	Outbound Connectivity	The terminal serves a broad network of regions, countries, ports, and trade lanes, but outbound TEU is concentrated in a smaller group of routes and destinations.
03	Workload Analysis	Operational workload is concentrated across selected vessel calls, operators, equipment units, and container profiles.
04	Equipment Dashboard	QC10 carries the largest visible workload, while validation-aware QC measures demonstrate that workload and productivity must be evaluated separately.
05	Container Flow and Dwell	Entry/exit flow, current active inventory, and completed dwell answer different operational questions and should not be interpreted as interchangeable measures.
06	Cargo and Commodity	The cargo and commodities generating the greatest TEU activity are not necessarily those associated with the highest completed dwell.
07	Weight Profile	Weight data provide useful workload and distribution patterns, but interpretation remains limited to the unconfirmed source unit.
08	Vessel Port Call	Pre-berth waiting is almost equal to berth occupancy within the detailed vessel-call dataset, making arrival-to-berth delay an important area for further operational investigation.
09	DT Container Detail	Aggregate analytical results can be traced to searchable container-level operational records.

10.2. Q1 Throughput and Efficiency Performance

The most important high-level result is the simultaneous increase in terminal entry activity and completed dwell between Q1 2025 and Q1 2026.

The Executive Overview reports that:

- TEU entered increased from approximately 82.85K in Q1 2025 to approximately 87K in Q1 2026, representing an increase of **4.5%**;
- container entries increased from approximately 47K to approximately 49K, representing an increase of **3.1%**; and
- average completed dwell increased from approximately 6.4 days to 6.7 days, representing an increase of **4.4%**.

These movements provide an important management interpretation. The terminal handled more incoming activity in Q1 2026, but the increase in throughput was not accompanied by an improvement in container time efficiency. Instead, average dwell increased at the same time.

This does not imply that additional volume directly caused longer dwell. It does, however, show that throughput growth and dwell performance moved in opposite operational directions: volume increased while the time required for completed container lifecycles also increased.

The Q1 result therefore suggests that future performance monitoring should not evaluate throughput independently. A more balanced terminal-performance view should consider at least:

- container and TEU volume;
- dwell distribution;
- equipment workload and validated productivity;
- cargo and commodity mix;
- vessel and berth performance; and
- current inventory conditions.

The management implication is that growth should be evaluated together with the operational resources and process conditions required to sustain that growth efficiently.

10.3. Container Flow and Current Inventory

The container-flow results provide one of the most important interpretation lessons in the project.

Across the displayed reporting context, the final dashboard reports:

- **95,325 container entries;**
- **89,221 container exits;**
- **168,345 TEU entered;** and
- **158,090 TEU exited.**

The difference between recorded entries and exits is:

$$95,325 - 89,221 = 6,104.$$

This value represents the difference between movement counts within the reporting context. It does **not** mean that 6,104 containers remain physically inside the terminal.

The current-state logic separately identifies only **19 active records** with a valid entry and no recorded exit in the latest available extract.

The distinction can be summarized as follows:

Table 10.3.: Difference between flow and current-state measures

Measure	Operational meaning
Container entries	Records entering during the applicable reporting and filter context.
Container exits	Records exiting during the applicable reporting and filter context.
Net container flow	Arithmetic difference between entries and exits in the selected reporting window.
Current inventory	Latest known records with a valid entry and no recorded exit.

This separation prevents a common analytical error in operational reporting. Period flow measures describe activity; inventory measures describe state. They cannot be substituted for one another without an explicit historical inventory model.

The result is particularly important for management reporting because a large cumulative difference between entries and exits may appear to indicate yard accumulation even when the latest open-record population is small. The final model therefore reports these measures separately.

10.4. Completed Dwell Performance

Completed dwell is one of the most important efficiency indicators in the project because it measures the elapsed time between valid container entry and exit timestamps.

The final dashboard reports:

- **4 days median completed dwell;**
- **6.6 days average completed dwell;** and
- **13.2 days P90 completed dwell.**

These three statistics reveal more about terminal behavior than the average alone.

The median of 4 days indicates that half of the valid completed records have dwell values at or below approximately four days. The average is considerably higher at 6.6 days, which indicates that longer-stay records are pulling the mean upward.

The P90 value of 13.2 days indicates that approximately 90% of completed records fall at or below that threshold, while the remaining high-dwell tail extends beyond it.

The relationship:

$$\text{Median Dwell} < \text{Average Dwell} < \text{P90 Dwell}$$

is consistent with a right-skewed dwell distribution.

Operationally, this means that the terminal's dwell challenge should not be treated only as an average-performance issue. A smaller population of long-stay containers can materially increase the average even when the majority of containers leave substantially sooner.

This supports an exception-based monitoring strategy. Rather than focusing exclusively on reducing the overall average, management may obtain greater operational value by identifying:

- commodities associated with unusually high dwell;
- operators with persistent long-stay patterns;
- destination ports associated with higher completed dwell;
- specific container types or transport profiles;
- records exceeding defined age thresholds; and
- individual long-dwell containers requiring operational follow-up.

The detail dashboard provides the final record-level path for investigating these exceptions.

10.5. Outbound Connectivity and Trade-Lane Concentration

The connectivity analysis demonstrates substantial geographic reach within the available operational records.

The final dashboard reports:

- **15 destination regions;**
- **21 destination countries;**
- **50 discharge ports;**
- **118 active trade lanes;** and
- approximately **158.09K outbound TEU.**

These indicators show that the terminal participates in a geographically diverse trade network. However, the route-level charts demonstrate that activity is not evenly distributed across all available connections.

A smaller set of POL–POD combinations accounts for a substantial proportion of outbound TEU. Leading visible routes include combinations involving Antwerp, Novorossiysk, Damietta, Piraeus, Hamburg, Rotterdam, and Jeddah.

The discharge-port analysis similarly demonstrates concentration. Piraeus and Novorossiysk appear among the leading displayed discharge ports, with Antwerp, Hamburg, Rotterdam, Ashdod, Mersin, and Salerno also contributing substantial visible activity.

This concentration has several operational implications:

- high-volume trade lanes may require greater planning attention because disruptions affecting a major lane can influence a substantial portion of terminal activity;
- equipment and yard planning may benefit from understanding the cargo profiles associated with the highest-volume routes;
- port and route concentration can be combined with dwell analysis to identify high-volume routes that also experience time inefficiency; and
- network-diversity measures should be interpreted separately from volume concentration.

A particularly important result is that the ports with the highest outbound volume are not necessarily those with the highest average completed dwell.

The high-dwell destination ranking includes ports such as Trieste, Hamburg, Nantong, Shanghai, and Shekou. This indicates that route volume and dwell pressure represent different operational dimensions.

A low-volume destination with unusually high dwell may require process investigation even if it does not appear among the highest-volume trade lanes.

Conversely, a high-volume port may be operationally important primarily because of capacity and workload rather than dwell performance.

The connectivity analysis therefore supports two distinct management questions:

1. Where is the greatest amount of traffic concentrated?
2. Where are the largest time-efficiency exceptions occurring?

10.6. Workload Concentration Across the Terminal

The workload analysis indicates that terminal activity is not distributed evenly across equipment, vessel calls, or operators.

The Workload Analysis dashboard displays approximately:

- **191 vessel calls;**
- **68K unique containers;**
- **96K container records;** and
- approximately **338.8K TEU activity** within its displayed measure context.

The page reveals concentration at several operational levels.

10.6.1. Equipment Concentration

QC10 leads the visible equipment workload ranking by container records. QC01, QC08, QC02, and QC07 represent additional major workload contributors.

This indicates that a relatively small number of equipment units process a substantial portion of the container-operation workload.

Such concentration is not inherently negative. A crane may handle more activity because of berth assignment, service pattern, equipment availability, vessel characteristics, or operational planning.

However, persistent concentration increases the importance of understanding:

- equipment availability;
- maintenance exposure;
- workload balancing;
- validated productivity;
- vessel assignment; and

- resilience if a heavily utilized crane becomes unavailable.

10.6.2. Operator Concentration

MSC appears as the leading displayed operator by TEU, followed by other major operators including COE and ONE.

The concentration of workload among a relatively small group of operators means that changes in their service frequency, cargo mix, vessel size, or scheduling patterns could materially affect terminal-wide performance.

Operator-level analysis is therefore useful for identifying where service coordination, dwell monitoring, equipment planning, and vessel-call review may have the greatest operational impact.

10.6.3. Vessel-Call Concentration

The highest-TEU vessel-call rankings show that individual calls can contribute substantially different amounts of workload.

This reinforces the importance of analyzing vessel calls using more than call counts alone. Two calls can each count as one vessel visit while generating very different container volumes, TEU workloads, equipment requirements, and berth-service demands.

10.7. Quay-Crane Workload and Validated Productivity

The equipment results provide one of the strongest examples of why analytical grain and validation are essential.

The Equipment Dashboard reports approximately:

- **95.3K handled containers;**
- **326.4K handled TEU;**
- **9 active quay cranes;** and
- approximately **51.9K validated QC handled records** within the displayed validation context.

QC10 carries the largest visible raw workload:

- approximately **24.0K container records;** and
- approximately **66K TEU.**

QC01 carries a smaller raw workload but leads the visible valid-QC-handled-record ranking.

This difference demonstrates that workload and productivity should not be treated as the same concept.

A high-workload crane can be heavily utilized without necessarily having the best productivity rate. Conversely, a crane with lower total workload may perform efficiently within its assigned operating periods.

The project therefore evaluates equipment through several complementary measures:

- container workload;
- TEU workload;
- validated QC handled records;
- validated QC working hours; and
- TEU per validated QC working hour.

The technical validation process groups source QC hours at vessel-call–equipment grain because the same hour value may be repeated across multiple container rows.

The project identified a substantial number of QC groups requiring validation. The final analytical baseline contains:

- **1,087 QC groups;**

- **838 validated groups;**
- **249 inconsistent groups;** and
- a validation rate of approximately **77.1%**.

The inconsistent-group share is therefore:

$$\frac{249}{1,087} \times 100 \approx 22.9\%.$$

This is analytically significant. Approximately one in every five QC groups contains working-hour information that should not automatically be accepted as a valid productivity denominator.

The result has two implications.

First, productivity reporting should continue to exclude unresolved inconsistent groups rather than silently averaging conflicting values.

Second, the quality of QC working-hour capture itself represents an operational data-governance opportunity. Improving the source process would increase confidence in equipment-performance reporting and reduce the proportion of workload excluded from validated productivity analysis.

10.8. Cargo and Commodity Results

The Cargo and Commodity dashboard reports approximately:

- **326K TEU;**
- **68K unique containers;**
- **9 cargo types;** and
- **52 commodities.**

The results reveal two distinct analytical patterns: **volume concentration** and **dwell concentration**.

General Dry and Refrigerated containers dominate the visible container-type TEU distribution. At the commodity level, Orange and Wood appear among the leading TEU contributors, with Accessories, Clothes, Vaccines, and Electrical Appliances also represented among the principal visible categories.

These results identify the cargo streams most important for workload and capacity planning.

However, the commodity dwell ranking is different. Commodities appearing among the highest average-dwell categories include:

- Reinforced Iron,
- Sugar,
- Potatoes,
- Sensitive Tools,
- Wood,
- Shrimp,
- Vaccines, and
- Special Chemicals.

The difference between these two rankings is operationally important.

High-volume commodities require planning for:

- yard space;
- equipment capacity;
- handling demand;
- reefer requirements where applicable; and
- expected throughput.

High-dwell commodities require a different investigation focused on:

- extended container residence;
- documentation or processing characteristics;

- cargo-specific requirements;
- transport availability;
- service scheduling; and
- individual exception records.

The analysis therefore demonstrates why commodity management should not rely on a single ranking. Volume and dwell represent different operational risks.

10.9. Weight Profile and Data Readiness

The Weight Profile dashboard reports:

- approximately **1.71 billion** total handled source weight;
- approximately **21.50K** median source weight;
- approximately **95.33K valid positive-weight records**; and
- approximately **17.94K** average source weight.

The result demonstrates strong apparent weight-data coverage within the analyzed records.

The source-defined Extra Heavy and Extra Light groups represent the largest visible weight-group categories, while COE, ONE, MSC, and BOR appear among the leading operators by total handled source weight.

Weight analysis provides useful information about workload composition, but interpretation must remain within an important limitation: the physical unit represented by the source WGT field has not been formally verified.

Accordingly, the analysis supports:

- relative comparison;
- operator workload ranking;
- cargo and container-type profiling;
- distribution analysis; and
- data-completeness monitoring.

It does not support conversion into tonnes, kilograms, verified gross mass, or another physical measure without source-system confirmation.

The *Weight vs. Dwell by Operator* scatter also provides an exploratory view of whether operators with higher average weight tend to exhibit different dwell behavior.

The visible points do not establish a causal relationship. Weight may be associated with cargo type, container type, operator mix, destination, or handling pattern. Further multivariable analysis would be required before drawing conclusions about the independent effect of weight on dwell.

10.10. Vessel-Port-Call and Berth Performance

The vessel-port-call analysis provides a detailed view of marine-side operational time.

The final dashboard reports:

- **177 detailed vessel calls**;
- **82 distinct vessels**;
- **46.4 hours average turnaround**;
- **22.5 hours average pre-berth waiting**; and
- **23.9 hours average berth occupancy**.

Table 10.5 summarizes these results.

Table 10.5.: Detailed vessel-call timing indicators

Metric	Displayed value
Detailed vessel calls	177
Distinct vessels	82
Average turnaround	46.4 hours
Average pre-berth waiting	22.5 hours
Average berth occupancy	23.9 hours
Approximate pre-berth share	48.5%

The average pre-berth share of turnaround can be approximated as:

$$\frac{22.5}{46.4} \times 100 \approx 48.5\%.$$

This is one of the most operationally significant findings in the project.

Nearly half of the average tracked turnaround occurs before the vessel is berthed. Average pre-berth waiting is also almost equal to average berth occupancy.

This indicates that vessel service performance cannot be evaluated only by what happens after a ship reaches the berth. The arrival-to-berth period represents a substantial component of the total port-call cycle.

Potential areas for investigation include:

- berth availability;
- vessel-arrival synchronization;
- overlapping call schedules;
- berth sequencing;
- resource readiness;
- navigational or tidal constraints;
- operator service patterns; and
- coordination between marine and terminal operations.

These represent investigation areas, not established causes.

10.10.1. Operator Variation

The operator-level charts show meaningful variation in turnaround, waiting, and berth occupancy across the selected vessel calls.

This variation suggests that not all vessel calls experience the same operational conditions.

Differences could potentially reflect combinations of:

- service schedules;
- vessel characteristics;
- cargo workload;
- berth assignment;
- arrival timing;
- equipment allocation; and
- call sequencing.

The dashboard is valuable because it identifies where variation exists and which operators or calls deserve closer review. It does not establish that operator identity itself is the cause of longer turnaround.

10.10.2. Vessel Size and Turnaround

The LOA-versus-turnaround scatter does not show a single tight visual relationship.

Larger vessels do not consistently exhibit proportionally longer turnaround within the displayed observations.

This suggests that vessel length alone is unlikely to explain the observed turnaround variation.

A more complete vessel-performance model would need to consider additional factors such as:

- containers or TEU handled per call;
- number of cranes assigned;
- validated crane hours;
- vessel type;
- cargo composition;
- berth assignment;
- arrival timing; and
- other operational conditions.

The vessel-size scatter should therefore be viewed as exploratory evidence rather than a predictive or causal relationship.

10.10.3. Scope of the 177 Vessel Calls

The 177-call KPI belongs specifically to `FactVesselPortCalls` and its detailed vessel-call records.

It should not be interpreted as the universal vessel-call count across the full project.

Other dashboards use different analytical structures and display different vessel-call values because their measures operate at different grains, periods, or filter contexts.

This is an important example of why semantic-model grain must be considered when comparing values across analytical pages.

10.11. Container-Level Traceability

The final detail dashboard provides the record-level foundation behind the aggregate analysis.

Users can search by container number or commodity and inspect operational attributes including:

- entry timestamp;
- exit timestamp;
- completed dwell;
- source weight;
- weight group;
- operator;
- cargo type;
- commodity; and
- loading-port information.

This capability is important because aggregate dashboards identify patterns, but operational follow-up often requires an individual record.

For example:

- a high-dwell commodity can be traced to the individual containers contributing to that average;
- an unusual operator pattern can be reviewed through the underlying container lifecycles;
- an unexpected weight value can be inspected at record level; and
- an aggregate port or cargo result can be verified against its contributing operations.

The detail page therefore provides an important quality-control function in addition to its operational reporting role. It closes the analytical loop between KPI, visual pattern, filtered result, and individual source-derived record.

10.12. Supporting Broader SQL Findings

The broader SQL exploration provides additional context across the available 2025–2026 analytical records.

The SQL analytical layer reports:

- **95,870 operational records;**
- **67,738 distinct containers;**
- **81 vessels;** and
- **190 voyages.**

The available records are distributed as:

- **52,096 operations** associated with 2025; and
- **43,774 operations** associated with 2026.

These broader counts should not automatically be interpreted as complete year-over-year terminal performance because the analytical records available for the two years may not represent identical reporting coverage.

The SQL analysis also identified several broader patterns:

- DELPHINUS appears as the most frequent vessel in the available SQL analysis with nine recorded visits.
- Full-container share remains relatively stable at approximately 70–72% across the available yearly records.
- Wood is a major import commodity, while Orange is a major export commodity in the broader operational analysis.
- Damietta represents the dominant discharge location within the analyzed port records.
- Piraeus, Novorossiysk, Antwerp, and Hamburg appear among the principal external port relationships.
- Egypt represents the largest discharge-country share in the available records.

These findings complement the Power BI dashboards by providing additional database-level exploration. They should be interpreted within the scope and grain of the SQL analytical layer rather than used to replace page-specific dashboard measures.

10.13. Integrated Operational Findings

When the results are considered together, several broader operational conclusions emerge.

10.13.1. Throughput Growth Was Accompanied by Time Pressure

The Q1 comparison shows increased container and TEU entry activity together with increased completed dwell.

This suggests that terminal performance should be evaluated using both volume and time measures. Increased throughput represents greater handled activity, but sustained operational improvement requires the terminal to maintain or improve service time while accommodating that volume.

10.13.2. Operational Pressure Is Concentrated

Equipment workload, operator TEU, trade lanes, cargo types, commodities, and vessel calls all show concentration.

This means that operational improvement efforts do not necessarily need to target every category equally. Prioritization can focus on the relatively small number of high-impact entities contributing the greatest volume, workload, dwell, or waiting time.

10.13.3. Time Inefficiency Appears at Multiple Operational Stages

The project identifies time-related opportunities in both major lifecycle domains:

- container dwell within the terminal; and
- vessel waiting before berthing.

These are different operational processes and should be investigated independently.

Long container dwell concerns the duration of container residence between entry and exit.

Pre-berth waiting concerns the duration between vessel arrival and berthing.

The fact that both areas represent meaningful portions of the operational cycle suggests that performance improvement should consider both landside/container processes and marine/berth processes.

10.13.4. Exception Monitoring Is More Informative Than Average Alone

Several results demonstrate the limitations of averages.

For container dwell, the median is 4 days while the P90 is 13.2 days.

For vessel turnaround, individual vessels and operators vary materially around the 46.4-hour average.

For equipment, raw workload and validated productivity produce different rankings.

These patterns support the use of:

- percentile measures;
- top-N exception rankings;
- long-stay buckets;
- operator and vessel comparisons;
- QC-validation status; and
- record-level drill paths.

10.13.5. Data Quality Directly Affects KPI Reliability

The QC analysis demonstrates that data quality is not only a technical concern. It changes which records can legitimately contribute to a management KPI.

The same principle applies to:

- unmapped TEU factors;
- invalid date sequences;
- unresolved weight units;
- unknown transport classifications;
- unmatched dimension values; and
- incomplete lifecycle records.

The project therefore treats validation as part of the analytical result rather than as an invisible preprocessing activity.

10.14. Interpretation Boundaries

The project identifies patterns, differences, concentrations, and associations within the available operational data. These findings should not automatically be converted into causal conclusions.

For example, the analysis can establish that:

- dwell increased;
- crane workload is concentrated;
- some operators exhibit higher vessel turnaround;
- some destinations have higher average dwell;
- pre-berth waiting is substantial; and
- cargo and trade mix changed.

The available data do not independently establish that these patterns were caused by:

- geopolitical events;
- customs performance;
- maintenance activity;
- yard congestion;
- berth shortages;
- shipping-line disruption; or
- any other specific external factor.

Such explanations remain hypotheses requiring additional operational or external evidence.

The results should therefore be used to determine **where further investigation is warranted**, rather than to assign unsupported causes.

10.15. Final Analytical Conclusions

The completed analysis provides a coherent picture of terminal performance across the available operational data.

The strongest conclusions are:

1. **Q1 throughput increased.** Container entries increased by approximately 3.1%, while TEU entered increased by approximately 4.5%.
2. **Time efficiency did not improve at the same rate.** Average completed dwell increased by approximately 4.4%, indicating that higher throughput was accompanied by longer average container residence.
3. **Long-stay containers represent an important exception population.** Median dwell is 4 days, average dwell is 6.6 days, and P90 is 13.2 days.
4. **Flow and inventory must remain analytically separate.** The reporting-period entry/exit difference of 6,104 is not current inventory; the latest active-state measure reports 19 records.
5. **Outbound activity is geographically broad but operationally concentrated.** The network spans 15 regions, 21 countries, 50 discharge ports, and 118 trade lanes, while a smaller set of routes accounts for a large share of TEU activity.
6. **Terminal workload is concentrated.** Selected operators, cranes, vessel calls, cargo categories, and commodities account for substantial portions of activity.
7. **QC10 carries the greatest visible raw crane workload, but workload and productivity are different measures.** Validated QC-hour logic is required before equipment-efficiency conclusions can be made.
8. **QC working-hour quality remains an important governance issue.** Approximately 22.9% of the analyzed QC groups are inconsistent and should not automatically contribute to validated productivity

calculations.

9. **Cargo volume and cargo dwell require different management priorities.** The commodities generating the greatest TEU are not the same as those exhibiting the highest average dwell.
10. **Weight data support relative operational profiling, but the source unit remains unresolved.** No physical-unit conversion should be applied without business confirmation.
11. **Marine-side waiting represents a major part of vessel turnaround.** Average pre-berth waiting is 22.5 hours compared with 23.9 hours of berth occupancy, meaning approximately 48.5% of average turnaround occurs before berthing.
12. **Vessel size alone does not explain turnaround variation.** The LOA-versus-turnaround analysis does not show a single strong visual relationship, indicating that additional operational variables should be considered.
13. **The analytical solution provides traceability from management KPI to individual records.** The final detail dashboard allows aggregate findings to be investigated at container-operation level.

Overall, the results demonstrate that terminal performance cannot be represented adequately by throughput alone. A complete operational view requires simultaneous consideration of **volume, time, workload, connectivity, cargo composition, vessel performance, and data reliability**.

The strongest opportunities for continued management attention are therefore not simply to increase handled volume, but to understand the operational conditions associated with increasing dwell, long-stay container exceptions, pre-berth waiting, concentrated equipment workload, and inconsistent QC working-hour records.

These findings provide the analytical foundation for the management recommendations and future improvement priorities presented in the following chapter.

11. Conclusion

This project demonstrates the complete development of a governed data analytics and business intelligence solution for container-terminal operations. Beginning with fragmented operational extracts, the project progressed through data understanding, profiling, cleaning, transformation, dimensional modeling, SQL exploration, quality assurance, semantic modeling, DAX development, validation, visualization, and final management-oriented reporting.

The resulting analytical environment goes beyond the production of individual charts or KPIs. Its principal contribution is the creation of a structured and traceable analytical framework in which operational measures are evaluated according to clearly defined business logic, appropriate analytical grain, controlled relationships, and explicit data-quality rules. This foundation improves confidence in the reported results and reduces the risk of drawing conclusions from duplicated values, incomplete lifecycle records, inconsistent classifications, or measures evaluated at the wrong level of detail.

A major technical achievement of the project is the separation of different operational processes according to their natural analytical grains. Container activity is represented through `FactContainerOperations`, validated quay-crane performance through `FactQCPerformance`, and detailed vessel timing through `FactVesselPortCalls`. Supporting dimensions provide consistent analytical context for vessels, vessel calls, operators, container specifications, cargo, equipment, loading and discharge ports, transport characteristics, damage conditions, and calendar dates. This structure allows multiple operational perspectives to coexist within the same Power BI solution without forcing fundamentally different processes into one analytical table.

The project also establishes several important governance principles. TEU conversion is controlled through an explicit container-length mapping; unsupported lengths remain identifiable rather than receiving assumed values. Entry and exit dates are treated as separate lifecycle roles. Completed dwell is calculated only for valid date sequences, while incomplete and invalid records remain distinguishable. Current terminal inventory is separated from reporting-period entry and exit flow. Source weight values are retained without assigning an unverified physical unit. QC working hours are validated at vessel-call-equipment grain before productivity is calculated. Vessel turnaround is decomposed into pre-berth waiting and berth occupancy and validated through timestamp-sequence and reconciliation controls.

These controls are important because technically plausible calculations can still produce misleading business conclusions when analytical grain or source quality is ignored. The project therefore treats validation as part of the analytical solution itself rather than as an isolated preprocessing activity.

The final nine-page Power BI dashboard suite converts this governed model into a progressive decision-support environment. The dashboard sequence moves from high-level management performance to increasingly detailed operational investigation:

1. **01 Executive Overview** summarizes throughput, dwell, vessel-call activity, and operator performance.
2. **02 Outbound Connectivity** examines destination geography, discharge ports, trade lanes, outbound TEU, and route-level dwell.
3. **03 Workload Analysis** identifies workload concentration across vessels, operators, equipment, and container characteristics.
4. **04 Equipment Dashboard** separates equipment workload from validated quay-crane productivity.

5. 05 Container Flow and Dwell distinguishes entry and exit flow, current inventory, and completed-dwell performance.
6. 06 Cargo and Commodity evaluates cargo composition, commodity volume, and commodity-linked dwell.
7. 07 Weight Profile examines weight-data completeness, distribution, and workload using the available source values.
8. 08 Vessel Port Call evaluates vessel turnaround, pre-berth waiting, berth occupancy, operator variation, and vessel characteristics.
9. 09 DT Container Detail provides record-level traceability for operational review and validation.

Together, these pages establish a clear analytical path from management-level indicators to operational detail. A performance change identified at the executive level can be investigated through connectivity, workload, equipment, cargo, dwell, weight, or vessel analysis and ultimately traced to individual container-operation records.

The final analytical findings present a balanced view of terminal performance. Q1 2026 recorded approximately **3.1% more container entries** and **4.5% more TEU entries** than Q1 2025, demonstrating increased handled activity. At the same time, average completed dwell increased by approximately **4.4%**. The project therefore shows that higher throughput did not automatically correspond to improved time efficiency.

The completed-dwell distribution provides additional operational context. A median of approximately **4 days**, an average of **6.6 days**, and a P90 of **13.2 days** indicate that long-stay records form an important exception population. This finding supports a management approach that combines overall dwell monitoring with targeted investigation of high-dwell commodities, operators, destinations, and individual container records.

The connectivity analysis demonstrates that the terminal operates across a broad network consisting of **15 destination regions**, **21 destination countries**, **50 discharge ports**, and **118 active trade lanes**. Nevertheless, activity is concentrated within a smaller number of routes and ports. The analysis also demonstrates that high-volume destinations are not necessarily those with the highest average dwell, reinforcing the need to distinguish volume management from delay-exception management.

Equipment analysis produces a similar distinction between workload and performance. QC10 carries the greatest visible container and TEU workload, but validated QC analysis produces a different productivity perspective. Approximately **22.9%** of the analyzed QC groups contain inconsistent working-hour information and therefore require exclusion or further validation before being used as productivity denominators. This is both an analytical finding and a data-governance issue: improving QC-hour recording would directly strengthen future equipment-performance reporting.

Cargo and commodity analysis further shows that the categories generating the greatest throughput are not necessarily those generating the greatest dwell pressure. This distinction allows capacity planning and dwell-reduction efforts to target different operational populations rather than assuming that the largest cargo streams automatically represent the greatest time-related concern.

The detailed vessel-call analysis identifies another important operational opportunity. Across the **177 detailed vessel calls** represented in FactVesselPortCalls, average turnaround is approximately **46.4 hours**, consisting of approximately **22.5 hours of pre-berth waiting** and **23.9 hours of berth occupancy**. Pre-berth waiting therefore accounts for approximately **48.5%** of average tracked turnaround.

This finding makes the arrival-to-berth period a significant area for further operational investigation. The dashboard does not establish whether the observed waiting originates from berth availability, vessel sequencing, schedule variation, resource readiness, navigational conditions, or another operational factor. Instead, it identifies where additional operational evidence and management attention may provide value.

The vessel-size analysis also demonstrates the importance of avoiding simplistic explanations. The

relationship between Length Overall and vessel turnaround does not form a single tight visual pattern. Vessel size alone therefore does not appear sufficient to explain turnaround variation. Cargo workload, equipment allocation, berth assignment, vessel service pattern, operating conditions, and other factors would need to be evaluated before stronger conclusions could be established.

Across the project, these findings reinforce a central analytical principle: **volume, efficiency, workload, inventory, connectivity, and data quality must be evaluated together**. Throughput indicates how much activity occurred, but it does not independently explain how efficiently that activity was handled. Average measures summarize performance, but they can conceal important exception populations. High workload identifies concentration, but it does not automatically establish productivity. Correlations and visual patterns identify areas for investigation, but they do not by themselves establish causality.

The project therefore concludes with a disciplined operational-improvement cycle:

Measure → Validate → Investigate → Act → Monitor

First, operational performance is measured using governed KPIs. Second, the underlying data and analytical grain are validated. Third, unusual patterns, concentrations, or exceptions are investigated using the appropriate dashboard and record-level detail. Operational actions can then be based on verified evidence, after which the same analytical framework can be used to monitor whether performance improves.

This approach provides a more sustainable foundation for terminal analytics than isolated reporting because the model, validation controls, and dashboards can continue to support future data refreshes and additional operational questions.

Final Project Outcome

The project transformed fragmented container-terminal operational records into a structured, governed, and traceable business intelligence environment. The completed solution integrates data preparation, SQL analysis, quality assurance, dimensional and semantic modeling, DAX measures, validation controls, and nine coordinated Power BI dashboards. The resulting framework supports both management-level monitoring and detailed operational investigation while preserving clear analytical boundaries around data grain, lifecycle logic, source quality, and unsupported causal interpretation.

Appendix A.

Data Dictionary and Field Mapping

The project data dictionary contains 75 documented field-definition rows covering the broader operational data landscape associated with the container-terminal environment. These definitions extend beyond the 33 fields implemented directly in the analytical model and provide additional business context for vessel operations, container characteristics, cargo conditions, yard and logistics activities, and route-related documentation.

A.1. Domain Coverage

For documentation and governance purposes, the 75 field definitions are organized into seven operational domains. Table A.1 summarizes the number of documented fields associated with each domain.

Table A.1.: Data dictionary coverage by operational domain

Domain	Field rows
1. Vessel & Voyage Data	12
2. Vessel Timestamps & Dimensions	13
3. Container Characteristics	11
4. Weights, Safety & Cargo Conditions	8
5. Stowage & Planning (Vessel Locations)	6
6. Gate, Yard & Logistics	12
7. Documentation & Route Verification	13

The distribution demonstrates that the data dictionary was designed to document a broad operational context rather than only the fields directly used in the implemented business intelligence model. Vessel and voyage information, operational timestamps and dimensions, container characteristics, cargo and safety attributes, stowage information, logistics processes, and route-verification fields are all represented.

This broader field coverage supports data governance, business interpretation, future model extension, and traceability between operational terminology and analytical implementation. Not every documented field is necessarily present in, or required by, the current analytical model; the dictionary therefore serves both as a reference for the implemented project and as a foundation for potential future development.

A.2. Representative Data Dictionary Entries

Table [A.2](#) presents a representative subset of the project data dictionary. These entries illustrate the broader operational information documented beyond the 33 fields implemented directly in the analytical model. The selected rows emphasize vessel-call identification, voyage references, arrival and departure timestamps, turnaround measures, and vessel physical characteristics.

The descriptions are written to clarify the operational meaning of each field and support consistent interpretation across data preparation, analysis, modeling, and validation activities.

Table A.2.: Representative entries from the project data dictionary

Column	Type	Source sheet	Description
Calling Seq.	Integer	Vessel Turnaround Time & Data Analysis	Calling Sequence: sequential identifier associated with a vessel visit or port call and used to distinguish individual call records.
Calling Year	Integer	Vessel Turnaround Time & Data Analysis	Calling Year: calendar year associated with the corresponding vessel visit or port call.
Vessel Code	VARCHAR / Code	Vessel Turnaround Time & Data Analysis	Vessel Code: operational alphanumeric identifier assigned to a vessel and used to support vessel-level identification and matching.
Vessel Name	VARCHAR	Vessel Turnaround Time & Data Analysis	Vessel Name: registered or operational name of the vessel recorded in the source data.
IMO Code	VARCHAR	Vessel Turnaround Time & Data Analysis	IMO Code: permanent vessel identification number assigned under the International Maritime Organization vessel-identification system.
MMSI	VARCHAR	Vessel Turnaround Time & Data Analysis	Maritime Mobile Service Identity: nine-digit identifier used in maritime communication and vessel-tracking systems.
In Voyage	VARCHAR	Vessel Turnaround Time & Data Analysis	Inbound Voyage Number: voyage reference associated with the vessel's inbound movement to the terminal.
Out Voyage	VARCHAR	Vessel Turnaround Time & Data Analysis	Outbound Voyage Number: voyage reference associated with the vessel's departure from the terminal.
User Voyage	VARCHAR	Data Analysis	User Voyage: internal operational voyage reference used to associate terminal records with a particular vessel voyage or call context.
User Voy	VARCHAR	Vessel Turnaround Time	User Voyage shorthand field used within the Vessel Turnaround Time dataset to represent the operational voyage reference.
2nd Vessel	VARCHAR	Data Analysis	Second Vessel: vessel identifier used in transshipment-related records to represent the subsequent vessel associated with the container movement.
2nd User Voyage	VARCHAR	Data Analysis	Second User Voyage: internal voyage reference associated with the second vessel in a transshipment or connecting-voyage scenario.
ATA	DateTime	Vessel Turnaround Time & Data Analysis	Actual Time of Arrival: timestamp representing the vessel's recorded arrival at the port or terminal area.
ATB	DateTime	Vessel Turnaround Time & Data Analysis	Actual Time of Berthing: timestamp representing the point at which the vessel is recorded as berthed.
ATD	DateTime	Vessel Turnaround Time & Data Analysis	Actual Time of Departure: timestamp representing the vessel's recorded departure following completion of the port call.
Turnaround Time	Decimal / Duration	Vessel Turnaround Time & Data Analysis	Vessel Turnaround Time: total elapsed duration between vessel arrival and departure, commonly derived as ATD - ATA.

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Table A.2 continued from previous page

Column	Type	Source sheet	Description
LOA (m)	Decimal	Vessel Turnaround Time	Length Overall: maximum vessel length, expressed in meters, and used as an important physical characteristic for berth and vessel-size analysis.
Width (m)	Decimal	Vessel Turnaround Time	Vessel Width or Beam: maximum width of the vessel, expressed in meters, and used as a supporting vessel-dimension attribute.

These representative entries demonstrate that the data dictionary extends beyond container-level operational fields and includes additional vessel, voyage, timing, and physical-dimension attributes. The complete dictionary contains 75 documented entries and is summarized in [Appendix A](#).

Appendix B.

SQL Query Catalog

The complete SQL workstream was supplied in a Word document rather than a native SQL file. The catalog below preserves every numbered task and flags known grain, assumption, or transcription caveats.

Table B.1.: SQL modeling, exploration, and analysis catalog

No.	Question/purpose	Technique	Status/caveat
1	Discover existing primary keys	INFORMATION_SCHEMA joins	Structural validation
2	Check User_Voy for nulls	NULL filter	Precondition for proposed key
3	Add FactVesselPortCalls primary key	ALTER TABLE / PRIMARY KEY	Implemented in SQL workstream
4	Audit table and column data types	INFORMATION_SCHEMA.COLUMNS	Structural exploration
5	Align join-key data types	ALTER COLUMN	Required before foreign keys
6	Create foreign and unique constraints	ALTER TABLE / FOREIGN KEY / UNIQUE	ERD construction
7	Delete two unusable rows	DELETE by operation key	Rows 48676 and 48677 in SQL layer
8	Count operations, containers, vessels, voyages	COUNT and COUNT DISTINCT	Overall exploration
9	Count core entities by year	GROUP BY calling year	Available-year comparison
10	Vessel visits overall	COUNT DISTINCT voyage	Vessel frequency
11	Vessel visits by year	GROUP BY year/vessel	Year-specific frequency
12	Operator workload overall	Distinct subquery + GROUP BY	Operator concentration
13	Operator workload by year	Distinct subquery + GROUP BY	Year comparison
14	Full/empty split	Distinct container-voyage + GROUP BY	Transport mix
15	Imported/exported commodities	Cargo/transport joins	Commodity mix
16	Commodities by year and mode	Multi-dimension GROUP BY	Year-mode comparison
17	Cargo-type operation counts	GROUP BY cargo type	Cargo mix
18	Cargo/commodity/container type by mode	Multi-dimension join	Detailed mix; transcription contains syntax gaps
19	Loading-port operations	Distinct subquery + POL dimension	POL concentration
20	Loading ports by year/country/region	POL master enrichment	Geographic analysis

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No.	Question/purpose	Technique	Status/caveat
21	Discharge-port operations	Distinct subquery + POD dimension	POD concentration
22	Discharge ports by year/country/region	POD master enrichment	Geographic analysis
23	Monthly design-capacity utilization	CTE + TEU factor	Uses 500,000 annual TEU assumption
24	Crane productivity by call	Move/hour ratio	Exploratory; raw repeated QC hours
25	Monthly crane productivity	Monthly rate aggregation	Exploratory; differs from Excel and governed Power BI
26	Average dwell by month and mode	AVG StackDays	Entry-month / completed-record view
27	Year-over-year dwell	CTE + CASE pivot	2025/2026 Q1 comparison
28	Monthly containers/voyages/calls/dwell	Multiple distinct counts	Operational summary
29	Vessel turnaround by month	AVG at vessel-call fact grain	Vessel performance
30	Vessel turnaround by mode	Fact-to-fact join	Grain fan-out risk documented later
31	Import/export share by year	Window function	Q1 trade mix
32	Destination-country share	CTE + window function	Country concentration

Appendix C.

Selected Formula and Code Library

C.1. Power Query M

Listing C.1: Common text-cleaning pattern

```
if [SourceColumn] = null or Text.Trim(Text.From([SourceColumn])) = "" then
    "UNKNOWN"
else
    Text.Upper(Text.Trim(Text.Clean(Text.From([SourceColumn]))))
```

Listing C.2: TEU mapping

```
let
    LengthClean =
        if [ContainerLength] = null then null
        else Text.Upper(Text.Trim(Text.Clean(Text.From([ContainerLength]))))
in
    if LengthClean = "20" then 1
    else if LengthClean = "40" then 2
    else if LengthClean = "45" then 2.25
    else null
```

Listing C.3: Distinct nonblank QC-hour count

```
List.Count(
    List.Distinct(
        List.RemoveNulls(
            Table.Column([AllRows], "QCWorkingHoursSource")
        )
    )
)
```

C.2. DAX

Listing C.4: Completed-dwell measures by exit date

```
Average Completed Dwell Days =
CALCULATE (
    AVERAGE ( FactContainerOperations[CompletedDwellDays] ),
    USERRELATIONSHIP (
        DimDate[Date],
        FactContainerOperations[OutDateOnly]
    )
)
```

```

)

Median Completed Dwell Days =
CALCULATE (
    MEDIAN ( FactContainerOperations[CompletedDwellDays] ),
    USERRELATIONSHIP (
        DimDate[Date],
        FactContainerOperations[OutDateOnly]
    )
)

```

Listing C.5: Final latest-year TEU comparison pattern

```

Latest Operational Year =
CALCULATE (
    MAX ( DimDate[Year] ),
    FILTER ( ALL ( DimDate ), CALCULATE ( [TEU Entries] ) > 0 )
)

TEU Entered Current Year =
VAR Y = [Latest Operational Year]
RETURN
CALCULATE (
    [TEU Entries],
    REMOVEFILTERS ( DimDate ),
    DimDate[Year] = Y
)

```

C.3. Python source extract

The full plain-text transcription is included at:
[source_extracts/Python_work_1_extracted.txt](#).

It is preserved as evidence, not represented as a re-executed notebook.

C.4. SQL source extract

The full extracted SQL workstream text is included at:
[source_extracts/SQL_Work_1_extracted.txt](#).

Syntax and commentary are retained from the source document and have not been silently repaired.